Modern Abattoir Practices and Animal Byproducts Technology

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Meat Animals

India is an agrarian country where more than 65% population lives in 5.8 lakh villages wherein about 73% people have their own livestock. Thus, livestock rearing constitutes an integrated and important segment of rural economy accounting for 14-40% of the total farm household income. In general, important meat producing species are largely those animals which consume food of plant origin. The flesh of carnivorous animals is rarely used as human food. In India goat, sheep, pig, buffalo and to some extent cattle are the principal meat animals. Of these, goat, sheep, buffalo and cattle meat constitutes nearly 75% of the total meat production. Meat of these species is often referred as red, whereas poultry meat is usually referred as white. Pork is intermediate in colour.

Our livestock population exceeds 500 millions. During 2001, nearly 4800 thousand MT of meat was produced (Table 1.1) from the slaughter of 107.64 million animals and 413.4 million chicken. Inspite of vast livestock resource, India produces only about 2% of the total world meat production. The export of meat and meat products are very meagre, i.e. 1.41% of the total world meat trade (Table 1.2). Frozen meat accounts for 65-75% of the total meat export. At present, access to markets in developed countries does not exist because of animal health issues.

The per capita meat availability in India is 4.26 kg per year which is further expected to increase in the coming years. The annual protein supply is 9.5g per head/day in our country against Asian average of 15.7 g and the world average of 24.6g. Thus, we have to take up enormous development programmes in meat production sector even to achieve the target of recommended intake of 11 g per head per day.

Fresh students should note that carcass yield of meat animals is calculated as a percentage of live weight.

$$\text{Dressing \%} = \frac{\text{Dressed weight}}{\text{Live weight}} \times 100$$

Table 1.1: Livestock population, meat production and slaughter rate during 2001

S. No.	Species	Population			Meat production				Slaughter rate (%)		Dressing yield (kg/animal)	
		India (million)	World (million)	Top ranking nation	India's rank	India 1000 MT	World 1000 MT	Market share India/ world	Market share within India	World	India	World	India
1.	Cattle	219.64	1351.79	India	1	1462	5664	2.58	30.60	20.52	6.47	204	103
2.	Buffalo	94.13	165.72	India	1	1426	3132	45.56	29.87	13.49	10.34	140	138
3.	Sheep	58.20	1056.18	China	3	230	7532	3.06	4.81	45.91	32.98	15	12
4.	Goat	123.50	738.25	China	2	469	3759	12.48	9.82	42.14	37.48	12	10
5.	Pig	17.50	922.93	China	9	595	91188	0.65	12.45	127.07	97.14	78	35
6.	Chicken	413.40	19859	China	6	595	69948	0.85	12.45	309.65	160.14	1.40	0.90

Source : www.fao.org

Table 1.2: Export of Livestock Products from India

Item	1998-99		1999-2000		2000-01	
	Quantity (000MT)	Value (Crores)	Quantity (000MT)	Value (Crores)	Quantity (000MT)	Value (Crores)
Buffalo meat	153.95	691.29	167.29	706.42	288.02	1375.04
Sheep and goat meat	8.64	78.47	12.41	90.89	11.90	78.16
Processed meat	0 .29	2.99	0.81	4.58	0.13	1.58
Animal casings	0.38	13.59	0.35	11.69	0.57	12.29
Poultry products	7.39	51.72	8.36	54.25	15.83	86.17
Dairy products	2.56	13.65	6.13	37.21	11.06	83.89

It varies with species, breed, age, sex and plane of nutrition. High dressing percentage is desirable only if the increase is brought about by muscular growth rather than deposition of excess fat.

Goat

Goat is referred as `*Asian Animal*' in the world livestock arena because almost 95% of goat population belongs to the developing countries of Asia. India ranks second in the goat population which is more than double to that of sheep. Though regarded as poor man's cow, goat has the distinction of being the most important meat animal of India because it provides choicest of all meats fetching maximum retail price in the home market. However, of late, relative contribution of goat and sheep meat to total meat production has gone down.

There are nearly 20 recognized breeds of goats, although it is very difficult to demarcate them into milk and meat breeds because of their dual nature. However, an attempt has been made to classify them on the following lines:

Meat breeds	*Fibre breeds*	*Dual purpose breeds (milk and meat)*
Sirohi	Gaddi	Jamunapari
Malabari	Marwari	Jakhrana
Black Bengal	Pashmina	Osmanabadi
Zalawadi	Chegu	Beetal
Sangamneri		Surti
Ganjam		Barbari

Jamnuapari, *Beetal* and *Jakhrana* breeds are large in size. They have comparatively high milk production also. But 80% of the goat breeds are medium sized. *Sirohi*, *Mehsana*, *Surti*, *Gaddi*, *Marwari* etc. attain 18 to 20 kg weight at 6 months of age which is ideal for optimum meat yield. The dressing percentage of an animal of good conformation varies from 43 to 48%. *Black Bengal* and *Barbari* are small sized breeds that attain about 12-13 kg of pre-slaughter weight. However, dressing percentage of Barbari is comparatively high (47 to 50%) because of its compact body and short stature. *Sirohi* and *Marwari* breeds of goat have a meaty conformation. Marwari males are specifically reared for meat.

Out of about 124 million goats, almost 38% are slaughtered yearly to contribute nearly 10% to the total meat production.

Goat meat (*chevon*) is light red in colour and is very firm in consistency. It imparts typical goaty odour. Goat carcass has an abundance of kidney fat which is white in colour. Chevon has practically no intramuscular fat. Due to low saturated fat content, the demand for chevon is on the increase.

Sheep

In India, sheep are primarily reared for wool and meat in arid, semi-arid and mountainous areas mostly unsuitable for crop farming. Besides, pelt (skin) and some times milk are also obtained from sheep. In fact, skins from sheep and goats are fairly valuable and about 90% are recovered from their slaughter. As far as meat production is concerned, almost 5% is derived from this species by slaughter of 33% sheep population every year.

India stands third in sheep population in the world with vast genetic resource of as many as 40 breeds. According to their utility, some important breeds could be classified as follows:

Apparel wool breeds	*Carpet wool breeds*	*Mutton and carpet wool breeds*	*Mutton breeds*	*Exotic breeds*
Kashmir-Merino	Chokla	Muzaffarna-gari	Nellore	Merino (Spain)
Niligiri	Nali	Deccani	Mandya	Ramboillet (France)
Hissardale	Pattanwadi	Ganjam	Lohi	Polworth (Australia)
	Gaddi	Malpura	Bandur	Suffolk (England)
	Rampur Bushair	Coimbatore	Hassan	Corriedale (New Zealand)
	Bhakharwal	Marwari	Madras Red	
	Gurej	Balangir	Trichy Black	
	Poonchhi	Chottanagpuri		
		Magra		

Hairy breeds of sheep are reared primarily for mutton in the entire Peninsular region of India. Of these, *Nellore breed* has small bones with well developed musculature and is good only for mutton. *Trichy Black* is a very good breed for mutton. *Madras Red* is reared specifically for mutton only. Out of dual purpose breeds, *Coimbatore* and *Marwari* breeds are small *whereas Deccani* and *Magra breeds* are large in size. Among exotic breeds, *Polworth* produces mutton of the most desirable quality whereas *Corriedale* is the most important dual purpose breed imported in India.

Table 1.3: Live weight and dressing percentage of some sheep breeds

Breed	*Weight at 6 months (kg)*	*Weight at 12 months (kg)*	*Dressing %*
Nellore	17	23	47
Mandya	12	21	45
Muzaffarnagari	15	25	48
Deccani	20	–	48
Malpura	13	21	47
Coimbatore	11	15	38
Chokla	13	18	48

In general, sheep weigh 13 to 16 kg at 6 months of age except for *Deccani* and *Magra* which weigh about 20 kg. At 12 months of age, sheep weight 18 to 22 kg except for *Muzaffarnagri* and *Magra* which weight 25 and 28 kg, respectively. A healthy sheep should give a dressing percentage of about 45 to 48 which may go upto 50% or more in exceptionally well bred stock. A sheep carcass having less than 40% dressing percentage (Table 1.3) is viewed with suspicion. Meat from sheep of less than 12 months of age is referred as *lamb* and that of more than 12 months of age is called *mutton*. It should be noted that in trade, *wether* refers to male sheep castrated early in life whereas *gimmer* is a female sheep which has not yet borne a lamb. Mutton has light to dark red colour with a firm and dense consistency. It has got a ammoniacal odour. Male sexual odour is sometimes felt. Sheep fat is white in colour with hard and firm consistency. Sheep carcass has very less quantity of fat around its typical bean shaped kidneys.

Pig

Pigs are reared for the production of pork. These are highly prolific producing two litters per year, each time farrowing upto 10-12 piglets. Their growth is also very fast. Pigs have the capacity to convert inedible feeds and garbage into nutritious and costly meat. The species is largely maintained by the economically weaker sections of the society. In fact, pig rearing has traditionally provided livelihood to millions of landless and downtrodden people where it served as a backyard enterprise. Indigenous or *desi* pig formed the basis of farming. These are

small in size. However, in the last three decades, breeding with the two imported breeds—*Yorkshire* and *Landrace* have been widely used for crossbreeding in India and the breeding programme has shown very good result around bacon factories. Further breeding of desi pigs with these crossbreds has brought about a significant improvement in the overall stock within the country. Middle sized crossbreds now form the basis of pig production in our country. Sound piggery units are now being established because of support being provided by the government. Pig population has maintained an encouraging growth during the last over one decade. Pig rearing for meat is very common in North Eastern Region of the country. Almost every rural household rears them and thus pig farming is playing a vital role in improving the socio-economic conditions of the rural population.

Pigs should be slaughtered at 6 to 7 months of age. A ideal slaughter weight is approximately 50 kg for desi pigs and 60-70 kg for crossbred pigs. Dressing percentage varies from 65 to 70 in case of desi pigs and 70 to 75% of the live weight in case of crossbred pigs. The yield is more from pigs as compared to other species due to the presence of skin on the carcass. Almost 98% of pig population is slaughtered every year which contributes to about 12.5% of the total meat production. Pork is grey pink in colour with very soft consistency. It emanates a urine like odour. There is a huge subcutaneous deposition of fat which is white in colour and soft greasy in consistency. Consumption of pork is prohibited for Muslims as a religious restriction.

Buffalo

India ranks first with 59% of the world buffalo population. Indian *water buffalo* is primarily a milk animal and as such there is no organized rearing of buffaloes for meat. However, unproductive and uneconomic buffaloes after 5-6 lactations find their way to the slaughter house. It is noteworthy that after independence annual growth rate of buffaloes far exceed that of cattle. The demand for buffalo meat is on the rise to meet the export requirement to Middle East countries and Malaysia. Male buffalo calves with proper feeding and management offer vast potential for good quality and better priced meat for export. A dressing percentage of about 50 is obtained from well maintained male buffalo calves of less than 3 months of age which are slaughtered for high priced veal.

Indian buffaloes show a wide variation in dressing percentage and generally figures vary from 50 to 55% of the live weight depending on the condition of the animals. Ideal age for slaughter of buffalo calves for meat has been recommended as 18 to 24 months. Out of a total of 90 million buffaloes, almost 11%, mostly unproductive and uneconomic animals are slaughtered every year but they contribute nearly 29% of the total meat production.

Buffalo meat is many times referred as *carabeef*. It is light red to bright cherry red in colour and firm in consistency. It is leaner than beef and has a reasonable degree of marbling. Buffalo fat is normally white in colour and firm in consistency.

Cattle

In India, cows are reared only for milk production. Slaughter of cow is banned by law in most of the states except the ones ruled traditionally by communists such as West Bengal and Kerala. However, slaughter of bullocks does take place at specific places having predominantly Muslim population. Carcass yield varies from 50 to 54% depending on the condition of the animal.

It should be noted that in trade *steer* refers to the bullock which is castrated at 6-12 weeks of age. *Stag* is also a bovine male but it is castrated in late in life allowing the development of a better musculature.

JUDGING AND GRADING OF LIVE MEAT ANIMALS

Meat industry aims at producing animals with large amounts of highly priced and palatable lean and least amounts of fat and bone. Judging and grading of meat animals ready for slaughter is a widely practiced art for evaluating their value at market. It is based on *visual appraisal* and *body measurement* in live animals for desired carcass characteristics. During judging, *type* defines all characteristics of the animal raised for specific purpose whereas *conformation* refers to the form and shape of the animal. A good judge should be familiar with the anatomy of meat animal, terminology and location of high priced wholesale cuts within the animal body. He should possess keen observation and analytical mind besides ability to make a comparative assessment and arrive at a logical decision.

A systematic method of judging and grading involves:

i. Initial observation of the meat animal from a distance,

ii. Determination of conformation of the animal by *degree of fleshing* and firmness of finish by having side, front and rear views,
iii. Making the animal walk for some distance and looking for qualities, commonly accepted for the standard grades
iv. Feeling the animal along the top of the shoulder, ribs, back, loin, rump and round etc.

Judging and Grading of Sheep and Goat

Meat animals are judged on the basis of type, conformation and finish. *Conformation* gives an overall assessment of muscling. A broad forehead, short thick neck, short legged deep wide blocky body with symmetrical conformation are the desired traits. *Flank fullness* and *firmness* is also an important parameter which is determined by taking hold of the flank muscle with hand. Special care is taken to look for wide chest, hind legs, wide and thick back and loin along with long rump and bulging stifle. Thigh and loin form the highest priced cuts on the mutton carcass.

Lamb and mutton quality grades may be prime, choice, good and utility whereas yield grades from the highest to lowest may be 1 to 5.

Judging and Grading of Swine

A meat type swine encompasses all the characteristics and contribute maximum in terms of valuable wholesale cuts. Swine judging card developed by USDA considers general appearance, form, finish, quality and expected dressing percentage. General appearance as observed from a distance should take note of general length, depth and width, strength of arch and development over back, loin and rump, fullness of ham and neatness of head and neck. Fairly high and uniform finish is important to get firm, high quality pork cuts. Back and loin should be of good length and moderately wide with uniform arch. An ideal ham should be wide from front to rear, plump, smooth, firm and trim at the base. A belly that is neat, trim, firm and smooth contributes to a good dressing percentage. Length of legs should be medium.

During grading, sex classes of swine need to be described several times. *Boars* and *sows* are older breeding animals. *Barrow* is the male pig castrated early in life whereas *stag* is the male pig castrated after it has developed certain boar like characteristics.

Gilts are younger females that have not produced any young ones. Grades for gilt and barrow are assigned on the basis of expected combined yield of four major cuts that in turn depends on degree of muscling and back fat thickness.

Judging and Grading of Buffaloes and Cattle

Conformation and age play major role when these animals are judged for meat production. Age has a significant effect on tenderness. *Veal* is obtained from young calves of 1-3 months of age whereas calf is from animals ranging between 3-10 months in age. Mature animals over 12 months of age produce beef or carabeef (buffalo meat). A desirable conformation of meat type animal should be long, trim, deep-sided body, heavy muscled forearm and maximum development of round, rump, loin and ribs. There should not be excessive fat on brisket and flanks. Besides, there should be no extra hide around the throat, dewlap and sheath. Highest USDA quality grade start from prime and yield grades from 1.

SOME OTHER MEAT SPECIES

Poultry

Indian poultry industry has made a phenomenal progress in the last few decades. It was transformed into a sophisticated industry quite rapidly after the establishment of several franchise hatcheries in 1970s. This growth coincides with the rapid demand for white meat in the wake of ongoing campaign for low fat, low cholesterol meat. Due to intensification of poultry industry, traditional breeds have lost their commercial importance and modern chicken is the product of several crossbreeding and strain crossing. The only exception is *White Leghorn* which forms the backbone of egg industry with hen house production of 260 eggs per annum. *Rhode Island Red* produces 220 eggs or so. *White Cornish, New Hampshire* and *White Plymouth* Rock have been crossbred at many places to produce broiler strain. These days live weight of broilers at the age of 6 weeks and 8 weeks are about 1.20 kg and 1.60 kg, respectively. The broilers have a dressing percentage of 65 to 68, whereas White Leghorn and Rhode Island Red have a dressing percentage of nearly 70. Broiler meat is a true delicacy while it is available at comparatively lower price than chevon. Cockerel raising is emerging as an alternative to persistent shortage of broilers. These birds can be raised to 10

weeks to get a desired weight of 1 kg or even more. Another species of poultry, Japanese quail is also growing fast. It gains a weight of about 125 g at 5 weeks of age. The dressing percentage is about 70%.

Rabbit

Rabbit is emerging as an alternate source of meat in the hilly tracks of India. These are prolific, grow rapidly and produce meat from cheap roughages. The potential breeds of broiler rabbit are *New Zealand White, Chinchilla, Californian* and *White Giant*. The average live weight ranges from 1.2 to 1.5 at 8 weeks and nearly 2 kg at 12 weeks. Carcass yield varies from 52 to 58% in different grades. Rabbit carcass contains 82% lean meat which is white in colour and soft in consistency. It is comparatively low in sodium and rich in potassium and phosphorus. The saturated fatty acids account for more than 60% of the total fatty acids whereas free cholesterol is also relatively low.

The meat production pattern of our country is quite different in comparison to the world. Besides, horse meat is not at all eaten in India. Deer comes in the category of protected wild life. So *venison* (deer meat) is also not to be served or eaten. We have to strive hard for increasing the productivity of principal livestock species to make them internationally competitive. A large market base is also required to be created to increase the export of processed meat products.

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Organization, Layout and Management of Slaughter houses

Animals meant for human consumptions are sacrificed at a place, premises or building called slaughter house. This is technically referred as *Abattoir*. Traditionally, these premises or buildings were called butcheries and due to unhygienic disposal of wastes, it generally emanated foul smell. Butcheries gave a negative feeling and public always raised a hue and cry over the establishment of slaughter house in their locality. It has been emphasized that such places should be maintained on hygienic lines and be conveniently referred as meat production plants. Landscaping and development of an abattoir area can make it an acceptable preposition. Development work may include provision of proper road, abundant water and electricity, effluent treatment plant alongwith production of methane as a source of energy.

PLANNING

Planning of a modern abattoir involves

i. Policy decisions
ii. Engineering aspects
iii. Cost consideration

Policy Decisions

These will necessitate decisions on site, size, trade practices and range of byprodcts required to be processed. Besides, it should also satisfy the sanitary and hygienic measures expected to be undertaken so as to produce clean and wholesome meat at a reasonable cost. The building must be clearly divided into clean and unclean sections.

Selection of Site for an Abattoir

1. Abattoir must be located in the outskirt of the town or city.
2. The building should be constructed at an elevated ground to provide easy drainage.
3. There should be provision of adequate and potable water supply.
4. It should have concrete road facilities where transportation could continue throughout the year. This road must have access to major roads into area.
5. There should be arrangement of uninterrupted electric supply.
6. There must be enough open space nearby for future expansion.
7. Abattoir has to be located far away from airport.
8. Although it is not always possible, due consideration should be given to the social and religious acceptance of atleast a part of the nearby population.
9. It should be free from pollution of industrial odours, smoke, dust, ash etc.
10. Technical and skilled manpower should be available in the nearby area.
11. There should be availability of sufficient number of good meat animals in the region.
12. The soil should be able to support good foundation and pillars of the spacious building.
13. The abattoir building should have a separate entity at a reasonable distance from any other building.

Size of Abattoir

Size of abattoir will depend on:

1. Number of animals of each species to be slaughtered every day and future projection. As far as species are concerned, each buffalo/cattle constitutes 1.0 livestock unit, pig constitutes 0.5 livestock unit whereas goat/sheep constitute 0.2 livestock unit. Area requirement :
   a. Small abattoirs slaughtering < 100 livestock units/day = 1-2 acres,
   b. Medium abattoirs slaughtering 100-200 livestock units/day = 2-4 acres
   c. Large abattoirs slaughtering > 200 livestock units/day = 4-6 acres.
2. Range of operations to be undertaken
   a. Whether slaughter and dressing only
   b. Whether carcass chilling also to be done
   c. Whether deboning and packaging also to be done
   d. Whether byproduct processing also to be undertaken
3. Availability of capital
4. Market to be covered—domestic or export or combined and marketing channels in vogue.

This is an era of specialized abattoirs for buffalo/cattle, sheep/goat and pig. It must be borne in mind that in India due to religious considerations, pig abattoir must be a separate building, preferably away from others.

Engineering Aspects

Layout Plan

The abattoir should have the following essential facilities:

1. Resting place for animals before slaughter
2. Ante-mortem examination
3. Humane slaughter
4. Flaying, dressing and washing of the carcasses
5. Hanging carcasses and edible offals
6. Handling byproducts
7. Inspection of meat and disposal of condemned carcasses or part thereof.
8. Staff welfare

A flow diagram for basic operations of an abattoir is shown in (Fig. 2.1) and typical layout plans for buffalo/cattle, sheep/goat and pig abattoir are presented in (Figs 2.2 to 2.4).

As a thumb rule, an abattoir should have a clearcut separation between clean and unclean sections. In continuous flow, the edible meat and offal should never come in contact with any inedible material or byproduct or waste.

ABATTOIR DESIGN AND CONSTRUCTION

The design of abattoir varies from situation to situation. There can be no single blue print for all the situations. The guiding principle should be to provide all related services under hygienic conditions at lowest cost. In general, the following basic guidelines should receive serious considerations:

1. Abattoir building should face either towards east or west.
2. The operational facilities should be so designed and located that a clear demarcation exists between clean and unclean

Fig. 2.2: General layout of a combined abattoir

Fig. 2.1: Flow diagram of basic operations of an abattoir
(Source: IS: 4393-1967)

Fig. 2.3: Layout of an abattoir for Sheep and Goat

Fig. 2.4: Rough layout of a Pig abattoir (sequence of operations)

sections. There should be only forward flow of operations without any possibility of reversal.

3. A modern abattoir building should have the following essential components or units :
   i. Lairage or livestock holding pens
   ii. Slaughter hall, with clearly demarcated
      a. Stunning section
      b. Bleeding section
      c. Skinning section
      d. Dressing section
      e. Evisceration section
      f. Postmortem section
   iii. Isolation block/emergency slaughter unit
   iv. Chill rooms
   v. Detained and condemned meat rooms
   vi. Hide and skin store
   vii. Gut and tripe room
   viii. Offal room
   ix. Meat cutting room
   x. Veterinary office and laboratory
   xi. Despatch room
   xii. Cloak room
   xiii. Manure bay
   xiv. Effluent plant
4. Building floors depend on the level of operations and the capital available. Single floor generally minimizes the handling cost. In a double storey building, upper floor is used for slaughter and clean operations whereas ground floor is used for offal processing and all unclean operations. All the floors should be made up of concrete, easily cleanable and non-slippery material. A gradient of about 2 cm per meter is generally recommended.
5. Abattoir ceiling should be at a height of atleast 5 meters. The walls should have white ceramic tiles to promote sanitation and light reflection.
6. The building should have efficient drainage with one drainage inlet for every 36 m^2 of floor space. The drains should have effluent flow in opposite directions from edible product flow. A slope of 2 cm per meter to drainage inlet is required for usual conditions. Sewage disposal system should be efficiently designed and operated. Each floor drain including blood drains should be equipped with a deep seal 'S' shape trap.

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7. Adequate ventilation must be provided to prevent excessive heat and steam condensation. Besides a good ventilation also prevents accumulation of odour and dust. The opening windows should be screened.
8. There should be provision of enough water to cope with both clean and unclean operations in the abattoir. As per FAO Bulletin (1978), the water requirements are :
   a. 1000 litres/buffalo or cattle
   b. 100 litres/sheep or goat
   c. 450 litres/pig
9. Electricity load of an abattoir should be assessed in advance. It is generally required for artificial lighting, exhaust fans, operational equipment, chilling etc.
10. There should be adequate natural light and provision of artificial light throughout the abattoir. The artificial light should not disrupt colours. It is recommended that the overall intensity of light should not be less than

    550 lux (50 ft candle) at all inspection points alongwith packing point.
    220 lux (20 ft candle) in all work rooms
    110 lux (10 ftt candle) in the chilling room.

    These intensities are measured at a height of 0.9 meters from the floor except for inspection points where the relevant height is 1.5 meters.
11. Mechanical handling systems and other type of equipment used in abattoir should be designed and located so as to permit easy and efficient cleaning and disinfection.
    In modern abattoirs, overhead rail system ensures the basic requirement of hygienic conditions during slaughter and dressing. These overhead rails conveniently move the carcasses and increase the efficiency of operations. Besides, these are aesthetic in appearance.
    The overhead rails may be either gravity rails or conveyor (feeder) rails. In our conditions, gravity rails are preferable. Hooks with shackle, chain or gambrel can be conveniently attached in such rails. The height of these rails (vertical distance between from floor) differ in different sections of the abattoir:

Height	*Buffalo/cattle*	*Sheep/Goat*	*Pig*
Bleeding rail	16 ft (4.8 m)	10.5 ft (3.15 m)	9 ft (2.7 m)
Dressing rail	11 ft (3.3 m)	10 ft (3 m)	7.5 ft (2.25 m)
Cooler rail	11 ft (3.3 m)	9 ft (2.7 m)	6.5 ft (1.95 m)

The overhead rails are made up of galvanized steel and require frequent cleaning and oiling with edible oil to prevent dusting. Conveyorised rails are very suitable for large abattoirs because

i. Carcasses arrive in different positions without any manual handling.
ii. Carcass distance is fixed eliminating the chances of cross contamination.
iii. Meat inspection is comparatively easy.

A patented continuous conveyorised method called *'Canpak'* system has become popular in developed countries wherein carcasses are suspended by heavy trolleys from overhead rail and no spreader or gambrel are used.

12. A boiler of suitable capacity has to be installed in the abattoir to generate steam which is required for heating water for clean-up operations (60°C), sterilization (100°C) and for rendering the byproduct.
13. Provision for compressed air is also made these days in the abattoir for running the pneumatically operated tools and for ultimate cleaning as it is capable of drying even the droplets.
14. A chilling room is an inevitable necessity of the modern abattoir. It should have atleast two chambers—one maintained at 7°C for carcasses and cuts and the other at 3°C for offal.
15. A separate drain for water and water containing manure is desirable. This waste water can be pumped in dry or wet pit where manure can be screened and disposed of suitably.

Building Components

Livestock receiving area In the unloading area, the animals should be handled gently. It should have facilities for watering and feeding. The animals are subjected to veterinary inspection before sending to lairages. Sick and injured animals are sent to isolation pen.

Lairages Animals have to travel long distances to reach abattoir and they need a rest of nearly 24 hours to restore their normal physiological condition. It has a definite bearing on the quality of meat. The lairage area is protected from heat, cold and rain. There should be arrangement for feeding, watering and antemortem inspection. Lairage facilities should be enough to hold nearly 2 days kill for large animals and 1 day kill for small animals. The space requirements are 2.8 m^2 for buffalo and 0.6 m^2 for small animals.

A distance of atleast 10 meters is kept between lairage and slaughter hall. This space is converted into a tapering narrow race.

Slaughter hall Stunning or knoking pen is required for humane treatment of the animals. It should restrict the forward and backward movements of the animals without any discomfort. The trap door should permit only one animal and it should not see the other animal being stunned or slaughtered.

Slaughter hall of a modern abattoir has the following equipment:

a. Dehorner
b. Hock cutter
c. Hide puller
d. Brisket saw
e. Aitch bone cutter
f. Carcass split saw

Isolation block/emergency slaughter unit It is a miniature abattoir. It has small lairage for few sick or suspected animals besides a slaughter pen with hanging facilities. It is situated adjacent to meat detention room.

Chill rooms Wherever hot (unchilled) meat consumption is in practice dressed meat and offal are despatched from the premises for immediate sale and cooking. In all other cases, carcasses should be chilled soon after the postmortem inspection is over. Chilling temperature should not exceed 7°C for carcasses and 4°C for offals as per EEC regulations. These conditions are applicable to the storage and transport also. The temperature of cutting room should not exceed 10°C. Rail spacing in the chill room should be 0.9m for buffalo, 0.7m for pig and 0.5m for sheep/goat carcasses. It will maintain the minimum space between carcasses at 0.3 to 0.4m, which is necessary to check cross contamination and allow free movement of cold air around the carcasses.

Condemned meat room This room holds the material unfit for human consumption before the despatch. It should have durable and covered containers for hygienic disposal. Weighing facilities are must.

On the basis of management systems, abattoirs can be broadly put into two categories:

Service Abattoirs

These abattoirs generally belong to municipal authorities which are responsible for the general upkeep and maintenance of the building. Meat Inspection is also undertaken by the Government Veterinary Officers. Such abattoirs can be of two types:

i. *Booth system* This system is prevalent in most of the Indian abattoirs. Here, the butchers/traders bring their animals and slaughter them on the floor. The municipal authorities charge a particular fee for the maintenance and operation of the slaughter house. They also raise some money through the sale of manure, blood and bile. Meat inspection is done by the government veterinarian. This system has the advantage of individual and quick approach but hygienic conditions remain poor and operations are congested.

ii. *Line system* This system is prevalent in some modern abattoirs (e.g. Debonair Abattoir, Mumbai). In this system, the traders hand-over their animals to the abattoir that undertakes the service of slaughter and dressing on the overhead rail/conveyor line and charge fee. The butchers are highly skilled. The meat inspection is done by the municipal veterinary officers. This system has the advantage of conducting hygienic operations but the time taken is more as compared to the booth system because the entire chain or sequence of operations is done on the dressing line. Besides charging slaughter fee, municipal corporations also raise some money by the sale of manure and byproducts.

Commercial Abattoirs

These abattoirs are owned and operated by some meat exporters in India (e.g. M/s Allana, M/s Al-Kabeer etc.). Their aim is to produce high quality meat at reasonable profit margin. They have modern plant and machinery conforming to the concept of modern abattoir. They maintain stringent quality control measures to sustain the name of their firm/brand in the market.

These abattoirs have a well organized functional set up with a board of management, meat inspectors, supervisors and skilled operating staff. They have to keep a close watch on the consumers choice, emerging markets and maintain high efficiency to remain in close competition within the trade. The financial returns come from the sale/export of fresh meat, offal, processed meat, byproducts etc. The problem generally faced by them is the control of raw material consistency due to highly variable food animals in India.

Semi-modern Abattoirs—A Workable Proposition

In India, slaughter and bleeding of small ruminants—sheep and goat take place on the floor but dressing takes place in the hanging position. Hence, the chances of further contamination are reduced to a large extent. In contrast to this, large ruminants—buffalo and cattle are generally slaughtered and dressed on the floor due to heavy carcasses. This practice allows heavy soiling and contamination of carcass/meat due to contact with blood and ruminal contents.

There are nearly 3600 licensed slaughter houses in our country. Most of them do not have a proper building. These are poorly lighted and ventilated. Water supply and drainage facilities are also not upto the mark. These slaughter houses are ill equipped to produce hygienic and wholesome meat. Government is also concerned about it and schemes are being launched to modernize many of these abattoirs in phases. However, it needs lot of finance. Even if modern line abattoirs are established with a heavy investment, it would not be easy to operate them for less than 100 animals per day. Besides, time factor is also important as far as present day Indian meat trade is concerned.

An intermediate concept between western type modern line slaughter and traditional floor slaughter is being advocated as the most satisfactory workable proposition for Indian conditions. Such semi-modern abattoirs or slaughter houses have facilities for slaughter and bleeding on the floor but subsequently entire dressing, washing, halving and quartering takes place on the rails in hoist position. The system involves provision of multi hoist points in the slaughter hall alongwith short rail facilities for mechanically lifting the carcasses at several points. Thus, slaughter hall will have so many slaughter pens with mechanical hoist facility. On the basis of demand for a particular day, the slaughter pens can be allotted to different butchers to carry out self controlled slaughter and dressing under hygienic conditions. Such a quick system can have better acceptability among the

butchers. Stunning of the animals can also be done by captive bolt pistol by a single skilled worker. Chilling facilities may not be necessary at such a slaughter house if hot or pre-rigor meat is purchased and cooked or processed within a few hours.

Tips for Efficient Abattoir Management

All local civic authorities—Town Area, Municipalities, Corporations etc. have their own laws and regulations to enforce hygienic conditions and management norms. Some of them have their Do's and Don'ts. Our discussion will confine only to general principles.

1. Thorough meat inspection has to be made a compulsory routine necessity in the abattoir set up.
2. Dressing of carcasses should not be done on the floor.
3. The number of slaughtering hours should be limited. After working hours, the premises should be sufficiently cleaned by detergents/sanitizers and pressurized hot water.
4. All doors, windows, ventilators etc. should be covered with wire net to check the ingress of flies and other insects. Electrically charged grill with UV light may be installed to control the menace of flies.
5. All the workers should be provided with clean aprons and head gear daily. There should be well defined storage bins for dirty ones. The workers should wear gum-boots and pass through phenyl water dip before entering the main building. Proper accommodation must be provided to store the protective and other clothing with locker facility. Smoking is not allowed within the premises.
6. There should be enough taps with sinks for hand washing with hot or cold water and these should be conveniently accessible to workers. In fact all movement are considered as wasteful. So hand washing facilities should be close to the work place.
7. Inflation or blowing of the carcass is a fraudulent practice and it should not be allowed. It misrepresents the body and muscular condition of the carcass.
8. Use of wiping cloth should not be permitted. After wiping a few carcasses, the cloth may become a source of contamination.
9. Waste disposal should be taken up as a routine on all working days.

10. The management should establish a good working relationship with the inspection staff.
11. Requirements of human resource in an abattoir:

Staff	*Buffalo 60 per day*	*Sheep/Goat 150 per day*	*Pig 100 per day*
Abattoir Incharge	1	1	1
Foreman	1	1	1
Clerks	2	1	1
Skilled workers	15	10	10
Maintenance fitter	1	1	1
Labour	2	1	1
Security guard	1	1	1

12. Efforts should be made to improve the surrounding environment by suitable plantation.
13. Effluent treatment should be done as a matter of routine.
14. Concrete steps should be taken to check the unauthorised slaughter by the unsc rupulous traders.

three

Preslaughter Care, Handling and Transport of Meat Animals

It is not enough to produce healthy meat animals, it is equally important to ensure that these animals reach the point of slaughter in sound condition. Preslaughter care and handling can markedly influence the quality and quantity of meat. Ways of loading and unloading, means of transportation and average distance covered by the animals from the point of production to the point of slaughter has a definite bearing on the keeping quality of meat. Excited, stressed, fatigued, suffocated, bruised and injured animals are not expected to yield wholesome meat.

The underlying principles for preslaughter care, handling and transport of meat animals are:

1. To avoid unnecessary suffering of animals during transport
2. To ensure minimum hygienic standards
3. To prevent spread of diseases

TRANSPORT OF MEAT ANIMALS

The mode of transport should be decided on the basis of ground situation. Unless price differences are significant, the animals should be taken to the nearby slaughter house or abattoir avoiding long journeys. It will protect the animals from possible injury during loading and unloading as also adverse weather and inadequate ventilation during transport. Various modes of transport may be:

1. Driving on hoof
2. Transport by road truck
3. Transport by railroad
4. Transport by sea
5. Transport by air

Driving on Hoof

Animals reared within 8-10 km from the point of slaughter can be driven on foot. This distance can be covered in 4 to 5 hours. The time can be adjusted in early morning during summer and late morning during winter season. This mode is especially suitable for animals accustomed to pasture grazing. It allows them to browse on indigenous grass or shrubs and take water en route. So the animals suffer minimum weight loss or shrinkage due to travel.

Transport by Road Truck

Distances upto 500 km or 12-15 hours journey may be negotiated by road truck. Such vehicles should have non-slip floor. It is also important to provide temporary or permanent protective overhead coverings, making provision for adequate ventilation. Proper partitions should separate unequal sized animals or different species. The partition height may vary from 75 to 120 cm depending on the size of animals. Before the commencement of journey, the animals should be offered adequate feed and water. If the journey is required to be continued after 12 hours, the animals should be unloaded and offered enough feed and water. Transportation by road trucks allows convenience of loading and unloading at the appropriate places.

Transport by Rail Road

For distances over 500 km, it is advisable to transport animals by railways. It is economic on maintenance and easy to handle. Besides, there is a saving on extortions and other incidental expenses. Animals should be provided ad lib water and feed at least for an hour before the journey commences. It will be beneficial if water troughs are provided within the roofed rail wagons. Railway wagons meant for this purpose should have a non-slip floor and a free flow of air. Arrangement can be made to unload the animals after about every 1000 km and offer feed and water before reloading. This mode ensures comparatively less losses due to shrinkage and death. In fact, shrinkage losses may come down to as low as 5 percent.

Transport by Sea

Ships are used only for international transport.

Transport by Air

It is very rare and used only for highly expensive animals.

EFFECT OF TRANSPORT ON MEAT ANIMALS

1. *Stress and fatigue* These conditions are inevitable sequel to transportation and do have a bearing on meat quality. These conditions may at times lead to shipping fever and transient tetany. Shipping fever develops due to pasteurella and requires proper treatment, otherwise virus may act as secondary invader and aggravate the condition.
2. *Loss of weight or shrinkage* Shrinkage takes place due to dehydration (loss of water) and depletion of muscle glycogen during the period of journey. In general, it ranges from 3 to 10 percent depending on the conditions and duration of transport.
3. *Bruises, torn skin and broken bones* Bruises are noticed in most of the species due to transportation. The instances are particularly high in sheep and pigs. Muscular bleeding may occur especially in pigs.
4. *Death* It may occur during long transportation. Sheep and pigs are particularly susceptible if animals of unequal age and size are loaded in road trucks without partitions due to suffocation. Sheep and goats could also die in long distance transportation by ship due to non-inflammatory diarrhoea.

Effect on Meat Quality

Stress and fatigue lower the quality of meat in several ruminant species due to depletion of glycogen in the muscle. Due to low acid production, the ultimate pH of the muscle remains high causing a condition called *dark cutting meat* or *dark, firm and dry* (DFD) meat. Thus, keeping quality of meat is reduced and it looks dark due to high water content. Such meat is unusually tender on cooking

In pigs, acute stress or excitement before slaughter causes another abnormal condition wherein low ultimate pH is achieved within 45 minutes due to rapid glycolysis even when the temperature of muscle is quite high. Such *pale, soft and exudative* (PSE) pork has higher drip and cooking losses.

In sheep, dehydration impairs with the skinning process and produces a sticky textured meat. Transportation injuries such as bruising affect the quality of carcasses in all species particularly sheep and pigs.

PRESLAUGHTER HANDLING AND CARE OF ANIMALS

Handling of animals should conform to humane standards at every stage. It will safeguard the animal welfare as well as meat quality. Rough handling of animals before slaughter can result in several physiological stresses. Loading and unloading operations have to be prompt. It is particularly important in hot weather when lot of heat builds up in stationary trucks.

Preslaughter shearing and washing of sheep is quite stressful and leads to bruises. It is advisable not to mix strange animals shortly before slaughter in order to avoid fighting amongst them, especially males.

Lairage serves as a resting ground for the tired and stressed animals. Resting period depends on the length and mode of journey, animal species, age, sex, condition. Undue holding is also not advised.

Lairage should have adequate litter and drainage to avoid faecal soiling of skin. The feed of animals should be withheld for 12-18 hrs before slaughter whereas ample drinking water should be made available during this period. It lowers the bacterial load in the intestine and facilitates easy removal of the hide or skin during dressing of carcass. Stunning is made more effective and brightness of the carcass is also improved.

four

Ante-mortem Examination of Meat Animals

Ante-mortem examination of meat animals awaiting slaughter is very necessary in order to produce wholesome meat and thus safeguard the health of meat consumers. It should be conducted 12-24 hrs before slaughter by qualified veterinarians in lairage pens. Hence, layout and construction of lairage should be such that it provides proper light and provision of an isolation pen for diseased and injured animals. It should be ensured that animals are not subjected to any kind of cruelty. A number of para-veterinary staff should also be at hand to help in the examination. All animals meant for slaughter should be rested atleast for 24 hours and should not be fed for atleast 12 hours before slaughter but they should be provided with plenty of water.

Objectives

1. Detection of animals suffering from scheduled infectious diseases which are communicable to man.
2. To detect certain diseases which are toxic or contagious and whose identification is either difficult or impossible during post-mortem, e.g. tetanus, rabies, farcy, listeriosis, septic metritis, sturdy in sheep etc.
3. To prevent food poisoning outbreaks e.g. in salmonellosis carcass or organs show little noticeable change on post-mortem.
4. To make postmortem examination more efficient, accurate and less laborious.
5. To protect the health of butchers and slaughter house personnel.
6. To prevent unnecessary contamination of building and equipment of the abattoir.
7. To implement disease control programme with more precision by tracing back the source of diseases.

कृपया रेषा मारुन
पुस्तक खराब करु नये.

Ante-mortem Examination Procedure

It should be carried out in two stages:

Stage I

General examination Meat animals should be observed in the lairage pens during rest as well as in motion. The general behaviour, reflexes, fatigue, excitement, gait, posture, evidence of cruelty, level of nutrition, symptoms of diseases or any other abnormalities should be closely observed.

Stage II

Detailed examination Suspected or diseased animals should be segregated for detailed examination. Their temperature, pulse rate and respiration rate should be recorded. Animals showing elevated temperature and systemic disturbance should be detained for further inspection and treatment in the isolation pen.

Animal	*Temperature*	*Pulse/ min*	*Respiration/ min*	*Gestation period*
Sheep and Goat	39.5°C	75	12-20	147 days
Pigs	39°C	75	10-16	112 days
Cattle and Buffaloes	39°C	50	12-16	280 days

Principles of Judgement in Antemortem Examination

1. *Fit for slaughter* Animals which are normal and free from any symptoms of disease should be sent for sacrifice.
2. *Unfit for slaughter* Highly emaciated, skin bound animals and those affected with tetanus or communicable diseases like rabies etc. or diseases which can not be treated should be declared unfit for slaughter.
3. *Suspects* All suspected animals need further attention. Some animals with localized condition and recovered cases should be passed for slaughter as suspect with instructions for careful postmortem examination.
   i. *Detained animals* Some animals need to be detained for specified period of time for treatment of disease or excretion of known toxic residues.
   ii. *Emergency slaughter* It is recommended in cases where the animal is in acute pain or is suffering from a condition where any delay in slaughter would be contrary to the welfare of animal. It is done under strict supervision so that there is no hazard to the consumer health.

Such conditions could be external injuries, recent fractures, tympany (bloat), prolapse of uterus etc.

Specific diseases encountered in different meat animals during ante-mortem examination and their judgement

S. No.	Disease	Salient symptoms	Judgement
1.	Emaciation	Pathological condition due to chronic illness, higher metabolic rate	Unfit for slaughter
2.	Rabies	Acute infective disease due to bite of an infected animal manifested by neurological disorder	Unfit for slaughter (to be destroyed)
3.	Anthrax	Acute manifested by fever, bloody diarrhea and red dark blood discharge from natural openings.	Unfit for slaughter (to be destroyed)
4.	Foot and Mouth Disease (FMD)	Most contagious, causing dullness, depressed appetite, lameness, salivation etc	Unfit for slaughter
5.	Actinomycosis	Chronic granulomatous disease diagnosed by lumpy jaw	Suspect
6.	Actinobacillosis	Marked development of fibrous tissue causing enlargement and hardening of tongue (wooden tongue)	Suspect
7.	Black Quarter	Acute infection characterized by severe inflammation of muscles followed by crepitant swelling on shoulder, neck, breast, loins or thigh	Unfit for slaughter (remove at once)
8.	Tetanus	Acute highly fatal infective disease characterized by spasmodic contraction of voluntary muscles especially masseter muscle often causing lock jaw condition	Unfit for slaughter (remove at once)
9.	Mastitis	Inflammation of teats	Suspect
10.	Tuberculosis	Chronic inflammation of lungs, swelling of retro-pharyngeal lymph gland	Generalised-unfit Localised – suspect
11.	Sheep Scab	Parasitic disease caused by sucking mites, crust formation on the skin coupled with progressive emaciation.	Suspect

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S. No.	Disease	Salient symptoms	Judgement
12.	Caseous Lymphadenitis	In sheep and goat, externally placed lymphnodes enlarged and contain a greenish yellow gelatinous pus	Generalized-unfit Localized-suspect
13.	Pneumonia	Lungs are severely affected	Suspect
14.	Swine Fever (Hog cholera)	Acute highly contagious disease. Symptoms of septicaemia in the form of multiple haemorrhages	Unfit for slaughter.
15.	Gut Oedema	In swine, incoordination of movement, staggering gait during walking with swelling of the eyelids	Suspect
16.	Swine Erysepalus	Characterized by different forms such as acute septicaemia, skin form, chronic arthritis and vegetative endocarditis	Suspect
17.	Atropic Rhinitis	In pig, snout becomes dished with wrinkling of skin over the affected area. Nasal septum distorted with bloody nasal discharge	Suspect
18.	White Scour	In calves, a large abscess in the abdominal wall near umbilicus which becomes hard and swollen	Unfit for slaughter
19.	Calf Diptheria	Characterized by evil smelling diptheric patches and ulcers in mouth, tongue, gums and pharynx	Unfit for slaughter
20.	Salmonellosis	Severe diarrhea with foul - smell, many contain blood, fever, loss of appetite, dullness, dehydration	Unfit for slaughter
21.	Listeriosis	Stiffness of neck, incoordinated movement of limbs, paralysis of muscles of jaw and pharynx	Acute case-Unfit for slaughter Recovered-Suspect
22.	Fluorine poisoning	—	Unfit for slaughter
23.	Selenium poisoning	Peeling of skin	Unfit for slaughter Recovered animal-Suspect

Antemortem examination of meat animals is of prime importance from public health point of view. It is the initial step in detection of any sign of disease, distress, injury etc. which helps in taking appropriate decision before slaughter of animal. It should be done properly and systematically by qualified and experienced veterinarian which will in turn help in maintaining high standards of meat quality.

five

Slaughtering and Dressing of Meat Animals

Slaughtering means putting the food animals to death and thereafter preparing the carcasses for human consumption. The essentials in the slaughter of food animals are that it should not cause unnecessary suffering to the animal and bleeding should be as efficient as possible. Besides, it should be safe for the handlers also.

There are two main types of slaughter methods:

I. Scientific or Humane slaughter.
II. Ritual slaughter

Scientific or Humane Slaughter

Such a slaughter avoids unnecessary pain and cruelty to the food animal and ensures as complete bleeding as possible. It also ensures speed of operation and safety of the personnel.

Stunning is a process employed to create a state of immobility or unconsciousness at the time of slaughter (usually 30 seconds to 1 minute). Immediately, the animal is hoisted and blood vessels on the neck are severed (sticking) to bleed the animal to death. In USA and European countries, stunning has been made mandatory by legislation in abattoirs, excepting those where ritual slaughter is followed.

Stunning Techniques and Devices

They fall into three main categories:

i. *Mechanical instruments* Instruments such as *captive bolt pistol (Fig. 5.1), percussion stunner or free bullet* cause damage to the brain so the animal immediately looses the consciousness. In Britain, use of captive bolt pistol is common for stunning cattle and sheep and several models of pistols are in vogue.

In captive bolt pistol, by using an explosive cartridge, the

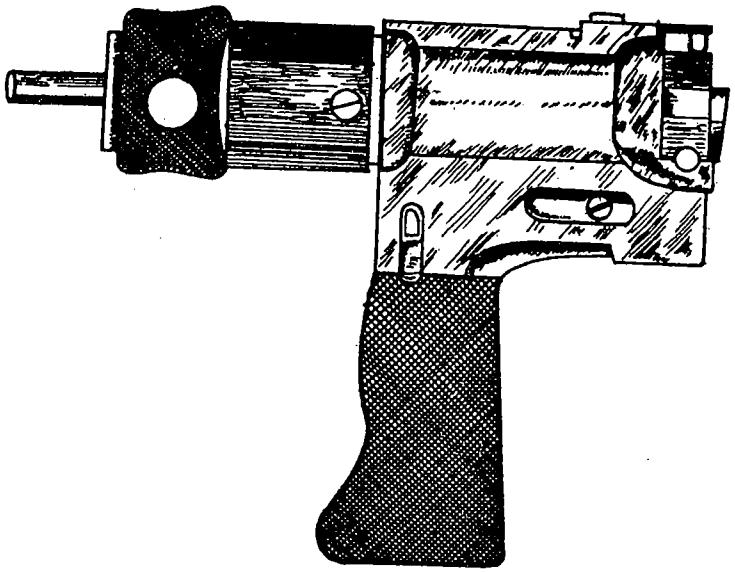


Fig. 5.1: Captive bolt pistol

bolt is driven in the frontal bone of the animal causing immediate immobility by destruction of the cortex and deeper parts of the brain. In pneumatic stunner, bolt is activated under pressure to cause concussion without penetrating the frontal bone. Use of free bullet pistol may become necessary for large bulls having very thick skull. Application point of captive bolt pistol differs with species. In adult cattle or if employed for buffalo, it should be positioned in the middle of the forehead where two lines drawn from the medial canthus of each eye to the base of opposite horn cross. In horned goat and sheep, the pistol point is placed just behind the ridge running between the horns. In effective stunning, the animal is immediately collapsed followed by tonic spasms and then movements of the hind legs.

Pithing is also done sometimes in adult cattle after stunning with captive bolt pistol. In this process, a thin steel rod (approx. 0.5 m) is inserted into the hole made in the skull by the captive bolt pistol. It will destroy the medullá oblongata and reflex muscular action will not take place

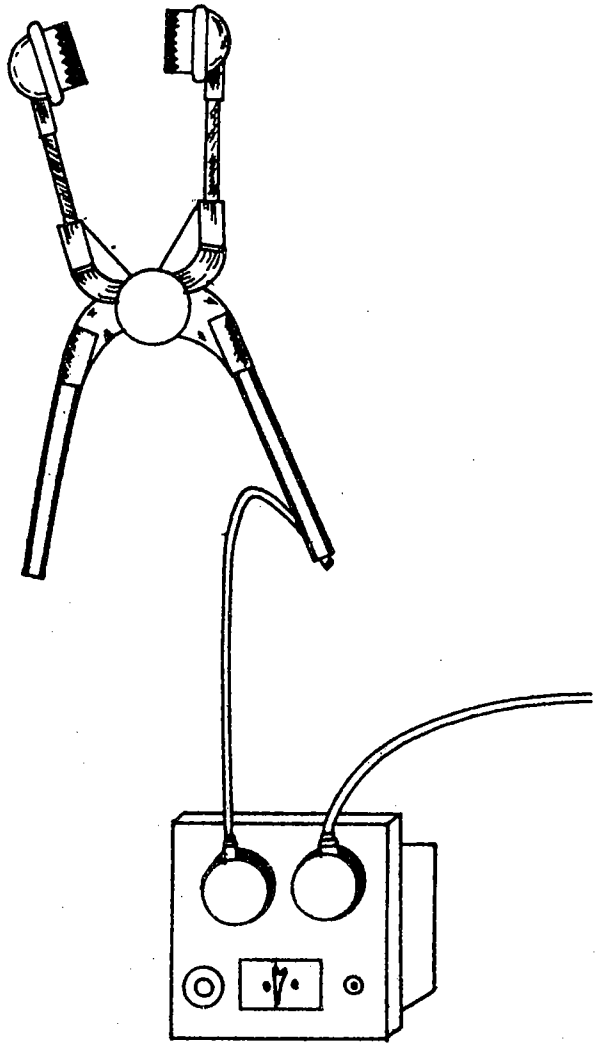


Fig. 5.2: Elther apparatus

during sticking and initial dressing. Thus pithing enables the slaughterman to proceed quickly and with safety.

ii. *Electrical stunning* It is conveniently employed in stunning of small ruminants, pigs and poultry. The instruments—Elther apparatus (Fig. 5.2) with a pair of tongs carries electrodes by which alternating current (AC) is passed through brain. A high frequency current (250mA) of comparatively low voltage of usually 75 volts is used for 10 seconds by way of application of electrodes or tongs at the base of the ears on either side of the head. The electrodes are kept wet in a 20% saline solution to enable the current to pass easily through hair and skin. It produces instantaneous unconsciousness for about 30 seconds. Bleeding is very efficient and the power consumption is extremely low. If the current remains low, *missed shock* may occur resulting in paralysis of the animal, although it remains fully conscious. It affects the quality of meat besides compromising the safety of the handler. On the other hand, too high a current may cause *splash*. It refers to the appearance of petechial haemorrhages throughout the subcutaneous tissue in pigs. The capillaries get ruptured due to excessive increase in blood pressure.

iii. *Chemical stunning* Carbon dioxide gas stunning is most suitable for pigs and is followed in many European countries. CO_2 is heavier than air and can be contained in a tunnel. The gas blocks the nerve endings. Animal is exposed to 65% carbon dioxide concentration through oval type tunnel, dip lift system or revolving wheel for 45 seconds. On exposure to gas, pigs become anaesthesized that are then shackled and bled. Bleeding is quite efficient since carbon dioxide stimulates respiration favouring blood circulation. However, it requires more space and is not fast.

Sticking or Bleeding of the Animal

It can be done by any of the two methods:

i. After hoisting on the overhead rail, carotid arteries and jugular veins of both the sides are severed across the throat region, caudal to the larynx.

ii. On the floor, skin is incised along the jugular furrow and carotid artery and jugular vein of one side are severed. The knife is then passed to the chest severing the anterior aorta and anterior venacava.

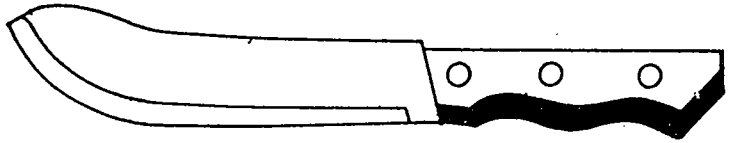


Fig. 5.3: Meat cutting knife

Sometimes, knife reaches too far in the chest puncturing the pleura and the blood may be aspirated into the thoracic cavity. This blood adheres to the parietal pleura especially the posterior edges of the ribs. This contamination of lungs is called *back bleeding* or *over-sticking*. It requires to be washed immediately.

The yield of blood is as follows:

Buffaloes and cattle	: 10-12 kg in 6 minutes
Sheep and goat	: 1-1.5 kg in 5 minutes
Pig	: 2-3 kg in 5-6 minutes

Ritual Slaughter

Slaughter without prior stunning of food animals as per religious rites are referred as ritual slaughter. These are widely practiced in many countries. In India and far East, practically all the meat animals are slaughtered in conscious state.

i. *Jewish method of Slaughter or Schechita* The regulations for Jewish slaughter came into existence around 500AD. Consumption of pork was strictly prohibited for Jews, probably as a safeguard against tapeworm.

   a. Animals to be slaughtered should be active. Moribund animals which lie quite and also do not get up even by striking with a stick must not be slaughtered. A blow on the head of the animal is strictly forbidden and perforation of the membranes of the brain constitutes a mutilation. Therefore, this method envisages slaughter of animals without prior stuning.

   b. The incision across the neck is made by single rapid thrust of sharp knife which severs the skin, muscles, oesophagus, trachea, carotid arteries and jugular veins. This incision should be completed **without** *pause, pressure, stabbing, slanting or tearing*.

c. The dressing of the carcass should expose the diaphragm and allow manual examination of thoracic organs.
d. The blood vessels must be removed before the retail sale of meat is undertaken. It is for this reason that only forequarters are normally eaten. Hindquarters containing lot of blood vessels are porged by highly skilled Kosher butchers (*Schochet*) and are thus rarely eaten.
e. Kosher meat must be sold and consumed within three days of slaughter.

Jewish slaughter (*Schechita*) is undertaken by a *Schochet* (cutter). Carcass fit for Jewish consumption is stamped with *Kosher* seal on the brisket while unfit is pronounced as *terepha*.

ii. *Muslim method of Slaughter or Halal* Some practices relating to slaughter of animals and consumption of meat by the members of Jewish faith match with those of Muslims. In both cases, consumption of dead animals, blood and pork is forbidden. The vertebral artery enclosed in the spint cord is not severed.

In this method, neck of the animal is severed by cutting the four major blood vessels-carotid arteries and jugular veins with a sharp knife. The spinal cord is left intact. So the nerve centres controlling the heart and lungs remain functional and an efficient bleeding is ensured. It also enhances the keeping quality of meat.

Unlike Schechita, Muslin method of slaughter is not controlled by a central board but it is overseen by the local Islamic authorities (*Muftis*). Here, stunning may be permitted in some cases if it permits the heart to beat for sometime. Since animal has to struggle for sometime before death, some people regard it as cruel.

In India, buffaloes and most of the sheep and goat are slaughtered by Halal method.

iii. *Jhatka or Sikh method of Slaughter* In this method, the head of the animal is chopped off with one big stroke of a sword. In some part of northern India, skilled workers can decapitate even buffaloes by a single stroke of a special sword. Since medulla oblongata is damaged in the process, bleeding remains incomplete. Due to presence of some blood in tissues, the keeping quality of meat may be affected.

iv. *Evernazine or Neck Stab method* It is followed in Spain, some parts of Italy, Mexico and some South American countries. In this method, cattle are slaughtered by plunging a short double edged knife (*puntilla*) into occipito-atlantal space at the nape of the neck severing the medulla oblongata.

DRESSING TECHNIQUES FOR DIFFERENT ANIMALS

Dressing techniques and sequence of dressing operations vary from place to place and are very much influenced by the equipment and facilities available in the abattoir. The present trend in organized abattoirs is towards line dressing whereby once the animal has been hoisted to the bleeding rail, it is not lowered to the floor till the entire dressing operation is completed. The carcass is conveyed by gravity or power driven along an overhead rail. Equipment such as brisket saw, hock cutter, hide puller, aitch-bone cutter etc. facilitate the dressing. The text hereunder describes the typical sequence of operations.

DRESSING OF BUFFALO/CATTLE

1. *After stunning, the animal is hoisted by one leg to the overhead rail. It is brought above bleeding trough or gully and an incision is made just in front of sternum cutting the main blood vessels.*
2. *A bleeding time of 6-8 minutes is allowed.*
3. A cut is made across the larynx, the weasand (oesophagus) is tied off and the head is skinned and detached at the atlas joint.
4. Now the forelegs or the shanks are removed.
5. The hind legs are skinned and removed while the carcass is hung by tendons on the spreader.
6. Deskinning (flaying) is carried forward from hind and forequarters and hide is now pulled with the help of a hide puller.
7. Brisket is opened along with the midline and the pelvic cavity is opened along the abdominal cavity. Evisceration commences and pluck as well as viscera are removed. These are taken to their respective offal rooms for cleaning and examination.
8. Now the carcass is sawn into two halves along the vertebral column.
9. Spray washing of the carcass.
10. Transfer to the chilling room.

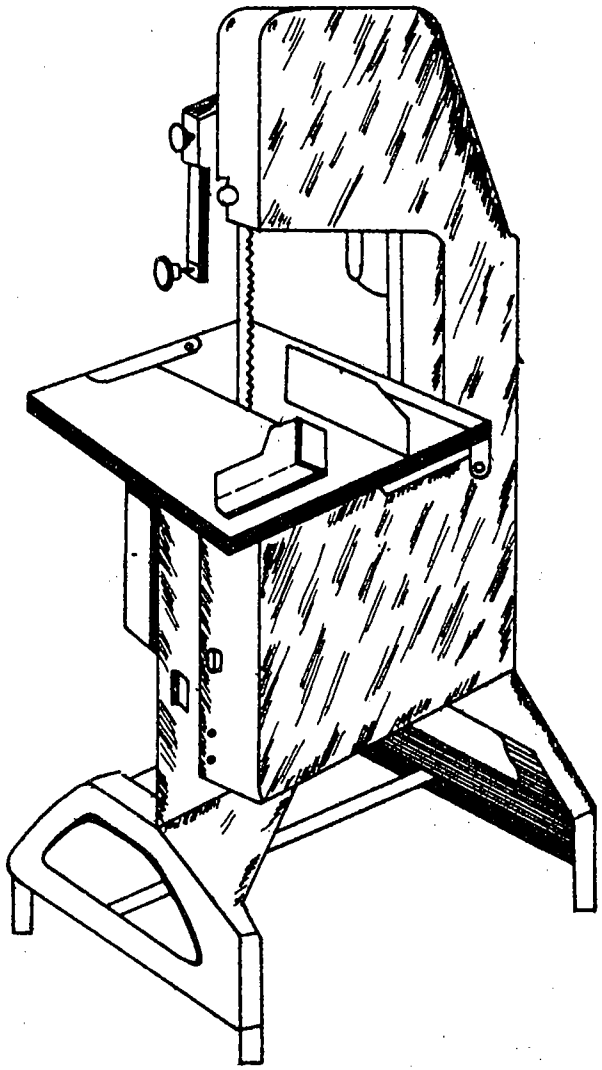


Fig. 5.4: Meat cutting machine (band saw)

Dressing of Sheep and Goat

1. *After stunning, the animal is hoisted to overhead rail and an incision is given in the jugular furrow near the head severing both carotid arteries. The head is jerked backwards to rupture the spinal cord, thereby checking the involuntary movement.*
2. *A bleeding time of 5 minutes is allowed.*
3. The forelegs are knuckled and a cut is made to the front. The forelegs are removed at knee.
4. The neck and cheeks are skinned alongwith the shoulder. The throat is opened up and oesophagus is tied.
5. The hindlegs are knuckled and a cut is made to the root of the tail. The legs are skinned.
6. The skin is incised in the middle of the belly and skinning proceeds towards the flank. Now skin is pulled down over the backbone and base of the head.
7. The head and hindlegs are removed.
8. A small cut in the abdomen is extended to the brisket and the breast bone is also split.
9. The pluck and viscera are removed. Kidney and its fat are left in the carcass.
10. Spray washing of the carcass is done followed by transfer to the chilling room.

Dressing of Pig

1. *Sticking* After stunning, the pig is hoisted to the overhead rail. An incision of 5 to 10 cm is made at the mid point of neck facing breast bone. The knife is inserted in this incision at an angle of 45° and is forced down and back atleast 12-15 cm to a point below the front of the breast bone. The knife is given a slight twist before it is withdrawn. Care should be taken not to insert the knife into the chest cavity.
2. *Bleeding* A bleeding time of 5 minutes is allowed.
3. *Scalding* After 6 minutes or so, the animal is dropped in the scalding tank maintained at a temperature of 60-62°C for about 6 minutes.
4. *Scrapping or Dehairing* Raise the animals on the overhead rail and pull of the dew claws and toes while hot. Scrap the loosened hair with the help of a hog scrapper or dull knife hindquarter downwards. Then rinse the carcass with warm water.
5. *Singeing* It is done with the help of a blow lamp in which a high temperature is achieved and all the remaining hair

are burnt. Besides, singeing sterilizers the cuticle and firms up rind giving it a better appearance and keeping quality.

6. *Removal of head* Done at atlas joint before the carcass is opened.
7. *Evisceration* Viscera is removed and examined. Leaf fat is also collected.
8. Splitting of the carcass.
9. Final inspection of head and carcass.
10. Spray washing of the carcass and transfer to the chilling room.

Apart from religious rituals considerations, the suitability and effectiveness of different slaughter and dressing methods vary from species to species and even the age within the same species of the animal. Regional variations are also quite common.

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Postmortem Inspection of Meat Carcasses

Many diseases and abnormal conditions not detectable on antemortem, necessitate a careful postmortem inspection. It should be conducted as soon as possible after carcass dressing is complete, as setting of carcass may render it difficult to expose and examine the lymphnodes afterwards. It has to be carried out in a hygienic manner avoiding unnecessary cuts. However, character and extent of disease lesions whether localized or general or whether condition is acute, sub-acute or chronic has to be properly determined. It will involve palpation of organs and tissues, incisions where necessary and laboratory tests in certain conditions. All these aspects are taken into consideration before passing the final judgement.

Objectives

i. To detect and eliminate abnormalities from the carcass and organs including contamination, thus ensuring that only meat fit for human consumption is passed for food.
ii. To check the efficiency of slaughter and carcass dressing.
iii. To diagnose disease conditions for control purposes.
iv. To ensure that carcasses, parts or organs marked unfit for human consumption are properly disposed of or destroyed.

Facilities

i. Inspection points should have sufficient and well distributed light of 540 lux or 50 foot candles.
ii. Adequate hand washing units with supply of hot and cold running water, liquid soap and towels.
iii. Meat inspection knives.
iv. Sterilizers for complete immersion of knives, saws, cleavers etc.

v. *Stamping ink*—A solution containing 1 to 2% of fuchsine in acetic acid is used as stamping ink for carcasses. Alternatively, it can be prepared according to the following formula : Methyl violet 35 g, Cane sugar 450 g, Ethyl alcohol 1363 ml and water 1636 ml. Sugar is first dissolved in water and methyl violet is added at the end. The stamp generally bears the head of the animal.

General Principles

Postmortem inspection involves visual perception and palpation of organs and tissues, incisions where necessary and laboratory tests wherever confirmation is required. It should always proceed in a systematic and hygienic manner. *Examination of lymphnodes is of paramount importance in postmortem inspection,* Since these glands drain different parts of the body, condition of specific lymphnode can given an indication of the condition of the particular part drained by it. A meat inspector must have the knowledge of topography and normal appearance of lymphnodes (colour, shape, size etc) in a particular animal. The size of a lymphnode generally depends on the area drained by it. A swelling or enlargement or discolouration of a lymphnode indicates a pathological condition. Some lymphnodes, which drain lymph from important muscles, are of special interest in meat inspection and these are referred as *'meat lymphnodes'*. In buffalo and cattle, these lymphnodes are prescapular, axillary, prefemoral, ischiatic and popliteal whereas in pig, these are prefemoral and popliteal. In postmortem inspection, lymphnodes are exposed and examined by palpation for size and consistency and if necessary, incised to observe colour and detection of lesions.

As a thumb rule, *viscera and head should remain identifiable* with the carcass until the inspection is completed. The carcass should not be subjected to any mutilation. The final decision about the fitness of the carcass and offals must rest only with the veterinarian. He should be armed with enough powers to stop the slaughter and dressing operations and seize the carcass, if there are sufficient ground to do so.

Based on the postmortem examination, meat inspector can given any of the four judgements:

a. Passed
b. Total condemnation
c. Partial condemnation
d. Conditionally passed

कृपया रेषा मारुन
पुस्तक खराब करु नये.

Procedure

In general, postmortem inspection of buffalo and swine carcasses should proceed in the following order:

Head

An examination of surface of tongue is done for FMD and other form of stomatitis. Palpation of tongue from dorsum to lip is carried out to detect actinobacillosis. Masseters are incised to examine for *Cysticercus bovis*. A detailed examination of retropharyngeal, submaxillary and parotid lymphnodes becomes necessary to detect tuberculosis and actinobacillosis. In case of pigs, the portions of muscles from tongue are examined under microscope or trichinoscope for trichinollosis and inspection of submaxillary lymphnode is done as a matter of routine.

Viscera

All viscera should be inspected as they are removed from the carcass. Every organ and associated lymphnodes are examined. In case of any abnormal condition, the organs are incised not to contaminate other organs or carcass parts.

Lungs Visual inspection followed by palpation are carried out for evidence of pleurisy, pneumonia, tuberculosis, fascioliasis and hydatid cysts. Bronchial and mediastinal lymphnodes should be incised to detect tuberculosis.

Heart Pericardium is opened and looked for tuberculosis pericarditis. Heart is incised to detect petechial haemorrhages or cysts.

Liver The surface and substance of liver is examined for fatty changes, actinobacillosis, abscesses, parasitic infections like Cysticercus bovis, fascioliasis, hydatid cysts or larval stage of Oesophagostomum.

Kidney Renal lymphnodes and adrenal glands should be examined before the removal of kidneys.

Spleen The surface and substance of spleen is inspected for anthrax, tuberculosis or presence of artifacts.

Stomach and intestine The serous surface of these organs could indicate tuberculosis or actinobacillosis while anterior part of reticulum could show penetration of a foreign body. Mesenteric lymphnode is incised as a routine to look for tuberculosis.

Urinary bladder The outer and inner surfaces of urinary bladder are observed for diseased condition.

Uterus and ovaries These are opened for examination of septic conditions.

Udder This is examined for septic mastitis or abscesses. Supramammary lymphnodes are incised for evidence of abscesses or tuberculosis.

Testicles The outer surface, substance and superficial inguinal lymphnodes are inspected.

Carcass

The carcass should be inspected for evidence of bruising or haemorrhage or discolouration, local or general oedema (dropsy), swelling or other abnormality of bones, joints or musculature. Age of a bruise can be determined by using bilirubin test. Inspection of thoracic and abdominal walls should be done for inflammation, abscesses or tuberculosis. Diaphragm should be lifted and examined carefully for tuberculosis.

After the carcass is split, cut surfaces of sternum, ribs, vertebrae and spinal cord should be examined. Incision should be made on each quarter in the musculature near the shoulder joint and near the pelvic bone for detection of cysts (*Cysticircus bovis* and *Cysticercus cellulosae*). Simultaneously, prescapular, popliteal and prefemoral, iliac and supramammary lymphnodes are also observed and if necessary, incised for inspection.

Postmortem Inspection of Sheep and Goat Carcasses

The examination is carried out on the above lines. However, sheep and goat do not require much detailed examination. The cutting of head, heart, pulmonary lymphnodes and body lymphnodes should be carried out only in cases of suspicion. Carcass is, therefore, examined for satisfactory bleeding, liver for fascioliasis and lungs for parasitic infections like nematodes or hydatid cysts etc.

Postmortem Inspection of Swine Carcasses (Some Specific Precautions)

Although PM examination follows the same pattern as in buffalo/cattle, skin lesions serve as important diagnostic feature of erysepalas, swine fever and urticaria, feet for abscess formation, udder for mastitis or actinomycosis and tail for necrosis. In places where *Cysticercus cellulosae* is common, the external masseter muscles and root of tongue are incised. If trichosis is known to occur in a particular area or suspected, muscle sampling should be done for appropriate examination. Besides, visceral organs should be examined for pneumonia, pericarditis, peritonitis and also submaxillary lymphnodes for tuberculosis.

Extended Examination

Carcasses of buffalo/cattle are further examined if affections with *Cysticercus bovis* (Measly beef) or tuberculosis are suspected. In case some cysts are noticed in the carcass, cheek muscles, tongue, diaphragm and heart should be incised and closely examined. If more than 10 clear or degenerated cysts are recovered accompanied by emaciation, the carcass is totally condemned; otherwise it is conditionally passed with instructions for proper cooking. Similarly, even if single TB lesion is present, the carcass should be split and vertebrae, ribs etc. are closely examined. Besides, additional lymphnodes such as lower cervical, presternal etc. are also examined. In generalized TB, the carcass is totally condemned. The lesions for tuberculosis are considered generalized if any two of the following organs and concerned lymphnodes are affected—lung, spleen, kidney, uterus, udder, ovary, testicle, adrenal gland and brain or spinal cord etc. Besides, lesions of tuberculosis in muscles, bones or joints are also bring it in generalized category. An organ or a part of carcass is condemned if the lesion is localized.

Swine carcasses are also subjected to extended examination when trichinosis or tuberculosis is suspected. Small pieces of tissue from diaphragm and muscles of larynx, abdomen and intercostal spaces are observed under trichinoscope after predigestion with trypsin. Carcasses showing heavy infection are totally condemned while those with mild infection are conditionally passed. For further examination of tuberculosis, the carcass and head are split and bone surfaces and spinal cord are closely examined. In addition to lymphnodes examined earlier, posterior cervical and lumber lymphnodes are also examined and generalized cases are totally condemned. In case of localized TB of head, it should be condemned and associated lymphnodes are also removed.

Important Diseases, Postmortem Lesions and General Regulations for Judgement of Meats

Sl. No.	*Disease*	*Causal agent*	*Animals affected*	*Salient postmortem lesions*	*Judgement*
1.	Anthrax	*Bacillus anthracis*	Cattle, sheep and pig	a. Presence of uncoagulated and dark blood at natural orifices of the body. b. Spleenomegaly is very prominent. c. Extensive petechial lesions and swelling of mucous membranes d. Kidney and liver are pale and friable. to be destroyed immediately.	Total condemnation. Carcass and offals Effective disinfection of the premises.
2.	Black Quarter	*Clostridium chauvei*	Cattle, esp. calf, sheep and goat	a. Heavy muscular areas e.g. thigh, loin, neck, breast etc. show crepilant swelling. b. Connective tissue surrounding swelling is infilterated with yellow gelatinous substance which may be haemorrhagic and permeated with gas bubbles. c. Adjoining lymphnodes are enlarged and haemorrhagic. d. Pleural and peritoneal cavity may be filled with blackish red fluid. e. Kidney and liver may be congested, enlarged and develop brown porous areas.	If diagnosed in lairage-slaughter is forbidden. Total condemnation in postmortem.

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S. No.	*Disease*	*Causal agent*	*Animals affected*	*Salient postmortem lesions*	*Judgement*
3.	Johne's Disease	*Mycobacterium paratuberculosis*	Cattle, sheep and goat	a. Due to diarrhea, intestinal mucous membrane becomes very thick and are corrugated. b. In sheep, ileum is thickened without corrugation, intestine is intense yellow and having calcified nodules. Mesenteric lymphnodes necrosed, caseated and calcified.	Total condemnation, if carcass is emaciated otherwise, intestines and mesentery are generally condemned.
4.	Brucellosis	*Brucello abortus* Brucella melitensis *Brucella suis*	Cattle, goat, pig	a. Viscid exudates mixed with pus between uterine mucous membrane and foetal membrane. b. Both foetal and maternal cotyledons are covered with grey and yellow purulent material. c. In male, necrotic and purulent foci in the testicle especially multiple abscesses in epididymis.	Causes undulating fever in man. But organism is destroyed by the lactic acid formed after slaughter of animal. Carcass may be passed for food after condemnation of genital organs, udder and associated lymph nodes.
5.	Actinomycosis (Lumpy jaws) and Actinobacillosis	*Actinomyces bovis Actinobacillus lignieresi*	Cattle	a. Actinomycosis lesions are confined to head and involve jaw bones which present a honey combed appearance on section of esp. lower jaw.	Affected organs such as head, tongue, stomach, liver, lung etc. are condemned.

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
	(Wooden tongue)			b. Actinobacillosis shows typical lesions in the form of ulcers and small nodules (1 cm diameter) which are easily seen and palpated on the dorsum and lateral surfaces of tongue. c. In chronic form, there is a marked fibrous growth causing enlargement and hardening of the tongue. d. Lesions of ulcer may also be seen in the reticulum and liver. e. Actinobacillus nodules have a typical bright yellow pus, which is granular. f. In contrast to above, tuberculosis lesions have less bright yellow colour and show calcification.	If lesions are generalized carcass and offal are totally condemned.
6.	Tuberculosis	*Mycobacterium bovis* M. tuberculosis *M. avium*	Cattle, sheep and goat, pig, birds	a. Destruction of lung tissue by caseation and liquifaction, caseous or calcified foci are common. b. Tuberculous pleurisy may show caseous masses like pearls or grapes. c. Liver may show pinhead or barley grain size encapsulated masses tending	If localized, whole organ and associated lymph-nodes are condemned and destroyed. Generalised tuberculosis extending to several organs warrant total

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
				to undergo caseation and calcification. d. Tuberculous spleen and peritonitis e. Bone tuberculosis most common in ribs and vertebrae characterized by formation of yellow granulation tissue f. Skin tuberculosis in cattle having hard, painless nodules.	condemnation and destruction of the carcass and organs.
7.	Rabies	*Rhado virus*	Cattle, pig, sheep and goat, other animals	a. Degenerative changes are found in brain. b. Histopathological examination reveal negri bodies in brain tissue. c. Inflammation of gastrointestinal mucosa.	Total condemnation.
8.	Foot and Mouth Disease	*Picornal group of virus with subtype A, O, C and Asia 1, Sat-1, 2 or 3*	Cattle particularly besides all cloven footed animal	a. Ulcerative vesicular type lesions are seen on tongue, lips, dental pad and feet. These are commonly extended upto udder and teats in cattle, In sheep, lesions are common on dental pad and feet. b. Vesicles in mouth have a clean, white appearance and contain yellow straw-like fluid in cattle, sheep and goat. c. Mouth lesions are not common in pigs but principally affect feet, lesions being found around coronet of both main and supernum erary digits.	Carcasses are totally condemned in acute febrile cases, otherwise conditionally passed after condemnation of head and tongue.

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
9.	Rinderpest or Cattle plague	*Paramyxo virus*	Cattle. Occasionally in sheep, goat and pig	a. Necrotic foci with red erosions are found in mouth, muzzle and digestive tract. b. Mucous membrane of mouth cavity is covered with soft, grey, plaques looking like sprinkled bran or soft cheese. c. Intestine are so congested that mucous membrane of particularly present a tiger-skin or zebra marking appearance.	Total condemnation.
10.	Haemorrhagic septicaemia (Pasteurellosis)	*Pasteurella multocida type 1*	Cattle	a. Subcutaneous tissues of head and throat are filled with a clear, yellowish fluid and lymphnodes show marked enlargement. b. Petechial haemorrhages all over the body, organs and lymphnodes. c. There is a marked enteritis.	Total condemnation.
11.	Salmonellosis (Paratyphoid)	*Salmonella typhimurium*	Cattle, sheep, goat and pig	a. Peracute septicaemia-shows multiple submucosal or subserosal haemorrhages and few cases show a necrotizing fibrinous enteritis. b. Acute enteritis showing watery contents or an intense haemorrhagic enteritis showing blood-stained contents.	Total condemnation.

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
				c. Mesenteric lymphnodes are enlarged, oedematous and haemorrhagic. d. In pig, patchy haemorrhages and ulcers in intestine, prominent petechiae in kidney.	
12.	Listeriosis	*Listeria monocytogenes*	Cattle and sheep. Also in goat and poultry	a. In septicaemic form, multiple minute greyish areas showing necrosis of liver. b. Haemorrhagic gastritis and enteritis. c. Haemorrhages in spleen and heart. d. In the encephalitic form shows congestion of brain and meninges. Cerebrospinal fluid becomes cloudy.	Total condemnation.
13.	Sheep Pox	*Pox virus*	Sheep	a. Eruption of vesicles and pustules around the cheeks, eyes, lips and inner side of thighs, belly and tail which degenerate to form crust. b. Subcutaneous tissue becomes oedematous	Total condemnation.
14.	Blue Tongue	*Orbivius*	Sheep, cattle and goat	a. Inflammation of mouth cavity with ulcers, erosion and necrosis esp. of dental pad. b. Tongue becomes blue and enlarged.	Acute cases with extensive lesions and emaciated carcass are condemned. Less severe cases

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
				c. Generalised oedema and necrosis of skeletal and cardiac muscles. d. Abomasum shows inflammation and ulceration.	are judged leniently.
15.	Bovine	*Prion* Spongiform Encephalopathy (BSE) (Mad cow disease)	Cattle	a. Degenerative changes in the central nervous system. b. Discrete round and ovoid vacuoles in the grey matter of brain. c. In sheep, brain, spinal cord, spleen and lymphnode show degenerative changes.	Total condemnation.
		Scrapie	Sheep	d. An essential feature is vacuolation of neurons in the spinal cord and medulla.	
16.	Tetanus (Lock jaw)	*Clostridium tetani*	Sheep, cattle	a. Pathogenic effects are produced by toxin which has special affinity for nervous tissue. b. Antemortem examination is more important since disease produces no characteristics postmortem lesions. c. Carcass shows evidence of asphyxia, petechial haemorrhages on the serous membranes or heart. Lung show congestion and oedema.	Total condemnation.

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
17.	Botulism	*Clostridium botulinum*	Cattle, sheep and goat	a. Since there is involvement of toxin producing bacteria prominent lesions may not be seen. b. Haemorrhages are seen in gut, heart and brain. c. Presence of carrier material in the gut may indicate botulism.	Total condemnation.
18.	Glanders or Farcy	*Actinobacillus mallei*	Horse, mule etc.	a. Lungs show grey nodules surrounded by red area of congestion. b. Ulcers are seen in nostril and mucous membrane of trachea. c. Chronic lymphangitis, nodules with purulent material.	Total condemnation.
19.	Calf Diptheria	*Fusiformis necrophorus*	Cattle, sheep and pig	a. Necrotic stomatitis, catarrhal or diptheric inflammation of the laryngeal or nasal mucous membrane. b. Foot rot of cattle and sheep showing inflammation.	If infection is local, a good carcass devoid of affected parts may be passed. Otherwise, total condemnation.
20.	Swine Fever (Hog Cholera)	*Toga virus*	Pig	a. In acute and per acute cases, haemorrhages are found on the skin, larynx, trachea, lung, kidney, heart, digestive tract, bladder etc. with swelling and	Total condemnation

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
				congestion of associated lymphnodes. b. In chronic cases, characteristic lesions in the intestine-submucosal haemorrhages, button-like ulcerations etc. which get more concentrated around ilco-coecal value.	
21.	Swine Erysepalas	*Erysephalothrix insidiosa*	Pig	a. Lesions of septicaemia i.e. formation of petechial haemorrhages in the lungs, kidney and on serous covering of stomach and intestine. b. Enlargement of spleen and lymphnodes. c. Skin discolouration and petechial haemorrhages are more obvious on the poorly bled animals. d. In the urticarial form, diamond-like areas on the side, back and buttocks and skin get necrosed. e. In chronic cases, pericardium contains serofibrinous exudate, petechiation of kidney and red granulatous tissue in hock and stifle joints.	Total condemnation in systemic involvement; otherwise conditionally passed after removal of skin and affected organs.

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
22.	Measly beef	*Cysticercus bovis*	Cattle	a. The cysts of this tapeworm are most commonly found in the masseter muscles, besides shoulder, tongue, diaphragm, oesophagus and heart. b. These are sometimes found in the lungs, liver, fat and lymphnodes.	Carcasses showing heavy infestation of tapeworm cysts should be condemned. If infestations is localized, the affected parts and organs should be condemned; remaining carcass may be conditionally passed with instructions for proper cooking.
23.	Measly pork	*Cysticercus cellulosae*	Pig	a. This cystic state of pork tapeworm (*Taenia solium*) are noticed in the muscles of heart, diaphragm, tongue, neck, shoulder, abdominal and intercostal muscles. b. The deep muscles of thigh such as triceps brachi are very often affected. Thus, cysts are scattered in pig muscles.	-do-
24.	Hydatid disease	*Echinococcus granulosus*	Sheep, cattle, pig	a. The cysts of this tape worm mould as per the shape and size of the organ in which they grow.	Infected liver and lungs must be condemned and destroyed.

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S. No.	Disease	Causal agent	Animals affected	Salient postmortem lesions	Judgement
				b. These are noticed in liver and lung of sheep with clear fluid in the central cavity. Hydatid infection in liver of sheep, cattle and pig is always associated with enlargement and connective tissue makes a fibrous wall around it. In sheep and pig, liver is most commonly involved whereas in cattle the involvement of lung is most common.	
25.	Pimply gut (Nodular intestinal worm disease)	*Oesophagostomum spp.*	Sheep, goat, cattle	a. Greyish white nodules of varying sizes from pin-head to pea form distinct elevations in the small intestine, colon and coecum mucous membranes.	Affected intestines are not used for casings and are condemned. Carcasses are passed for food.
26.	Trichinosis	*Trichinella spiralis*	Pig	a. The principal location of trichinae is the musculature of the pillar and costal portions of diaphragm, tongue, larynx and abdominal and inter-costal muscles. b. Macroscopically, numerous white specks are seen in meat, which get calcified and make a gritty sound when cut with knife. c. Microscopic examination of two strips from each organ is recommended. Fortunately, trichinosis is not reported in India.	Total condemnation.

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Besides above mentioned infectious and parasitic diseases, a veterinarian meat inspector is faced with several conditions and organ diseases on which he is required to give a scrupulous judgement regarding suitability of meat for human consumption.

Some of the important ones are listed hereunder:

27. Anaemia	Total condemnation if general emaciation or associated with some disease.
28. Bruising	Total condemnation if it is general, extensive and severe with or without gangerene. Otherwise local condemnation.
29. Dropsy	Total condemnation if general type associated with emaciation.
30. Jaundice	Total condemnation if pronounced, carcass showing toxic icterus with intensive yellow or greenish yellow discolouration, otherwise passed conditionally.
31. Lymphadenitis	Total condemnation, if caseous type.
32. Malignant tumours	Total condemnation.
33. Melanosis	Total condemnation, if generalized; otherwise passed unconditionally.
34. Metritis	Total condemnation if acute and septic; otherwise local condemnation.
35. Peritonitis	Total condemnation if acute, diffused-septic; otherwise local condemnation.
36. Pericarditis	Total condemnation if septic; otherwise local condemnation.
37. Pneumonia	Total condemnation if gangrenous. In chronic localized case, lungs must be removed.
38. Pyaemia	Total condemnation if joint ill or umbilical pyaemia; otherwise local condemnation.
39. Rickets and Osteodytrophia	Total condemnation if carcass is emaciated and off smelling; therwise passed unconditionally.
40. Sarcocysts.	Total condemnation if generalized in musculature and visible to the naked eye; otherwise local condemnation.
41. Poisoning	Total condemnation.

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Evaluation of Dressed Carcasses

Grading and evaluation play an important role in marketing and merchandising of carcass. Evaluation is a broader term than grading alone because it encompasses value differences with greater precision. Grading segregates carcasses into standardized groups with minimum common characteristics whereas evaluation is an attempt to predict yield, processing, palatability characteristics and overall quality of meat.

Several carcasses characteristics, which generally form the basis of grading and evaluation, need to be properly defined.

i. *Conformation* of carcass is the most important term, which refers to the general build, form, shape, contour or outline of the carcass, sides or cuts. A conformation, which yields greatest quantity of edible meat, is most desired. Superior conformation implies short necks and shanks, thick backs with full loins and plumb legs. It should have a larger proportion of most desirable value cuts, thus a higher meat to bone ratios.

ii. *Finish* refers to the quantity, amount, colour and distribution of fat. Within reasonable limits, fat increases palatability especially flavour, juiciness and tenderness besides adding to the general appeal of the carcass. Superior finish implies abundant marbling (intramuscular fat) and a smooth, thin and even covering of firm fat over most of the outer surface of the carcass.

iii. *Quality* in reference to carcass evaluation includes several factors, which influence the palatability of cooked meat. It may even necessitate examinations of several inner and concealed parts of the carcass. In general, several important indications of quality are-uniformity of carcass, balanced finish, fine texture of muscles, bright colour and lusture of cut surfaces, firmness of lean devoid of watery or dry appearance, white and flaky fat and red porous bones.

Different countries have their own grading pattern and detailed specifications to suit their market and consumer requirements. Most of them have been adopted from USDA and modified as per local needs.

Several carcass evaluation methods have been in vogue. *Visual judging* is the oldest and most widely used method. It is performed qualitatively by eye judgement and hence a fairly satisfactory evaluation is done in comparatively less time. It is still prevalent in most of the countries. However, possibility of error can not be ruled out. Score card appraisal has been suggested as an alternate method based on carcass measurements. Although this method eliminates the chances of personal preferences of judges, it is too cumbersome and time consuming. Still another method of evaluation is by fleshing index, which relates carcass weight and length. A gross fleshing index gives an estimate of the amount of meat (muscle and fat), which a carcass carries relative to its length. The drawback of this method is that it does not discriminate between relative development of muscle and fat whereas over fatness is a disqualification in the modern world.

EVALUATION OF GOAT AND SHEEP CARCASSES

Specifications of Bureau of Indian Standards for evaluation and grading of goat and sheep carcasses mainly rely on fleshing in relation to length, thickness of back, fullness of legs and flank and amount of fat in intercostal muscles. There are four grades as under:

Prime Grade

i. Carcass moderately wide and thick in relation to length
ii. Plum and full legs
iii. Wide and thick back
iv. Abundant fat in intercostal muscles
v. Full and firm flank

Choice Grade

i. Carcass narrow and thick in relation to length
ii. Thin and tapering legs
iii. Thin and narrow back
iv. Slight fat in intercostal muscles
v. Slightly full and firm flank

Utility Grade

i. Carcass very narrow in relation to length
ii. Slightly concave and thin legs
iii. Slightly thin and narrow back
iv. Traces of fat in intercostal muscles
v. Soft and watery flank

Cull Grade

i. Carcass extremely angular in relation to length
ii. Extremely thin and concave legs
iii. Extremely thin and narrow back
iv. Traces of fat in intercostal muscles
v. Soft and watery flank

Raw meat (chilled/frozen) grading and marking rules (1977) stipulated by Government of India for export quality control specify following special characteristics in goat and sheep carcasses:

Choice Grade

The dressed carcass shall have:

i. Wide, deep, smooth, compact conformation
ii. Thick and neat shoulders, broad and thick back and plump legs
iii. Firm and evenly distributed finish
iv. Sufficient deposit fat
v. Dressed weight not less than 8 kg

Standard Grade

As per choice grade except following deviations:

i. Carcass slightly deficient in above characteristics
ii. Lack of trimmable fat
iii. Back of dressed carcass shows a decided prominence of bones
iv. Has enough prominent shoulders and bar ribs.

Grade X

As agreed to between the purchaser and exporter against firm order.

EVALUATION OF BUFFALO CARCASSES

Specifications of Bureau of Indian Standards for evaluation and grading of buffalo carcasses mainly rely on conformation, finish and quality, although description is in detail to discriminate in six grades as follows:

Prime Grade

i. Carcass and wholesale cuts are thick fleshed, blocky and compact.
ii. Fat covering is firm, brittle and somewhat waxy uniformly distributed over the carcass.
iii. Cut surface of meat has smooth, velvety appearance with liberal marbling.
iv. Meat colour is red, uniform and bright.

Choice Grade

i. Carcasses are moderately thick flesh, blocky and compact.
ii. Fat covering is thin in young animal to moderately thick in mature animals.
iii. Marbling in the rib eye varies from moderate to moderately abundant depending on the age of the animal.
iv. Meat colour is light red to slightly dark red.

Good Grade

i. Carcasses are slightly thick fleshed, slightly compact and blocky in appearance.
ii. Fat covering may be somewhat soft or slightly oily.
iii. Marbling in the rib eye varies from a slight or modest amount.
iv. Meat colour varies from light red to slightly dark red but may be two-toned or slightly shady.

Commercial Grade

i. Carcasses vary widely in conformation, finish and quality. Young animals are angular and slightly thin fleshed whereas mature animals are slightly thick fleshed but irregular in contour.
ii. Fat covering may be thin or thick depending on the age but is patchy. It is moderately soft or oily in young but usually firm in mature animals.

iii. The rib eye muscle is soft and watery in young animals with little or no marbling. In mature animals, marbling may be coarse.

Utility Grade

i. Carcasses are from aged animals usually angular or irregular in conformation. These are thinly fleshed.
ii. Fat covering varies and is usually soft.
iii. There is practically no marbling.
iv. Meat colour is two toned or shady.
v. Bones are usually hard and white.

Cutter and Canner Grade

Carcasses are from very aged animals with angular and irregular in conformation. The meat is tough but wholesome. It is generally cured, canned or used in making sausages.

Raw Meat (chilled/frozen) Grading and Marking Rules (1977) stipulated by Government of India for export quality control specify following special characteristics in buffalo carcasses:

Choice Grade

Bright glistening appearance, fullness of muscle with short and thick neck, shank and legs, fineness in grain and velvety touch, firmness in consistency, minimum connective tissue, fine marbling, thin covering of fat, dry, firm and flaky fat, good finish, bright red meat colour, pinkish white bone marrow.

Good Grade

As per choice grade except slight deviations such as lack of uniform conformation, slightly deficient in fullness of muscling, slight concavity in hip and thighs.

Commercial Grade

As per good grade except slight deviations such as, thick and coarse muscles, rough and irregular contour, coarse deposition of fat.

X Grade

As agreed to between purchaser and exporter against firm order.

कृपया रेषा मारुन पुस्तक खराब करु नये.

EVALUATION OF SWINE CARCASSES

Primary factors in establishing the class of swine carcasses are carcass length, dressing percentage, yield of four lean cuts, loin eye area, back fat thickness, meat colour, firmness and marbling. Modern meat consumers have a clear preference for lean, so the carcass should have a high lean to fat ratio. A good carcass length is a desired characteristic. It is measured from front edge of the first rib to front edge of aitch bone. *Dressing percentage* is calculated by dividing carcass weight by live weight and multiplying by hundred. Ham and loin are the highest priced cuts of a pork carcass in that order. Boston butt and picnic, the top and bottom portion of shoulder are also highly priced. The percentage of these four lean and highly priced cuts in a carcass constitutes an important parameter for evaluation.

Loin eye (*Longissimus dorsi*) muscle in the thoraco-lumber region is directly proportional to the development of musculature in a carcass and is included in most carcass evaluation procedure. Loin eye area is measured by tracing the cross section of this muscle at 10th rib on a parchment paper and then determining the area with a polar planimeter.

Back fat of a carcass has to be very moderate. The thickness of back fat including the skin is measured at three places – first rib, last rib and last lumber vertebra and an average is taken. It is an important factor in grading because carcasses have a normal distribution of fat and development of muscling for their degree of fatness.

Standards for Evaluation of 87 to 104 kg Carcass in Western Countries

Parameter	*Excellent*	*Average*	*Poor*
Carcass length (cm)	> 72.5	71.25	67.5
Dressing percentage	> 73	71	69
Ham and loin percent in a carcass	> 40	35	30
Percent of 4 lean cuts	> 58	50	44
Loin eye area (sq. cm)	> 31.35	23.12	18.75
Back fat thickness (mm)	< 30	40	> 50

Grading of carcasses in most countries is based on degree of conformation, finish and quality. USDA grades for barrows and gilt carcasses are given on the basis of expected yield of four lean cuts alongwith other characteristics such as *degree of muscling* and *back fat thickness*.

Grade	Expected yield (%)	Degree of muscling	Back fat thickness (inches)
US No. 1	> 53	Thick	1.3–1.6
US No. 2	50–52.9	Moderately thick	1.6–1.9
US No. 3	47–49.9	Slightly thick	1.9–2.2
US No. 4	< 47	Thin	> 2.2

Quality score based on colour, firmness and marbling of the cut muscle surface can discriminate pork carcasses into:

Traits	Superior	Acceptable	Unacceptable
Colour	Pinkish-red	Greyish-pink	Very pale
Firmness	Very firm	Reasonably firm	Very soft and watery
Marbling	Small to moderate marbling	Slight marbling	No visible marbling

It can be seen from the above text that carcass evaluation and grading requires basic knowledge of traits desired in a particular species. While net *fleshing index* can be of value in case of goat and sheep carcasses, it is the percentage of four lean cuts which is of significance in the evaluation of swine carcasses and one has to rely on conformation, finish and quality with respect to buffalo carcasses.

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Chilling and Ageing of Dressed Carcasses

As the slaughter process completes, the animal carcasses have a temperature of 30 to 39°C which makes them highly perishable. This carcass heat has to be removed rapidly to check decomposition and weight loss due to evaporation. It is a general practice in meat industry to chill meat carcasses before wholesale distribution and even before freezing.

Chilling of Dressed Carcasses

Chilling is a process of prolonging the shelf-life of meat with the help of low temperature. It retards the microbial growth in meat and slows down the enzymatic and chemical reactions which may contribute to deterioration and spoilage. As soon as the post-mortem inspection is completed, the carcasses are passed on to chilling room on the overhead rails in the suspended state itself. Chilling room is maintained at a temperature of –1 to +3°C. For effective chilling an air speed of 0.5 to 0.75 meter/second is generally used. Besides, relative humidity of chilling room is kept at about 90% so that moisture loss or shrinkage is minimum. Warm carcasses will inevitably increase the temperature of chilling room, so care should be taken not to overload it.

The rate of carcass chilling depends on several factors-the temperature of chilling room, the size and fat covering over the carcass and the speed of air circulation in the chilling room. In general chevon, lamb, pork and veal carcasses require 24 to 36 hours to attain an internal temperature of 5°C or below whereas large animal carcasses require 48 hours or more to attain the same temperature. The chilling time can be reduced by 25% if high velocity air is passed through the carcasses. Two chilling rooms are sometimes used for heavy buffalo or beef carcasses. It should be noted that rapid chilling of hot carcasses may result in undesirable toughness of muscles. This phenomenon is called *cold shortening*. The initial chilling room, which receives warm

carcass sides or quarters, is maintained at about 7°C whereas the latter is maintained at a temperature between –1°C and 2°C, air speed being 0.75 meters/second in both of them.

Due attention is needed to keep the moisture loss or shrinkage during chilling to minimum in order to retain quantity as well as quality. Excessive moisture loss can affect the colour and appeal of meat surface making it dark, dry and shriveled. A relative humidity of 88 to 92% has been found to be ideal because higher relative humidity may encourage slime and mold growth. Even with these precautions, shrinkage loss of 1.5 to 2% is inevitable.

Guidelines for Proper Chilling of Dressed Carcasses

1. Chilling room temperature should be maintained between –1 to 3°C and it should be monitored on a regular basis.
2. Relative humidity of the chilling room should invariably be kept at 88 to 92%.
3. Hot carcasses should be hung separately from cold carcasses in the chilling room.
4. Carcasses should not be allowed to come in direct contact with each other.
5. Chilling room should not be overloaded with carcasses.
6. Chilling room door should have a self-closing device so that it remains closed as much as possible.
7. Temperature of chilling room should always remain above –1.4°C, since meat freezes at this temperature.
8. Chilling room equipment has to be safeguarded against excessive condensation.

Chilling of Dressed Poultry

As soon as the poultry carcasses are eviscerated, they are subjected to chilling. The purpose is to lower the temperature of carcasses from about 37°C to below 7°C. It retards the growth of microorganisms and increases the shelf life of meat. Due to increase in intramuscular osmotic pressure, water-holding capacity of meat is enhanced. Besides, there is increase in tenderness of meat due to rupture of chemical bonds and associated changes.

Chilling of poultry carcasses is accomplished with the help of chilling water or chilled air. Care is taken to avoid extremely rapid cooling so as to rule out cold shortening of muscles and the resultant toughness of meat. Due to this reason, slush ice (water and ice mixture, 50:50) has been widely accepted as an ideal medium of chilling. Immersion chilling may be either still water chilling or mechanically agitated water chilling. Cross contamination of carcasses through chilling process can be checked by adding chlorine in the chilling water to the extent of 25 ppm. It is also very important to control the uptake of water by the carcasses to the satisfaction of regulatory authorities and the consumers in general.

Ageing of Dressed Carcasses

In the absence of microbial spoilage, the holding of unprocessed meat above freezing point is known as *ageing*. It is also frequently referred as *conditioning* and sometimes *ripening*. During this period of holding at 0 to 3°C i.e. above freezing point several changes occur in meat at a subtle rate. Atmospheric oxidation proceeds very slowly in the dark. Bacterial action is retarded to a large extent. Proteolytic enzymes (proteases) within muscle fibres remain active and fragment myofibrils in natural course. Cathepsins or autolytic enzymes also play their role. Maillard reaction also proceeds to a varying degree. A combination of these alterations bring about desirable changes in the sensory attributes of meat system especially increase in tenderness, flavour and to some extent in the juiciness. Increase in tenderness is relatively rapid during first 3 to 7 days postmortem and tenderization rate diminishes after that.

Two types of postmortem ageing procedures are commercially practiced-dry and wet ageing. Dry ageing is the traditional procedure in which entire carcass or wholesale cuts, without any packaging, are hung in the chilling room at 0 to 1°C for 3 to 4 weeks at relative humidity of 86% and air velocity of 0.5 meter/sec. These conditions can vary widely at commercial level. Wet ageing is the predominant commercial practice these days wherein wholesale or primal cuts are put in vacuum bags and held at 0 to 1°C for 7 to 10 days. In such a situation, humidity and air velocity provisions become superfluous.

Some of the significant changes in the meat system during ageing are:

i. *Protein denaturation* Denaturation refers to physical rearrangement of chemical bonds in the amino acids of protein polypeptide chains without involving any hydrolysis. During postmortem ageing myofibrillar and

sarcoplasmic proteins denature to a varying degree. There is detachment of actin filaments at Z-lines resulting in the fragmentation of myofibrils. It enhances tenderness, although muscle proteins manifest some loss of water holding capacity. However, connective tissue proteins-collagen and elastin do not undergo denaturation.

ii. *Proteolysis* Denatured proteins are particularly susceptible to the action of proteolytic enzymes. So myofibrillar proteins are very prone to these enzymes. During ageing, sarcoplasmic reticulum looses the capacity to retain Ca^{++} ions and their release initiate a water-soluble enzyme called calcium-activated sarcoplasmic factor (CASF). This factor degrades desmin (a Z-line protein), connectin (gap filaments), troponin T (above pH 6), tropomyosin and M-line proteins causing tenderization of meat.

Cathepsins or lysosomal enzymes become active at low ultimate pH or comparatively high temperature and bring about degradation of myosin and actin to fragments. Besides, they also degrade cross-links of non-helical telopeptides of collagen and mucopolysaccharides of ground substance. Some lysosomal enzymes operate during postmortem ageing to cause hydrolysis of sarcoplasmic proteins to peptides and amino acids. Proteolysis, thus, brings about some improvement in water holding capacity of meat.

Some collagen fibres appear to swell during ageing suggesting partial damage to crosslinks in perimyseal and endomyseal collagen and solubilization to a limited extent.

iii. *Flavour enhancemennt* During postmortem ageing, ATP is broken down to mononucleotides-AMP and IMP which produce inosinic acid and hypoxanthine, enhancing flavour of meat. Besides, there is production of some flavour compounds such as hydrogen sulphide, ammonia, acetaldehyde, acetone and diacetyl etc. by microbes, most particularly by yeast and in long term ageing by molds.

Due to break down of protein and accumulation of free amino acids and presences of traces of soluble carbohydrates contributing carbonyl groups, maillard reaction can also take place during later part of ageing. This non-enzymatic reaction can form brown compounds, which may cause some discolouration and may impart some bitter taste to meat.

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Disposal of Carcasses and Condemned Parts of Animals Died due to Notifiable Diseases

Notifiable diseases of animals are those diseases, the break out of which on the farm must be reported to the veterinary or other appropriate government authorities. This is necessary in order to undertake speedy control and preventive measures that are deemed fit by the authorities. These actions may be restriction in the movement of animals from one area to another, mass vaccination of animals in the affected area, educating the farmers about isolation of diseased animals, disposal of dead animals, disinfection of the premises etc.

The list of notifiable diseases of animals varies in different countries, depending largely on the range of enzootic and epizootic diseases in a particular country and the likely risk of diseases being imported or emerged.

A list of notifiable animal diseases in India is produced below:

1. Anthrax
2. Black Quarter (BQ)
3. Foot and Mouth Disease (FMD)
4. Haemorrhagic Septicaemia (HS)
5. Rinderpest (RP)
6. Rabies
7. Johne's Disease (JD)
8. Tubercullosis
9. Glanders and Farey
10. Epizootic Lymphangitis
11. Surra
12. Pourine

The action with respect to disposal has to be based on the reasons for condemnation of the carcass or a part thereof, which

in turn depends on antemortem examination and postmortem inspection including laboratory tests. A rational judgement is always the most difficult part of meat inspection.

Disposal of Carcasses

It is utmost important to properly dispose of the carcasses of animals died of notifiable disease in order to prevent the spread of disease and to prevent human infection in case of zoonotic disease. In fact, an animal died of contagious disease should be removed from the shed as early as possible, since it is visually unsightly, may give offensive odour within few hours and may become a disease hazard. The carcass should not be dragged because discharge of its body fluids and blood during dragging may cause infection in the other livestock. It should preferably be lifted by mechanical means such as loader etc.

Carcasses should never be disposed of near the flowing water, otherwise it will become a potential source of infection in the areas receiving subsequent water supply. It is also not advisable to open carcasses without the approval of a veterinarian even for flaying. The handling of the carcasses has to be restricted to bare minimum. The method of disposal of the carcasses has to be decided on the basis of disease responsible for death.

Isolation of Carcasses and Related Materials on the Spot

The carcasses of animals died with symptoms of high risk infectious diseases should be immediately isolated and their related materials such as used feed, excreta etc. should be disinfected on the spot. Before removal, carcasses should be kept in dry place and covered with a polyethylene sheet. It will be wise to isolate the spot from the nearby environment by checking open feed as well as water supplies and drainage system through this spot.

Veterinary Advice

Veterinary authorities should examine the suspected and exposed animals and diagnosis of high-risk infectious diseases should be made by them. They should also render advice concerning the mode of transportation and methods of disposal of carcasses besides suggesting protective measures to be undertaken. Further, it should be indicated as to what equipment, items etc. need to be disinfected and which method will be more suitable.

Transportation

Carcasses or condemned parts thereof should be transported to the place of disposal with the utmost care. There is no need to collect such high risk infectious material from different locations at one place. Persons involved in handling such materials have to be provided protective clothing and materials. Such carcasses and condemned parts should be lifted and loaded by mechanical means. Transportation should be done in vehicles which are exclusively used for carrying carcasses. Such vehicles are properly covered from all sides. These vehicles should also be subjected to proper cleansing and disinfection after every use to prevent spread of contamination into the environment.

Methods of Disposal of Carcasses

i. *Burial method* This is the most common method and is fairly safe if the burial pit is dug 2 meter deep or highest part of the carcass is atleast 1.5 meter below the level of surrounding terrain. Besides the carcass, left over feed by the dead animal, its bedding, excreta and top 5 cm soil floor are also buried with the carcass. Deep burial will prevent the jackals from digging up the carcass and insects from carrying the bacterial spores to the surface. Once the carcass is in grave, the skin is slashed and drenched with crude phenol. Then the carcass is covered on all sides with quick lime and filled with mud and topped with some concrete objects. In case of anthrax, before removing the carcass for disposal, all its orifices are plugged with cotton soaked in 5% cresol and body is wrapped in similarly soaked bag. The byre should be disinfected with 5% cresol using long handled brushes and then washed over with freshly chlorinated lime-wash.

ii. *Burning or incineration method* Carcasses can be effectively destroyed by burning. It can bee conveniently done in incinerator where a temperature of 600-800°C is reached and all the organisms are destroyed. This method is quite suitable for animals that have died due to heat resistant microorganisms such as spores of *Bacillus anthracis*. If incinerator facility is not available, dead animals and other materials can be burnt in a pit, preferably dug near the site of death. The pit or trench (about 0.5 meter deep) is first filled with wood, making air spaces with the help of cross

iron bars. Then the carcass is placed and ignited with the help of kerosene. After complete carcass is burnt, the trench is filled with mud.

iii. *Chemical treatment* If the animals die from diseases which do not pose a potential health hazard, the carcass can be disposed of by usual means. Then the premises are treated with lime water (1:20) or with a suspension of bleaching powder (1:20). An aqueous solution of cresol or phenol is preferable for metal parts.

Disinfection

It is not enough to dispose of the animal died due to notifiable disease. All the related items and materials, which could serve as contaminating agents, are disinfected. Bedding straw, manure etc. should be buried, burnt or disinfected by mixing with slaked lime. Liquids such as blood, urine etc. should be disinfected with a 30% suspension of chloride of lime. For walls, floors, doors and tools, cleansing with lime-water (1:20) or with a suspension of bleaching powder (1:20) may be used. Metal tools or instruments of the abattoir may be disinfected by immersion in boiling water for this purpose. A 3% solution of washing soda (sodium carbonate) can also be used. Besides, hooks, handles, covers etc. should be scrubbed in boiling water or solution of washing soda.

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Maintenance of Slaughter houses and their Sanitation

The contamination in slaughter house is mostly carried by animals. Overcrowding and improper cleaning of lairages further increase the possibility of carcass contamination. In the slaughter house bacteria spread as a result of contact with personnel, clothing, equipment and surfaces. Besides insects, vermins, birds etc. also contribute to the spread of bacteria. If the hides or fleece are heavily contaminated, it will transfer a lot of contamination to the carcasses during dressing. So, it is utmost important to keep the initial contamination on the animals to as low as possible and then maintain strict hygienic conditions at all stages of abattoir operations. Besides above mentioned factors, abattoir hygiene will depend on its location, layout, construction, equipment, awareness among employees and facilities as well as routine of cleaning operations.

Sanitation is not only important from public health point of view but it also enhances the shelf life of the product, promotes consumer acceptance and increases the efficiency of workers. In fact, cleaning and sanitation denote a way of life. Cleaning refers to freedom from all foreign material. In order to be considered clean, the place, plant and equipment should look, smell and feel clean. On cost considerations also, sanitation should be considered as a necessary investment and not a luxury.

Building

The area surrounding the slaughter house building must be well maintained. It should be properly drained leaving no scope for water logging. Inedible material and manure should be collected in closed containers and removed regularly. Building proper should be vermin and fly proof. The junction between ceiling and walls should be rounded for convenient cleaning. Floor angles and corners should be imperviously sealed. Paint should be lead-free. The machinery installed in the building should be

smooth and its functional surfaces should be easily accessible for cleaning. Wood fittings are not allowed. There must be adequate washing facilities for personal hygiene. Adequate facilities for disinfection of knives and tools should be there in the plant. Eating and smoking should be banned in the abattoir building.

AGENTS USED IN SANITATION

1. Water Water is a general solvent. It wets the surface and floats away loose debris and dirt. It also washes away the traces of cleaning compounds. Water used in cleaning should be soft, otherwise dissolved salts may cause *'water spotting'*. Hot water (80-90°C) and steam alone or in combination can be used for cleaning utensils, food contact surfaces and small equipment. Steam is a suitable sanitizer for the conveyor belts. Cleaning with water alone is not enough. It becomes necessary to combine detergent/sanitizer with water for better results.
2. Detergents or surface active agents They act by reducing the surface tension, thereby lowering the amount of mechanical energy required for cleaning. By definition a *detergent* is a cleansing substance which acting in combination with water can remove debris or dirt from surfaces. It may be natural or artificial and may be in powder or liquid form. In fact, water itself can be referred as natural liquid detergent but it has a poor wetting power or surfactancy due to high interfacial tension. A detergent reduces the tension at water surfaces. It floats the dust particles, breaks down the particulate and greasy dirt and keeps it in suspension with the help of agitation and internal electric forces. It may not have any bactericidal activity.

Properties of a good detergent

i. It should be a good surfactant or wetting agent to be able to remove dirt and dust from the surface.
ii. It should ensure maximum rinsing.
iii. It should be capable to effect maximum suspension and deflocculation of greasy dirt.
iv. It should not be toxic.
v. It should not have a strong off odour.
vi. It should be able to combine satisfactorily with sanitizers.
vii. It should not be corrosive to metals.
viii. It should be biodegradable.
ix. It should be economic to use.

There are mainly two categories of detergents:

i. *Soapy detergents* are prepared by heating animal or vegetable fat with caustic soda.
ii. *Soapless detergents* are prepared by action of trioxide on mineral oil. These are available as powder or liquid.

Recently enzyme-based foam detergents have been suggested for cleaning fat and protein of animal origin.

Detergents can also be classified into four main groups—*anionic* (e.g. soap), *cataionic, non-ionic* or *amphoteric* detergents depending on their ability to produce negative, positive, nil or both type of ions (depending on the pH) in the solution. Among these, anionic and non-ionic detergents in the ratio of 2:1 are most suitable for meat industry. When used with hard water, anionic detergent can form scum or scale. To prevent this, chelating agents such as EDTA are used to sequester or lock up calcium and magnesium ions.

Detergents have also been classified as:

a. *Organic detergents* e.g. soaps; synthetic or lather producing material
b. *Inorganic detergents* e.g. carbonates, phosphates, sulphates, sulphonates etc. of sodium or potassium.

There is another classification for detergents—alkaline detergents act when the pH is more than 7 e.g. sodium carbonate (pH range 7-10), sodium metasilicate (pH range 10-12), sodium hydroxide (pH range 8-13). These are very effective for the removal of organic substances such as fat, grease, protein etc. Their use is limited due to corrosive nature. Acidic detergents act at lower pH (phosphoric acid, acetic or citric acid). These are effective against non-organic dust and so rarely used in meat industry.

Formulation of a general purpose liquid detergent

Phosphoric acid	: 35%
Surfactant	: 1%
Water	: 64%

Formulation of a general purpose powder detergent

Sodium metasilicate	: 5%
Sodium carbonate	: 30%
Sodium tripolyphosphate	: 30%
Alkyl aryl sulphonate	: 5%

5 parts of this powder are added to 1000 parts of water.

Sanitizers or Disinfectants

These compounds decrease the number of microorganisms to a level which is generally considered as safe.

Properties of a good sanitizing agent

i. It should be capable of rapidly killing microorganisms.
ii. It should be effective against gram positive and gram negative bacteria as well as majority of spores.
iii. It should be readily soluble in water.
iv. It should not be toxic.
v. It should not have an offensive odour.
vi. It should be stable during long and short term storage in concentrated and dilute forms, respectively.
vii. It should not be corrosive or staining to metallic equipment.
viii. It should be economic to use.

The sanitizing agents of common use in meat industry can be classified into four groups:

i. *Halogens* Among the halogens, chlorine and iodine are the main compounds used as sanitizer in meat and milk industry respectively. Chlorine compounds like sodium hypochlorite (concentrated liquid) when dissolved in water produce hypochlorous acid and hypochlorite ions. Out of these, hypochlorous acid has better efficacy. The recommended limit of chlorine as a sanitizer is 100 ppm for carcass washing, 15-200 ppm for abattoir sanitation and 250 ppm for clean equipment. However, prolonged exposure of chlorine is corrosive to metals.
ii. *Quarternary ammonium compounds* These are cationic detergents such as (a) cetyl trimethylammonium bromide (CTAB), (b) lauryl dimethyl benzyl ammonium chloride etc. which are bacteriostatic mainly against gram positive bacteria. These compounds are not very suitable for meat industry because these are most effective at neutral or alkaline pH. Their efficacy will be markedly affected in combination with anionic detergents or soaps or hard water. Further, these are expensive as compared to halogens.
iii. *Amphoteric compounds* Long chain substituted amino acids or betains are surface active agents. These are less corrosive and their activity is not affected even with hard water. However, these are quite expensive, so not of much use in food industry.
iv. *Detergent-sanitizers* Several compatible and complementary ingredients make the cleaning and disinfection possible in a single operation. These are quite effective against a wide spectrum of soils and microbes. Some important detergent sanitizers are:
   a. Sodium hydroxide plus quarternary ammonium compounds.
   b. Hypochloric acid plus iodophores.
   c. Phosphoric acid plus hypochlorite

ORGANIZATION OF CLEANING SCHEDULE

The entire sanitation team should be organized under a sanitary supervisor. All the persons working in the team should be trained in personnel, plant and environmental hygiene. The supervisor should workout a cleaning schedule with clear instructions for all areas and equipment. An intensive cleaning programme should be undertaken daily by the specially trained 'hygiene gang' immediately after the slaughter house operations are over. Hence, there should be provision of enough water, hose pipe points, high impact pressure water, detergents and sanitizers.

Sanitation in abattoirs and meat processing plants can be carried out by adopting a six step cleaning process:

i. Physically pick up all the waste material from floor and equipment.
ii. Dry sweep the floor to remove all the particles of meat, fat, dung, scraps etc.
iii. Pre-rinse the floor and equipment with water (50°C).
iv. Wash with appropriate detergent at high pressure.
v. Rinse again with hot water (80°C).
vi. Apply appropriate sanitizer.

It should be noted that equipment are required to be rinsed again with hot water (80°C) before use to remove the sanitizer.

Lot of emphasis is being given these days to foam and gel cleaning solutions which are very effective in cold and hot water, respectively. The foam adheres to the surface and increases the contact time and hence the effectiveness. The foam spreads to large and even inaccessible areas on the equipment and surfaces. These are economical to use, besides saving labour and time. These are biodegradable in nature.

Automated Cleaning Systems

i. *Central cleaning system (CCS)* This system is suitable for abattoir and meat processing plant. It has a central pump

which supplies cleaning solutions under pressure to remote areas in the plant. The unit works at very high pressure and flow rate.

ii. Cleaning-in-place (CIP) system : It is widely used in liquid food industry but has limited application in meat industry. It can be used to clean internal surfaces of equipment such as large mixers, choppers, processors, tanks etc. which can remain assembled. Cleaning is done by serial circulation of water, detergent and sanitizer through pipelines.

iii. Self contained cleaning (SCC) system : It comprises of a pump and chemical spray unit. Steam or hot water may be connected with foam production facility. It has better flexibility.

Personal Hygiene—An Essential Training

A sound training programme has to be devised with focus in personal hygiene for the employees of a slaughter house. This is absolutely essential in India where meat trade is in traditional hands and handling practices are passed from one generation to the next generation. The syllabus should include information on contaminants, their sources, replication; perishable nature of meat; cleanliness, sanitation, sterilization, wholesomeness, hygienic practices; use of relevant tools and equipment, their upkeep; protective clothing occupational hazards, food poisoning etc. The objectives can be better achieved with the help of audio-visual aids—films, charts etc.

The employees have to be medically examined at the time of recruitment. They must be vaccinated against contagious diseases. An awareness for cleanliness has to be created in the butcher community as a whole. The focus should be on remodeling their entire attitude and behaviour. They have to be groomed for cleanliness, hand and hair care, toilet hygiene and prompt treatment of cuts, abrasions and other skin lesions. Knowledge of first aid and slaughter house regulations should also be imparted to them.

Sanitation in an abattoir is an essential effort in the preservation of public health by production of clean, safe and wholesome meat. It is a prerequisite in maintaining the strict hygienic condition in the plant. It also enhances efficiency of workers and marketability of the product.

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Abattoir Effluent Treatment

A large quantity of water is utilized in the abattoir. Water used for cleaning blood from slaughter section, ruminal/stomach contents from dressing section, pen manure and similar material contain lot of organic matter as suspended solids. This waste water has great pollution capability and should not be directly connected to municipal sewer line. However, water from cooling towers and toilet lines should be directly connected to sewer lines.

The load or concentration of organic matter is measured in terms of *biochemical oxygen demand* (BOD) and is expressed as ppm or mg/litre. The BOD is the amount of oxygen required to decompose the organic matter completely by aerobic biological action in certain period at 20°C (BOD_5 in 5 days). BOD of normal domestic sewage is 250-300 ppm (mg/litre) whereas it is about 5 to 6 times in abattoir effluent (1600-2000 ppm). Thus abattoir effluents have a high potential for polluting the water courses and on-site treatment of this effluent is must before connecting it to municipal sewer line.

General Principles of Treatment

Although several effluent treatment systems are in vogue, the basic principle involves screening of solids, removal of fat particles and then decomposition of organic matter by bacteria either aerobically or anaerobically in the tanks or ponds. The entire process is carried out in two stages. In first stage, gross solid particles suspended solids and fat particles/grease are separated. It reduces about 65% of the solids and 90% of the fat. BOD is also reduced by 35%. In the second stage, remaining organic matter is subjected to biodegradation and it is reduced to harmful or even industrially useful substances (Fig. 11.1). Thus, BOD of the waste water is drastically reduced. It can now be connected to sewer line or dried up in pits as per the situation.

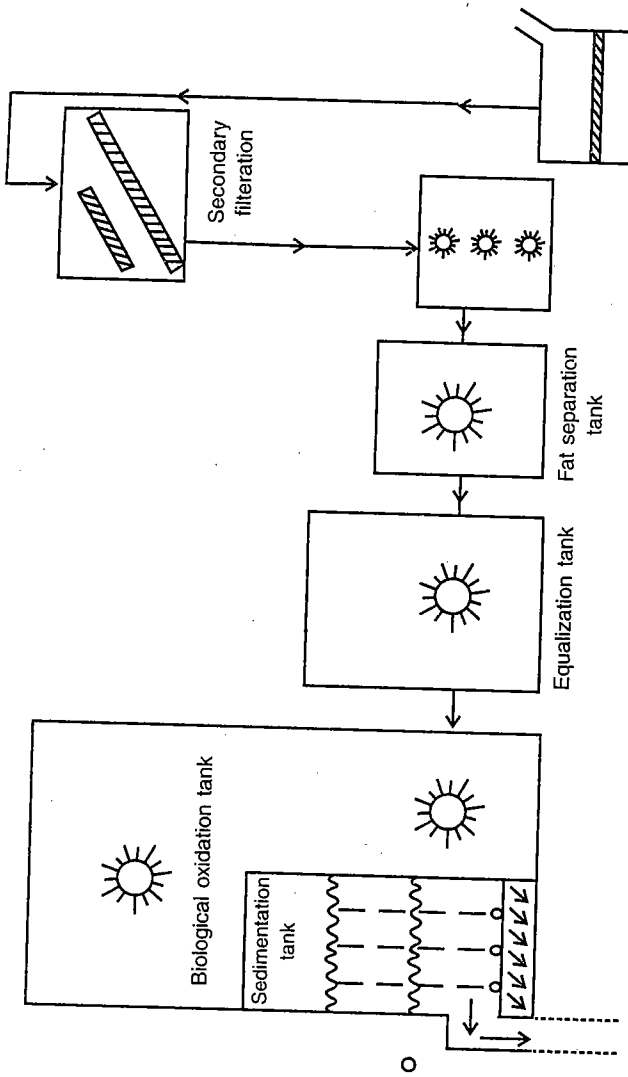


Fig. 11.1: Effluent treatment plant

In general, abattoir effluent treatment involves the following steps:

1. *Primary Filteration* It is carried out in a tank constructed below the ground level. It is divided by a partition of strong steel meshes. The main trunk line drainage of meat plant or abattoir opens into the first part. The gross solid materials like bits of fat, flesh, stomach, intestine, hide etc. are filtered through the mesh. The waste water free of gross solids is pumped to the secondary filteration unit.
2. *Secondary Filteration* This unit is erected at the first floor level and contains two vibrating screens with fine mesh which are arranged at an angle. This unit separates the suspended solids. Subsequent treatment is done in tanks erected at the ground level.
3. *Fat Separation Tank* It is a specially designed tank where waste water is agitated by pumping the air at several points. The separated fat rises to the top and is skimmed off at regular intervals.
4. *Equalization Tank* It is a large tank fitted with floating level mechanical aerator. Here waste water is continuously agitated to have a uniform quality for proper biological oxidation. As a biological stimulant, a small quantity of activated sludge is also recycled into this tank.
5. *Biological Oxidation Tanks or Ponds* Further treatment of waste water depends on the availability of land or open space.

   If enough land area is available, **aerobic process** is adopted for biological oxidation. Several shallow ponds of about 1 meter depth are filled with waste water, which is agitated in the presence of ample supply of air. The ponds are periodically replenished with biologically active sludge from the sedimentation tank. An efficient biological degradation of organic matter present in waste water in the presence of oxygen brings about substantial reduction in BOD.

   In limited land area, **anaerobic process** is adopted. Deep tanks of about 4.5 to 5 meter depth are filled with waste water which is continuously agitated. Some quantity of activated sludge is replenished at periodic intervals from the sedimentation tank. This anaerobic biodegradation of organic matter also brings about nearly 60% reduction in BOD of treated water.

It has been found that a combination of aerobic/anaerobic methods are suitable for treating the effluents of abattoirs and meat processing industries.

6. *Sedimentation Tank* It is a specially designed tank with rows of corrugated PVC sheets followed by baffles. The unoxidized organic matter and other suspended material is removed here through gravitational settlement. The clear supernatant fluid is directed to drain connected with the regular sewer line.

Besides above operational sections, sludge accumulation tank and filter press can also form a part of effluent treatment to further increase its efficiency.

Present waste handling situation in India is in very bad shape in most slaughter houses. The abattoir waste water is either directly connected to municipal sewer line causing blockages and pollution at many places or it is passed through poorly constructed open drain channels to stagnate in the surrounding area. It leads to the presence of vultures, stray dogs etc. in those areas. The entire area emanates foul smells and this pollution is one of the major causes of stigma against the establishment of abattoirs. With the growing concern for clean environment, it is hoped that our civic authorities will change their mindset and make determined efforts to treat the abattoir effluents at the origin itself.

twelve

Meat Eating and Animal Welfare

Meat is a highly nutritious food. It is almost fully digestible. It is appealing to the senses of taste and alofaction. The nutritive value of meat is attributed to its abundant high quality proteins, essential fatty acids, some important minerals and vitamins.

Meat is a concentrated source of superb quality proteins with very high biological value. It is rich in essential amino acids. Meat fat contains ample amount of essential fatty acids. In general, meat is a good source of all important minerals except calcium. It is rich in iron. Lean meat is an excellent source of B-complex group of vitamins. It has only traces of fat soluble vitamins.

It is a *paradox* that most of the principal meat species themselves are plant eaters. They are living reservoirs of human food. By any ethics, they are required to be treated well. *Since we are exploiting their potential for our benefit, we must look after their comfort and well being during their limited life span.* In fact, our interests are inter-related.

Animal welfare relates to the physical fitness and mental well being of the animals. In good welfare conditions, the animal is able to cope up with its environment with ease. In stress conditions, the animal gets alarmed, heart rate is increased and normal behaviour is also changed. Good management practices and enough housing space provide good animal welfare. In a herd high mortality rate indicates poor welfare. Removal of the animals from the acclimatized environment also evokes a stress response. Besides, transportation to long distances in crowded space and sudden jerks may result in physical injury, bruising or broken bones causing lot of pain to the animals. Prevalence of bruises is also an indicator of poor welfare.

Here it will be pertinent to note that cruelty against animals is punishable under PCA Act (1960). The main provisions of the Act are reproduced hereunder:

The Prevention of Cruelty to Animals Act, 1960

Short Title:

1. This act may be called the Prevention of Cruelty to Animal Act 1960.
2. Definitions : In this act, unless the context otherwise requires.
   a. "Animal" means any living creature other than a human being.
   b. "Board" means the animal welfare board established under section 4.
   c. "Captive" animal means any animal (not being a domestic animal) which is in captivity or confinement.
   d. "Domestic" animal means any animal which is tamed.
   e. "Local authority" means a municipal committee, district board or other authority.
   f. "Phooka or Doom De" includes any process of introducing air or any substance into the female organ of a milch animals.

Animal Welfare Board

1. Constitution of the board consists of
   a. The inspector general of forest, government of India, ex. officio.
   b. The animal husbandry commissioner to the government of India, ex. officio.
   c. One person to represent such association of veterinary practitioners.
   d. Two persons to represent practitioners of modern and indigenous system of medicine.
   e. One person to represent each of such three societies dealing with the prevention of cruelty to animals.
2. Term of office and conditions of service of the members of the board: Three years.

Cruelty to Animals

3. *Treating animals cruelly*. If any person
   i. Beats, kicks, overrides, over drives, over loads, or otherwise treats any animals so as to subject it to unnecessary pain.
   ii. Promotes or takes part in any shooting or competition wherein animals are released from captivity for the purpose of such shooting, he shall be punishable with fines which may extend to fifty and one hundred rupees or with 3 months imprisonment or with both.
4. Penalty for practicing of "Phooka or Doom De" he shall be punishable with fine which may extend to one thousand rupees or with imprisonment for 2 years or with both.

Experimentation on animals: Penalties – if any Person

a. Contravenes any order made by the committee under section 19 or
   Institutional Animal Ethics Committee set up under Breeding and Experiments on Animals (Control and Supervision) Rules, 1998 of the Act require location of animal houses in a quiet atmosphere and premises to be kept tidy, hygienic and animal protected from drought and extremes of weather. Animal cages for small animals and stables for large animal shall be such that animals can live in comfort and over crowding is avoided. It requires animal attendants to be suitably trained. It also stipulates that if an animal is suffering from abnormally severe pain at any stage of experiment, it shall be painlessly destroyed. Institutional Animal Ethics Committee is also required to maintain a record of the animals under its control and custody in the specified format.
b. Commits a breach of any conditions he shall be punishable with fines which may extend to two hundred rupees.

Performing animals: Restriction on exhibition and training of performing animals

Applied to be registered under this act which when not entitled to be so registered, he shall be punishable on conviction with fine which may be extended to five hundred rupees.

Tips for a Good Animal Welfare

1. Animals should be housed in such a space that these are able to perform their normal behaviour patterns. European Union has specified the space requirement for pigs as 0.65 m^2 per animal and for caged hens as 550 cm^2 per bird to live in.
2. At the farm, animals should be handled in a humane way. Husbandry practices should be adopted keeping in view the overall comfort of the animal in mind. The flooring and fence design should reduce the injuries to the animals.

3. Design of the restraint system should be evolved keeping in mind the highest considerations for animal welfare.
4. During transport, care should be taken to avoid any unnecessary suffering of the animals. The loading and unloading should be made smooth with the help of low slope wooden ramps. Proper mode of transport, overhead protection, provision of enough space and ventilation should be taken care of.
5. In the lairage, it is advisable not to mix strange animals, especially males. If possible, animals from the same herd should be kept together with adequate straw bedding. There should be provision of ample drinking water.
6. In the slaughter house, an animal awaiting slaughter should not be allowed to view the sticking of the other animal. Fear triggered by strange movements, noises and smell leads to severe physiological stress. It is further enhanced by the separation from fellow animals.
7. A suitable stunning procedure that can keep the animal in unconscious state for about a minute, should be followed. The sticking should be completed well before the animal is expected to regain the consciousness. The techniques should conform the socio-religious needs, least painful slaughter and good meat quality at the same time.

Meat is a highly nutritious human diet. The debates on animal welfare should not be allowed to come in the way of its popularity. In fact, it will be appropriate to direct the efforts to improve the well being of food animals and the conditions at the slaughter house simultaneously in order to popularize meat in the human diet. It should be borne in mind that excited, stressed, fatigued, suffocated, bruised and injured animals can never yield wholesome meat. So, human welfare is also linked to animal welfare.

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Scope for Utilization of Animal Byproducts

India has a vast livestock resource. Animal raising is subsidiary to agriculture for nearly 67% of the Indian population residing in rural areas. Instead of being a purely commercial activity on farm lines as practiced in western countries, it is part of their way of life. Every agriculturist is a small livestock raiser also.

Presently, we have a principal livestock population of more than 500 million (Table 13.1). About 100 million animals are slaughtered every year in nearly 3600 slaughter houses where besides meat, a large quantity of byproducts or offals obtained. In liberal terms, byproduct or offal includes every part of a slaughtered animal except for the dressed carcass. Yield of byproducts from large animals and small animals are presented in (Tables 13.2 and 13.3), respectively. In tropical countries, a major part of this material is spoiled before it could be gainfully utilized. Some of it is under-utilized because of poor knowledge, skill or equipment. Thus, a good profitability in livestock sector can be brought through proper utilization of animal byproducts and waste. Our slaughter houses lack facilities for timely collection and preservation of byproducts. Besides, it is still more difficult to collect fallen animal carcasses from the scattered rural areas. Meat industry can be made more competitive if byproducts are effectively utilized. It has been estimated that sale of byproducts such as skin, pluck, head, feet, gut etc. can recover about 30 and 35% of the live animal prices of goat and sheep, respectively. Value addition of principal byproducts is also very important. Hides and skins-a principal animal byproduct was being preserved in salt and exported to various countries for less than 1000 crores a decade back. However, further processing into finished leather and leather goods and then export fetched valuable foreign exchange of Rs. 6756 crores in 1998-99 in this sector alone.

Table 13.1: Principal livestock population and slaughter in India

Species	*Population (Million)*	*Slaughter (Million)*	*Carcass Weight (kg)*
Cattle	219.64	14.20	103
Buffaloes	94.13	10.34	138
Goat	123.50	46.90	10
Sheep	58.20	19.20	12
Pigs	17.50	17.00	35

Source: www.fao.org dated the 27.4.2002

Table 13.2: Approximate yield of meat and byproducts from large animals (Buffalo and Cattle)

Sl. No.	*Meat/Byproducts*	*Liveweight : 350 kg*	
		Kg	*%*
1.	Meat (Boneless)	100.00	28.57
2.	Bone, head and feet	80.00	22.85
3.	Paunch content and waste	56.00	16.00
4.	Hide	26.50	7.57
5.	Other offal (genitalia, udder, tail end etc.)	14.00	4.00
6.	Blood	11.00	3.14
7.	Drivers (urine, body fluid, bile, dung etc.)	10.00	3.50
8.	Casings	7.00	2.00
9.	Lungs and oesophagus	7.00	2.00
10.	Fat	6.00	1.71
11.	Liver	5.00	1.42
12.	Stomach	2.50	0.71
13.	Heart	1.50	0.42
14.	Head meat and brain	1.00	0.28
15.	Tongue	1.00	0.28
16.	Spleen	1.00	0.28
17.	Kidneys	0.50	0.14

Source : Chatterjee (1981)

Classification of Animal Byproducts

According to Use as a Food

Edible byproducts generally include liver, kidney, heart, brain, intestine, thymus gland (neck sweet bread), tongue, spleen etc. Besides meat trimmings and edible fats, caul fat, back fat, leaf fat) are also included in this class.

Inedible byproducts generally include hides and skins, ear, lips, snout, gall bladder, faetus, hoofs, horns, hair, bristles etc. All parts of head animal or condemned meat and organs also come in this category.

This classification is not a rigid one. The basic criterion of division between edible and inedible byproducts is governed by a number of factors such as purchasing power of the

कृपया रेषा मारुन
एस्तक खराब करु नये.

Table 13.3: Approximate yield of meat and byproducts from small animals (sheep and goat)

Sl. No.	*Meat/Byproducts*	*Liveweight : 25 kg*	
		Kg	*%*
1.	Dressed weight	10.00	40.00
2.	Paunch content	5.50	22.00
3.	Skin	2.30	9.20
4.	Gut and tripe	2.00	8.00
5.	Head	1.40	5.60
6.	Blood	0.75	3.00
7	Fatty tissues	0.50	2.00
8.	Lungs and oesophagus	0.25	1.00
9.	Liver	0.30	1.20
10	Hoofs	0.20	0.80
11.	Heart	0.10	0.40
12.	Kidneys	0.10	0.40
13.	Spleen	0.05	0.20
14.	Pancreas	0.04	0.16
15.	Shrink (3%)	0.75	3.00
16.	Unaccounted	0.76	3.04

Source : Chatterjee (1981)

consumer, his customs and traditions, his food habits, religious outlook etc. Likewise, a group of byproducts which fall on the border line are blood, testicles, lungs, spleen, uterus etc. These are edible in a particular situation and inedible in other.

In US terminology *variety meats* include all the organs that are segregated, chilled and processed under sanitary conditions. These include liver, kidney, heart, ox tail, tripe (stomach), chitterlings, fries (testicles) etc.

According to Ultimate Use

Agricultural byproducts	Meat meal
	Bone meal
	Fertilizer
Industrial byproducts	Gelatine
	Glue
	Casings
Pharmaceutical byproducts	Insulin
	Pepsin
	Bio-chemicals
	Hormones

According to Origin

Principal byproducts are directly harvested from the animals. These are hides and skins, bones, blood, hoofs and horns.

Secondary byproducts are derived from principal byproducts e.g. blood may yield blood meal, haemoglobin, fibrin, plasma, serum etc. as secondary products or Hides may yield leather shoes, bags, belts, clothings etc. as secondary byproducts.

Table 13.4: Principal and secondary animal byproducts

Sl. No.	*Principal Byproduct*		*Secondary Byproducts and Uses*
1.	Hide and skin	:	Leather products such as shoes, gloves, belts, bags etc. Gelatin and glue Fertilizer Tallow/grease Stock-feed
2.	Hair, bristles and wool	:	Pelting, brushes, fabrics and yarn, blankets, carpet, pillow, lanolin, fertilizer
3.	Blood	:	Blood pudding, sausage mix, serum (pharmaceuticals) and plasma, albumen, fibrin, Hb, weather proof, glue, textile sizing, industrial blood products, coagulated blood products, dye setting, stock feed, fertilizer, liquid blood products etc.
4.	Bones	:	Bone meal, fat, gelatin, glue, mineral sup plement, buttons, cutlery handles and bone articles. Ostocalcium tablets from bones.
5.	Hoof and horns	:	Buttons, combs, hair pins, novelties, toys, hoof and horn meal, fertilizer, pith used for gelatin and glue etc.
	Feet	:	Neats foot oil
6.	Intestines	:	Casings, catgut—suturing material, tennis, strips, musical instrument strings, tallow, muscularis layer for stock feed, rennin from calf stomach.
7.	Blood	:	Haemoglobin, serum, fibrin, plasma etc.
8.	Condemned meat, inedible offal, trimmings and residues	:	Tallow for soap, machine oil, leather dressing, candles, meat and bone meal etc.
9.	Glands and tissues	:	Gall bladder—Bile salts, gall stones Liver—Liver extract Lungs—Heparin, peptone Pineal gland—Melatonin hormone Pituitary gland—Anterior (GH, LH, FSH, prolactin) and post-pituitary lobe hormone (oxytocin, vasopressin)
		:	Thyroid gland—Thyroxine
		:	Stomach—Pepsin, Rennin, Heparin, Rennet
		:	Spleen—Melt
		:	Parathyroid—Parathormone

contd.../

contd.../

Sl. No.	*Principal Byproduct*		*Secondary Byproducts and Uses*
		:	Suprarenal gland—Cortex and medullary hormones
		:	Thymus gland (Sweet bread)—Extract increases body resistance and calcification of bones.
		:	Pancreas—Bates (in bading of leather) -trypsin—Insulin, pancreatin, glucagon
		:	Testes—Testesterone, hyaluronidase
10.	Ruminal and intestinal ingesta	:	Recycling as stock feed, compost manure, production of methane for light, heat and power.

Advantages derived from utilization of slaughter house byproducts are as follows :

1. *It protects the environment from pollution.* Hygienic disposal of slaughter house byproducts checks the pollution.
2. *It ensures better returns to the livestock producer.* Production of value added secondary byproducts will influence livestock pricing and livestock farmer will be the ultimate beneficiary.
3. *It helps in augmenting the supply of highly nutritive feed for more productive livestock.* Meat meal is very rich in protein whereas bone meal is a good source of phosphorus.
4. *It helps in setting up secondary rural industries.* Small-scale industries based on animal byproducts particularly tanning, glue, soap, brushes, pet foods, stock feed, fertilizers etc. can be set up.
5. *It can generate lot of new employment.* Conversion of principal byproducts into value added secondary byproducts is a cumbersome process and it has enormous employment potential.
6. *It can increase the crop production.* Very simple equipment can be fabricated to convert raw and cheap inedible buffalo into fertilizer rich in nitrogen, calcium, phosphorus etc. Additionally, methane can be produced for use in lighting and heating.

Several requirements for profitable processing of byproducts which need to be attended are:

1. To ensure adequate supplies of raw material collection.
2. Provision of cheap and satisfactory storage facilities for perishable materials.

3. Development of practical commercial processes for manufacturing derived or secondary byproducts and availability of suitable equipments and chemicals etc.
4. Availability of potential market outlets for the new proposed products.
5. To arrange for technical training of personnel for proper and scientific functioning.

Thus, utilization of animal byproducts is a desirable economic and sanitary necessity. If adopted seriously, it is a road from waste to wealth. Due to extensive range of secondary byproducts, it can provide employment to millions of marginally educated people.

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Plan and Layout of Byproducts Utilization Plant

Due to our vast livestock population, it is very important to harvest animal byproducts for their economic value as well as to provide employment to marginally educated masses. In our country, livestock density demands the setting up of atleast one animal byproducts processing plant at each 50 km range. Establishment of animal byproducts processing plant within or in the vicinity of each abattoir will also alleviate the pollution problem. The resultant waste from both could be utilized by the biogas production unit for the mutual benefit.

Important considerations during the establishment of a byproducts utilization plant are:

1. In tropical and subtropical countries, a byproduct plant should be established adjacent to the slaughter house or it may be a part of the same building connected through chutes or gravity pipes. In earlier case, the passage between slaughter house and byproduct plant must be constructed with concrete, stones or bricks and the slope should be towards byproduct plant.
2. The establishment should preferably be away from inhabited area.
3. As far as possible, there should be provision of overhead rails from slaughter house to the byproduct plant.
4. A byproduct plant should have its own screened drainage with a gulley fitted with individual grease traps.
5. The byproduct plant should have clear clean and unclean sections. Charging platform or raw materials inlets form a part of unclean sections whereas processed and sterilized byproducts are prepared, stored and disposed of from clean section having separate exit point.
6. The floor, walls and ceiling should be made up of smooth and concrete material so that these can be easily and frequently washed. A floor slope of ½" per ft. is generally recommended.

7. A byproduct plant should invariably have a hide/skin salting room, renderers, tripery, boiler, manure bunker, store besides common amenities.
8. The plant building should be open and ventilated to prevent humidity build up leading to corrosion of equipment and building. It will also check the growth of moulds and bacteria in the building. The roofing material may preferably be corrugated asbestos sheeting. There should be provision of exhaust fans from each partition.
9. The equipment to be installed should cope up with the byproduct or offal accumulation expected each day. Renderer, fat settling tank, fat expeller, blood press, grind mill etc. are essentially required. A boiler of suitable capacity is inevitable for processing as well as cleaning operations. It should have a steam pressure of > 80 psi.
10. At the initial stage, processing of one or two byproducts can be taken up. The operations can be further extended in stages.

A byproduct plant (Fig. 14.1) should have the following main sections:

1. *Hides and skins section* with salt store.
2. *Tripery/Gut processing section* with cleaning tables, water taps etc.
3. *Carcass utilization plant* with renderer, bone digester, fat settling tank, fat expeller and adjoining milling room and store.

Besides boiler, worker lockers, amenities and adjacent manure pit are other important requirements. The boiler should be of suitable capacity keeping in view the availability of raw material and consumption of renderer. The requirement of steam has been worked out to be 1.25 lb per lb of raw material to be processed.

Transportation of Dead or Fallen Animals

A large number of animals die in developing and under-developed countries because of natural causes, such as diseases etc. Besides, slaughter of cows for food is banned in countries like India. This factor also increases number of fallen animals. Sometimes, condemned material from slaughter house is also available for byproduct utilization. It is necessary to hygienically dispose of these fallen animals to avoid decomposition and formation of obnoxious gases. It is also important to process the byproducts of such animals to secure the economic returns.

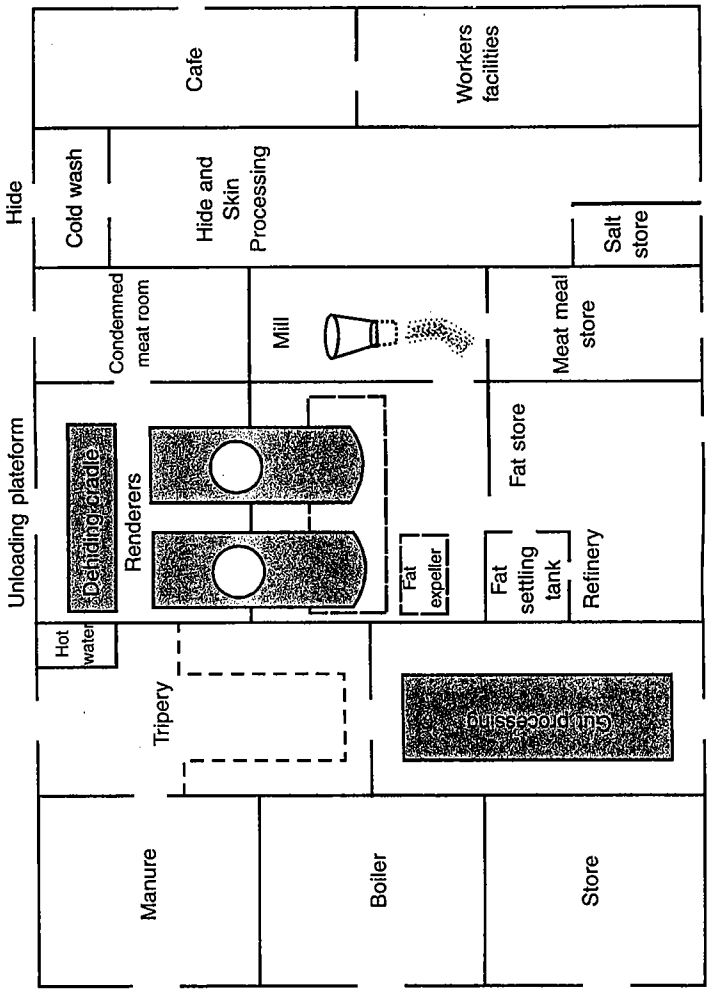


Fig. 14.1: Layout of a byproducts plant (adopted from Mann (1962)

Transportation of dead animals can be efficiently done by using road transport modes such as bullock carts or tractor trolleys. These modes have the advantages of being flexible, economic, convenient and fairly quick. Bullock carts are most commonly used in our country for carrying the dead animals, if the distance to be travelled is short, or else tractor trolleys are also quite convenient. The floor of these vehicles should be bedded with straws and be made free from any projections, otherwise bruising of the hides or skins of the dead animals lower the cost of material. The transportation of condemned material from slaughter house should be done in closed vehicles to the byproduct plant. The vehicles may be labelled as carrying as inedible animal byproducts.

Utilization of Dead or Fallen Animals

Dead or Fallen Animals				
Hides and Skins (Fallen)	Carcasses			
Leather and Leather articles	Carcass meal	Meat meal	Technical fat	Bone meal

The raw material for the carcass utilization plant generally comprises of carcass pieces, bones, heads, feet, lungs, trachea, spleen, intestine (without contents), trimmings, genitals, udder, faetus and sometimes even tripes. It should not have any hoofs, horns, hair, blood, stomach or intestinal contents.

Production of Stock Feed by Carcass Utilization Plant

The main equipment of carcass utilization plant (Fig. 14.2) are:

1. Renderer (dry or wet or twin model)
2. Bone digester
3. Fat settling tank
4. Fat expeller
5. Grinder or Mill

If carcass utilization plant is a separate unit, it should have its own boiler. Besides, provision of trolleys and overhead rails from the source of raw material to the charging platform of renderer is also important.

In general, *rendering* refers to the recovery of fat from animal material by heating. However, in context to carcass utilization plant, rendering refers to the processing of carcass with heat or steam so as to obtain a nearly sterilized material without any loss of nutrients. Technical fat is also recovered in this process.

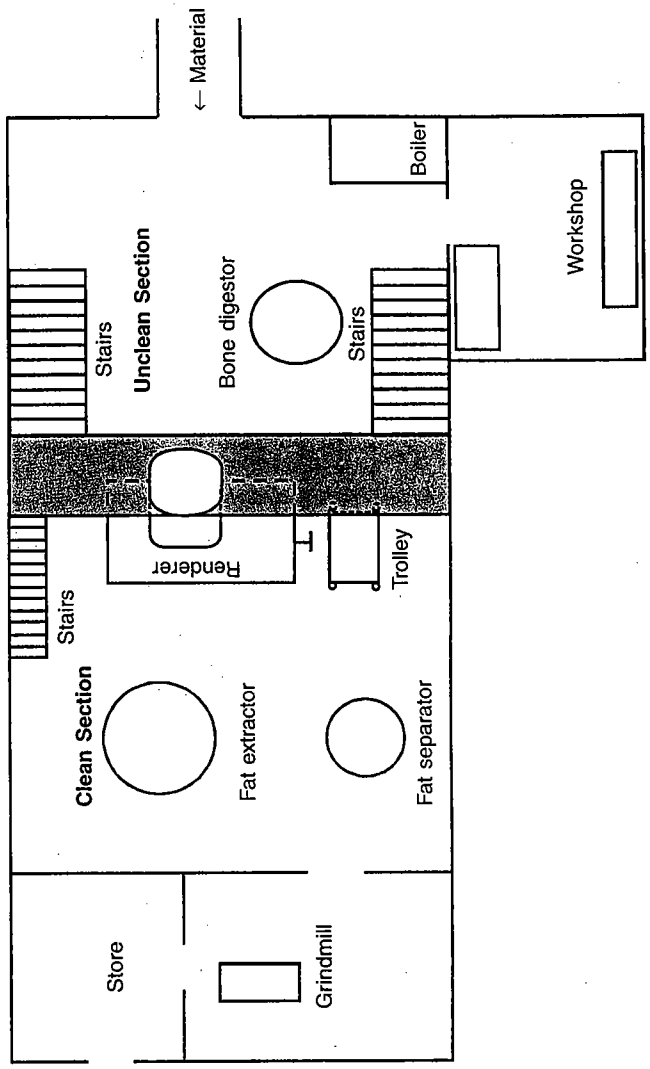


Fig. 14.2: Layout of carcass utilization plant

The resultant dried and ground material serves as a very good stock feed.

Basically, there are two types of renderers :

1. Dry Renderer
2. Wet Renderer

1. *Dry Renderer* A dry renderer (Fig. 14.3) is made up of a horizontal steam jacket with raw material or charge load capacity of 8 to 10 quintals. Two or three stirrers keep the raw material continuously under motion for uniform heat distribution to avoid charring. The steam remains in the outer jacket and does not come in direct contact with the material or charge which is cooked at 75 psi for 3-4 hours in its own moisture. There is no loss of nutrients in this process. The resultant material called cracklings is taken out. The cracklings should be gritty, fibrous and non-slippery. It still contains lot of fat, which should be removed to as low as 5% for two reasons-to economize on labour and to increase the keeping quality of the material. Removal of fat is done from cracklings inside the centrifugal expeller and fat settling tank. Long bones from crackling are sent to bone digester. Dried and nearly fat free cracklings are ground to recover meat meal. A cyclone and sacking unit allows direct filling of bags. Dry rendering process is advantageous as yield is nearly 20% higher than wet rendering. Besides, there is saving on labour and steam also.

2. *Wet Renderer* This is generally a vertical type equipment in which steam directly comes in contact with raw material or charge. The material is cooked at 40 psi for 4-8 hours. After processing, the tankage or slush is allowed to settle for 2 hours. Then grease or fatty material which floats on top is removed first, followed by water and finally by the tankage with digested meat and bones. The bones are sent to bone digester. Recovery of the fat is better in wet rendering.

3. *Bone Digester* Long bones are crushed and processed in steam at 60 psi for 2 hours. The steam separates most of the protein and fat. After extraction of fat and gelatin, the digested bones are chalky and soft and can be easily ground to bone meal which has about 32.5% calcium and 15% phosphorous.

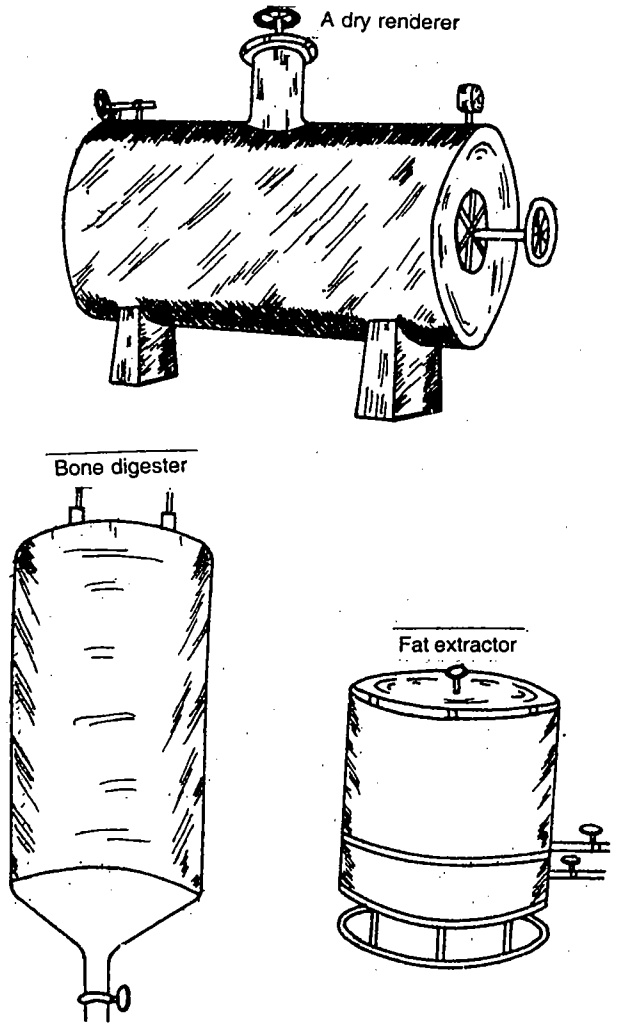


Fig. 14.3: Important Equipment of C.U.P.

Yield and Conversion Ratio

From a carcass of 100 kg live weight

Deduct 15% for ruminal and intestinal contents, hoofs and horns.

Less 7% for hide.

Less 2/3 weight will be moisture loss in drying.

Remaining 1/3 material will be meat and bone meal.

Average conversion of raw material to dry meal = 3:1 in dry rendering / 4:1 in wet rendering

Average conversion of raw blood to blood meal = 5:1

Carcass Utilization Plant (Sequence of Operations)

Carcasses
↓
Skinning → Hide/skin section → Curing → Storage and disposal
↓
Cutting into pieces (5 to 7 kg) except horn, hoofs and viscera
↓
Loading in trolley
↓
Hoisting to the charging platform of renderer
↓
Manual loading in the renderer
↓
Dry Rendering (75 psi for 3 hrs)
↓
Cracklings
↓
Manual screening → Long bones → Bone digester (60 psi for 2 hrs) → Crushing and grinding → Powedered bone meal
↓
Centrifugation for fat extraction (30 min) → Rendered impure fat → Settling tank → Techinical fat
↓
Screening of meat pieces from bones → Meat meal → Milling → Powdered meat meal
↓
Meat with bones → Milling → Powdered meat and bone meal

Typical Operations of a Dry Rendering Unit

Dehiding and cutting the carcasses into small pieces (approximately 5 kg)
↓
Receiving carcass pieces and offal on the loading platform
↓
Loading into the dry renderer
↓
↓Sterilization and drying (75 psi for 3 hrs)
↓
Cracklings dumped into the percolating tank
↓
a. Steam turbine centrifuge or fat extractor
↓
b. Hydraulic press
↓
c. Fat expeller
↓
d. Solvent extraction
↓

Defatted cracklings	Technical Fat
↓	↓
Milling	Fat Settling Tank
↓	↓
Sacking	Fat Refining Tank
↓	↓
Labelling	Packed in Drums and Labelled

Composition/Nutrients of Various Stock Feeds

Meat Meal	Protein	>55%
	Protein and minerals	85%
	Moisture	7%
	Crude fat	8%
	Phosphorus	4.5%
Carcass meal (Meat-cum-bone meal)	Protein	<55%
	Phosphorus	> 4.5%
Bone meal (steamed)	Protein	7%
	Calcium	32.5%
	Phosphorus	15%

Protein	> 80%
Calcium	Low
Phosphorus	Low

Mobile and Makeshift Carcass Processing Units

In the interior areas which experience period famines and epidemics, either mobile or simple workable equipment can be put in operation.

Mobile carcass processing unit consists of a wet renderer which operates with the help of steam. Bones are loaded first followed by carcass pieces. Only 2/3 of the renderer tank is filled with the raw material alongwith sufficient water to fill up the cone shaped bottom. Now a steam pressure of 40 psi is applied for 4-6 hours. Fat is first to be recovered after release of steam. Tankage containing stick water is rich in protein and nitrogenous substance. The defatted material is dried, milled and stored as carcass meal.

A simple equipment consisting of large kettle or split-up oil drum can also be used to process inedible offal and condemned parts from a small slaughter house or utilizable carcass parts of a fallen animal. After loading the raw material, 1½ times water by weight is added to it and material is boiled for 1 hour. After cooling, meaty portions are scrapped off the bones. Now, bones are removed and meaty mass is added with equal quantity of bran and cooked for another half an hour to a thick consistency. The resultant feed mass is rich in fat and protein and should be fed to the animals within one or two days.

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Utilization of Blood

An animal body contains nearly 5-7% blood of its live weight. However, it is not fully recovered during slaughter and some of it is always lost by spilling etc.

Average yield of blood/animal

Buffalo/cattle	:	10-12 kg
Goat/sheep	:	1-1.5 kg
Pig	:	2-3 kg
Poultry	:	30-50 g

Animal blood is used in several ways and its collection method also depends on the specific end use. Some of the uses of blood are listed below:

1. **Human food**
   a. Blood sausages
   b. Blood pudding
   c. Fibrin
   d. Peptone or lecithin
   e. Plasma—protein booster in foods
      —binder or stabilizer in meat products
2. **Industrial uses**
   a. Plasma is used as water proof adhesive in plywood industry
   b. In tanneries as light coloured protein finisher and to give gloss
   c. Lithographic coating solution which contains albumen and ammonium dichromate
   d. Textile dyeing
   e. Stabilizer for feed mixtures and pet foods
   f. Foam compounds as fire extinguishers
   g. Blood albumen as a substitute for egg albumen
   h. Ceramics
   i. Cosmetic base formulations

3. *Livestock feed*
   a. Blood meal b. Mixed blood meal
   c. Lysine supplement
4. *Fertilizer*
   a. Compound fertilizer containing nitrogen and superphosphates.
   b. Seed coating
   c. Soil pH stabilizer
5. *Biochemicals and Pharmaceuticals*
   a. Amino acids—lysine, leucine, histidine and phenylalanine
   b. Thrombin and prothrombin as blood clotting agents
   c. Immunoglobulins
   d. Serotonin, peptone, plasma extenders
6. *Laboratory and Bacteriological Media*
   a. Blood agar
   b. Tissue culture media
   c. Albumin and globulin
   d. Sphingomyelin
   e. Catalase

BLOOD PRODUCTS

Plasma and Serum

Blood consists of plasma in which erythrocytes, leucocytes and thrombocytes (platelets) are suspended. Plasma, in turn consists of serum and fibrinogen. Action of thrombin on fibrinogen separates out fibrin. Serum is a rich source of albumen and globulin. Thus, plasma is recovered from unclotted blood by centrifugation, whereas serum is obtained from clotted blood. It may be noted that normal blood clotting time of most domestic animals is 3 to 6 minutes.

Blood for human use can be collected only in those abattoirs where slaughter is conducted on bleeding rails. Trocar knife (with hollow handle) and canula are used for hygienic recovery of blood which is collected in clean and sterile stainless containers. An ideal vessel should have a diameter of 45 cm and a depth of 15 cm. It can be washed with hypochlorite or steam sterilized. To prepare plasma, blood collection is done quickly in an anticoagulant (Trisodium citrate 4%, 1 ml for each 10 ml blood, otherwise oxalate, EDTA or heparin is used). Blood is then centrifuged which will separate plasma from cell mass. Plasma is stored in bottles or polyethylene bags in frozen condition.

To prepare serum, blood is collected carefully and allowed to clot and subjected to further processing in a cold room. Chilled clotted blood is cut into smaller lumps to bring about quick contraction of clots. Serum collected in first 12 hours is generally clear except for some suspended RBCs. It is centrifuged in a bucket centrifuge at 1000 rpm and filtered through Seitz filter. The yield of serum is about 10-12% of the weight of whole blood. It keeps well at 4-5°C for one month and at-20°C for six months.

Dried blood serum is referred as blood albumen and is used as the cheapest substitute for dried egg albumen powder in the industry. To prepare it, clear yellow serum is added 0.05% phenol on weight basis. It is sprayed or vacuum dried to soluble fine powder. The yield of albumen is 10-20% of the weight of the serum. It is stable in air tight containers in cool places for several months.

Fibrin and its Products

i. *Fibrinogen* Blood collected with anticoagulant is centrifuged to obtain plasma. This plasma is buffered with sodium citrate acetic acid buffer with simultaneous addition of aqueous ethanol until pH of the solution becomes 7.2 and ethanol concentration reaches 8%. Now fibrinogen is precipitated and removed by centrifugation. It is filtered, sterilized, freeze dried and stored.

ii. *Fibrin foam* Plasma is suitably treated with citrate buffer of pH 6. The solution is clarified by centrifugation and diluted to contain less than 2% fibrinogen and pH is adjusted to 6.25. The solution is vigorously beaten to obtain fibrinogen foam. This is converted to fibrin by the addition of thrombin. The fibrin foam is frozen, cut into cubes and then freeze dried.
Fibrin foam is white to cream coloured spongy material. It can soak up liquid upto 30 times its weight and is very useful in nerve as well as arterial surgery and prostate operations. It does not inhibit the action of antibiotics and eventually digested in the body system.

iii. *Fibrin powder* It is used to arrest bleeding where coagulation is delayed in skin injuries.

iv. *Fibrin bioplasts* These are moulded materials resembling plastic which are prepared from fibrin powder. They can be moulded even as bone joints and need not be removed from the body.

Blood Meal

Most of the animal blood require proper disposal from the slaughter house for various reasons:

i. Blood will clot within few minutes and choke the drainage.
ii. Decomposition of blood will affect the sanitation and spread foul smell in the premises.
iii. It will attract flies, crows, rodents etc.

Preparation of stock feed is one of the best methods for the twin purposes of slaughter house upkeep and generating substantial income. It involves the following process :

i. *Blood collection* Blood should be collected in such a way as to avoid floor washings, detergents, insecticides or any other extraneous material.
ii. *Storage and transport:*
   a. Whole blood may be properly mixed with equal amount of rice bran which will absorb it. Such a mass can be transported in gunny bags in rainy or low sun season, whereas it can be dried in summer or hot sun on a concrete platform to nearly 10-12% moisture.
   b. To whole blood, 1% quick lime is added by weight which will give it a black rubber like consistency. This mass will not adhere to the transport containers, keep well for 24 hrs and it will not attract flies. Addition of lime also increases the calcium of the end product.
   c. Alternatively, common salt may be added @ 20% of the blood by weight to increase its storage life and enable safe transport to the plant.
iii. *Heat treatment or cooking* Blood mass is cooked, stirring continuously either directly over fire or in a double jacketed boiler by passing steam (100°C for 30 minutes) to sterilize it and to reduce the moisture content.
iv. *Pressing* Blood mass is collected in Hessian bags or any other porous bag, hung and squeezed to remove 40 to 45% of moisture. It is advantageous to have a screw press as drying time is greatly reduced.
v. *Drying:*
   a. *Sun-drying*—In hot and dry climate, pressed blood mass can be dried by spreading in shallow trays.
   b. *Cabinet drying*—In pilot plant, a cabinet drier with steam coils at the bottom and exhaust fan at the top can accommodate many trays at a time and remove the moisture efficiently.

कृपया रेषा मारुन
पुस्तक खराब करु नये

vi. *Cooling and milling* Dried blood is cooled to ambient temperature and ground to a consistency of floor.
vii. *Fumigation* Blood meal can be disinfected with methyl bromide, ethylene oxide etc. for long term storage.
viii. *Packaging* Blood meal is packed in polyethylene bags or air tight containers. It is advised to avoid undue exposure to high temperature and moisture during storage. Lime treated blood meal has a storage life of several months

Blood meal is a dark brown granular product. It contains about 80% protein and 10-12% moisture. It has a characteristic iron smell. The yield of blood meal is about $1/5^{th}$ of the initial weight of blood. It is an excellent source of essential amino acids, lysine being 6-9%

Foam Compounds

Protein foam compounds derived from blood are the cheapest foam type fire extinguishers. These are quite effective in dealing with fire involving petrol, paints, fat, naptha etc., although generally unsuitable for spirit or alcohol related fires.

Coagulated blood stock + 2% saturated NaOH
↓ Heated with constant stirring in steam jacketed vessel
Maintained at 90-95°C for 4.5 hrs
pH adjusted to 6.8 with HCl
↓
Solution heated gently to 60-70°C
Concentrated to a specific gravity of 1.2
↓
Add ferrous sulphate at 2.5% (w/v) as stabilizer
Add 0.2 to 0.5% Cresol (v/v) as Preservative
↓
Storage in 5 litre drums

Organic Fertilizer

Spilt and soiled blood is collected in drums and transported in closed vehicles in 4-6 hours for processing as fertilizers. This blood is preserved by treating with 2% formaline or 2% Lysol solution and dried in sun. Dried blood contains about 12% nitrogen with traces of phosphorus, iron and copper. It is

generally used as compound fertilizer after supplementing with super-phosphates.

Thus, animal blood can be subjected to multiple uses. The list is so varied that it not possible to cover the preparation and details of these products in this book.

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Utilization of Intestines

Intestines of domestic animals, as a slaughter house byproducts, have found various uses. When used as food containers, these are called *casings*. Non-edible uses of intestines include cat-gut, string for rackets and musical instruments etc.

Casings, besides intestines, are also obtained from weasands (oesophagus), urinary bladder etc. Although, artificial casings made up of plastic materials and hydro-cellulose have become popular in western countries, animal casings continue to be in demand because of their unique properties. A comparison between them is presented hereunder:

Sl. No.	*Property*	*Natural Casings*	*Artificial casings*	
			Collagen	*Cellulose*
1.	Strength	Variable	High and uniform	Good and uniform
2.	Length	Variable	Standardized	Standardized
3.	Cost	Low	High	Low
4.	Diameter	Variable	Uniform	Uniform
5.	Holes	Suspected	None	None
6.	Smoke penetration	Best	Very good	Good
7.	Presentation	Salted	Dried	Dried
8.	Printability	None	Limited	Best
9.	Refrigeration storage	Yes	Yes	No
10.	Shelf life	Limited	Fair	Very good
11.	Microbial quality	Suspected	Good	Good
12.	Tenderness	Most tender	Less tender	Peeled
13.	Breakage during processing	Most likely	Less likely	Least likely
14.	Machinability	Least	Less	Best

India has been exporting *salted sheep casings* to many developed countries like Japan, Germany, Spain, Italy, Switzerland, Denmark etc. after due inspection from Directorate of Marketing and Inspection under Animal Casings and Marking Rules, 1964. This is due to the fact that many countries are not able to meet

their demand for casings of small caliber. *Dried cattle casings* are exported mainly to Gulf countries. During 2000-01, India exported 574 tonnes of animal casings fetching Rs.12.30 crores.

Intestine as a Raw Material for Casings

Intestines of sheep, goat, pig or cattle/buffalo is four layered structure (Histological structure of intestine) (Fig. 16.1).

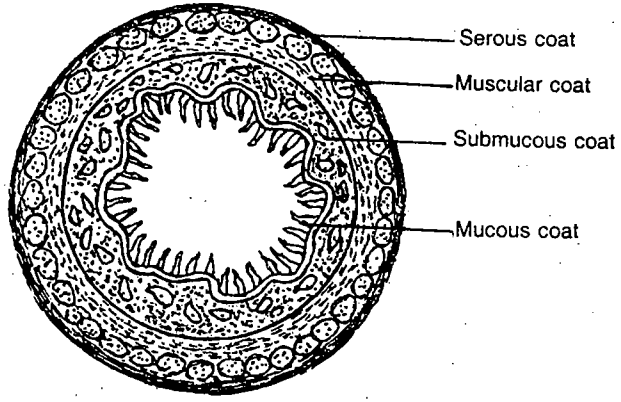


Fig. 16.1: Cross section of an intestine

a. Serous coat (*Tunica serosa*)—outer most
b. Muscular coat (*Tunica muscularis*)—longitudinal and circular
c. Submucous coat (*Tela submucosa*)—composed of dense connective tissue with elastic network. It has fine structure, sufficient flexibility, strength as well as delicacy. It is rich in collagen. Casings are made from this layer only.
d. Mucous coat—inner-most soft coat with lots of velvety projections.

Terminology Used in Casing Industry

Casings — Generally refer to sheep and goat intestine alone.
Rounds — Casings from small intestines of sheep, goat or pig.
— Called runners in case of cattle.
Middles — Casings from large intestine of cattle or sometimes pig.
Cattle casings — Casings from runner, middle, weasand (oesophagus), bung (caecum) or bladder of cattle and buffalo.
Sheep casings — Casings from small intestines of sheep.
Goat casings — Casings from small intestines of goat.
Maws or stomachs — Casings from cleaned pig stomach
Chitterlings — Casings from a part of large intestines of pigs.

Sheep and goat casings are generally stored wet and salted whereas cattle casings (Fig. 16.2) are generally dried in processing and sprinkled with insecticide to safeguard during storage period. Casings are measured in hanks :

a. For salted sheep and goat casings 92 meters in length = 1 hank
b. For dried cattle casings, 1 hank = 180 meters in case of runners
= 90 meters in case of middles.

Approximate length of casings from animals

Species	*Type*		*Length*
Cattle	Runner	90-135 ft	25– 40 meters
	Middle	20-25 ft	5.5 – 7.5 meters
	Bung	4-4.5 ft	1-1.5 meters
	Weasand	18-20 ft	5.5 meters
	Bladder	8-15 inches wide	20-35 cm wide
Sheep	Round	90 ft	27 meters
Goat	Round	75 ft	22 meters

Basic Requirements for Gut Processing

1. Casings should be manufactured from gut of healthy animals which have been properly slaughtered, inspected and passed.
2. Intestines for processing should not have ulcers and heavy parasitic infestations with nodules.
3. Sanitary handling conditions should be maintained from slaughter to final packing. The product should be clean with good colour and without objectionable odour. It should have proper length and be sufficiently cured.
4. The casings be graded as per Indian Standards IS:1981 (1962). Only clean, sound and wholesome pieces free from defects should be selected for packing.

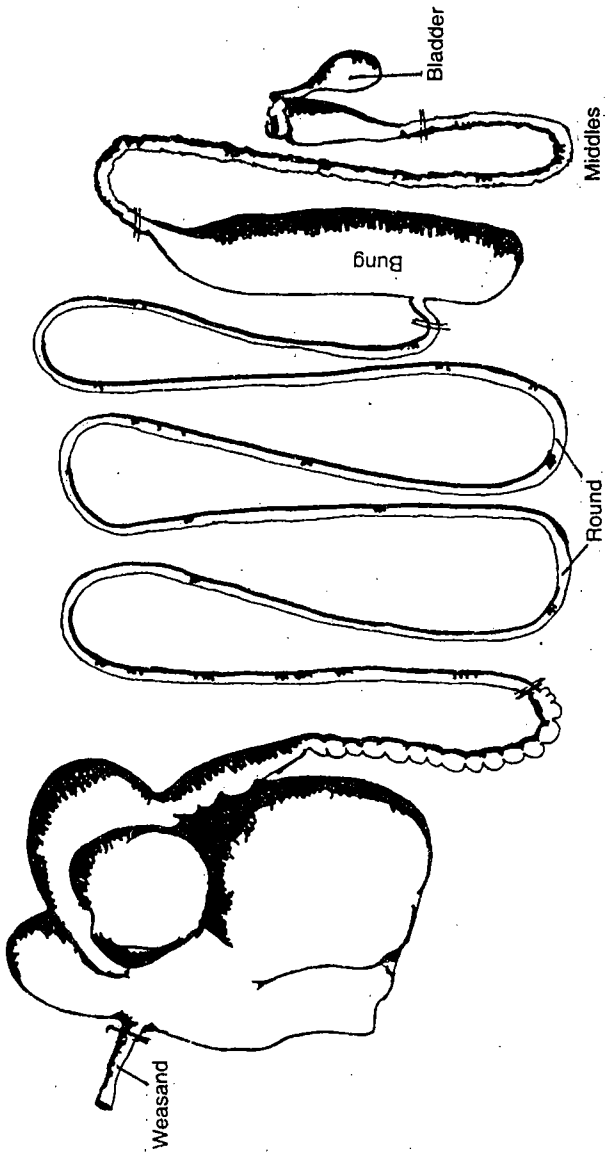


Fig. 16.2: Parts of intestinal tract used as dried cattle casings

Equipment

The equipment are different for small processing unit and processing plant attached to large abattoir.

Small Processing Unit

a. Scrappers made up of wooden or plastic knives
b. Gut cleaning table with a central drain pipe
c. Two stainless vessels one for fermentation and the other with chilled water to collect cleaned intestines
d. Table for grading and salting
e. Storage containers

Large Processing Plant

a. For salted sheep casings
   i. Casing stripper with rubber rollers
   ii. Casing crusher with rollers of bronze
   iii. Casing finisher with smooth and fine rollers
   iv. Casing flushing table with tap
   v. Inspection and grading table
b. For dried cattle casings
   i. Cattle casing stripper and fatters
   ii. Casing turning tank
   iii. Cattle casing crusher and slimmer
   iv. Cattle casing cleaner
   v. Equipment for inflation and drying

STEPS IN THE PREPARATION OF ANIMAL CASINGS

Casings should be processed as early as possible after slaughter and evisceration as follows:

a. *Removal* The intestinal tract should be removed carefully to avoid damage due to cuts. A ligature is made to check the escape of contents.
b. *Pulling* The intestines are detached from mesenteric attachment and fat by manual pulling. The same process is done with the help of knife in thick and strong cattle intestines and is called running.
c. *Chilling* The "green" intestines may be chilled at 10°C if the processing is not possible immediately.
d. *Stripping* The contents of intestines are then stripped off by drawing the gut between first and second fingers with uniform gentle pressure.

e. *Flushing* Now, tap water is forced through the gut in order to flush off the remaining intestinal contents.
All the operations upto this stage can be undertaken in the slaughter house itself.

f. *Fatting* All the fat adhering to the intestines must be removed. It can be done by scrapping in case of sheep and goat intestines whereas in buffalo, it may require tearing, pulling and even use of knife.

g. *Fermentation* Sheep, goat and pig intestines are allowed to ferment for 1-2 days in winter by immersing in water at 20°C. This facilitates easy removal of mucous membrane and slime due to bacterial action. However, in summer addition of a bit of salt is required to retard fermentation. This process is not required in cattle or buffalo intestines where judicious use of knife can accomplish cleaning on turned intestines.

h. *Turning* Buffalo and cattle intestines only are required to be turned inside out for efficient sliming. Water flowing by gravity from a tap can very well be used to turn the casing.

i. *Sliming* It is the removal of mucous lining of the casing and is done with the help of a sliming stick or plastic knife or sippi or a wooden knife. The knife is placed at an angle of about 30° with mild scrapping movements, the sliming is carried out. This process becomes easier if casings are dipped in a sliming solution consisting of 0.2% sodium pyrophosphate and 1% sodium chloride for 10-15 minutes. Properly cleaned casings are white and nearly transparent.

j. *Measuring* The width of water filled (sheep and goat) or inflated (cattle) casings is measured with the help of a calibrated casing gauge and length is measured in hank.

k. *Inspection and grading* The casings are examined for cleanliness, colour, pinholes, scores or cuts, blood spots etc. These are then graded in their respective caliberations.

l. *Preservation* Sheep and goat casings are preserved wet by salting with clean fresh salt of medium thickness. These casings are turned in salt several times and even gently rubbed by hand. Cattle casings are inflated with air and then dried in protected, fenced and fly-proof area. These are later deflated, pressed with a wooden roller and cut in required lengths.

m. *Packing* Wet salted sheep casings are stored and transported in metal tins which are completely lined from inside with food grade polyethylene. Dried cattle casings are coiled on wooden wheels and packed in barrels with recommended insecticide.

Defects Generally Encountered in Indian Casings

i. *Dull colour* (grayish or greenish instead of white or milky) and defective cleaning.
ii. *Nodules* due to oesophagostomum (a roundworm) in the intestines of sheep, goat and pigs.
iii. Holes and lacerations caused by negligence or rough handling of guts during processing.
iv. *Salt burns* due to long storage of guts in the salt or packed loosely leaving the air inside. Such patches may also develop due to defective salt which has calcium and magnesium ions.
v. *Defective grading*.

Besides, cicatrices (scars of healed up intestinal wounds), domestics (small grease spots in casing), kink (twisted loop in the casing), rust (black spot caused by putrefaction due to bacterial or fungal action) are also observed.

GRADING OF ANIMAL CASINGS

According to Bureau of Indian Standards IS:1981-1962, adopted by Agmark also, as *Animal Casings Grading and Marking Rules*, 1964, there are three grades for salted sheep and goat casings while there are four grades for dried cattle and buffalo casings.

Salted Sheep and Goat Casings

The calibration ranging from 12 to 26 mm in steps of 2 mm e.g. 12 to 14 or 14 to 16 mm is indicated.

Prime Quality (PQ) or Grade I

a. The casings shall
   i. be of natural colour throughout without any discoloration.
   ii. be free from defects like holes, blisters, laceration, nodules and cicatrices.
   iii. be intact and not torn or lacerated.
   iv. be free from salt burns, rust, domestics, black nodes, slime, mucous, dung or mould infestations.

v. not burst when filled with air or water to its normal capacity and slightly pressed.

b. The rings or hanks shall have been cured properly with common salt.

Grade II

As per Grade PQ except that a slight deviation shall be allowed in respect of colour and/or strength and wall. The material should be fit for use in the preparation of sausages.

Grade III

As per grade II, except that nodules will be permitted.

(For exports only, an additional grade X exists under Agmark for agreed requirements between the purchaser and exporter).

Dried Cattle and Buffalo Casings (Runners and Middles)

The calibrations in flat measurement ranging from 35 mm to 60 mm in steps of 2 or 5 mm is indicated.

PQ Grade

The casings shall

i. be of uniform natural colour, lustrous throughout without any spot or mark and be free of discolouration.
ii. be intact, free from any tear or laceration.
iii. be perfectly rolled.
iv. be free from black nodes, lacerations or cicatrices.

Grade I

As per PQ except that slight variation in colour and folds and a few black nodes may be permitted.

Grade II

Casings not conforming to PQ or grade I due to defects in rolling or having large black nodes, rough texture or a few streaks of fat.

Grade III

Short pieces of any or all the above grades.

Thus, animal casings are an important byproduct of meat industry which have long been used as containers for the production of sausages. Besides intestine, other organs such as urinary bladders and weasands (mucosal layer of oesophagus) are also used in the production of sausages. Only a negligible proportion of production is exported. Hence, there is ample scope for augmenting the exports.

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Utilization of Bones

Bones constitute almost 15% of the weight a dressed carcass, although yield may vary according to the breed, age, health status etc. On live weight basis, bones form 12-30% in cattle, buffalo and pig and 20-30% in sheep and goat.

Green or fresh bones are heavy and contain comparatively high moisture, fat and protein content. Their average composition is as follows:

Moisture	: 50%
Red and yellow marrow	: 15%
Organic matter	: 12%
Inorganic matter	: 23%

It is obvious that organic and inorganic matter are in a ratio of 1:2, respectively. Bone collagen called *ossein* is the main constituent of the organic matter. Inorganic matter consists of nearly 33% of calcium and 15% phosphorus, small quantities of sodium, potassium and magnesium and traces of copper, zinc, iron, cobalt, manganese etc. Almost 96% of red and yellow marrow is fat.

Desert bones belong to fallen animals. These decomposed bones remain exposed to bacteria, insects and other atmospheric actions for a long time and are devoid of meat, fat and tendons which are found on green bones These are light in weight and contain only dried ossein, calcium, phosphorus and lesser minerals. Bones can be processed as a source of following:

1. Gelatine and glue
2. Bone meal
3. Fertilizer

GELATINE AND GLUE

Manufacture of gelatine is commercially most remunerative utilization of bones. *Gelatine* is a derived protein of albuminoid class which has both—edible and inedible (technical) uses. Edible gelatine is manufactured from fresh bones obtained from slaughtered and inspected animals under strict hygienic conditions. Chemically, there is no difference between gelatine and glue. However, *glue* is a low grade gelatine with comparatively dark colour and has only inedible uses.

Pure gelatine is an amorphous and transparent substance devoid of any colour, taste and smell. It is brittle when dry, softens on heating and then decomposes with burnt hair smell. It swells in cold water absorbing 5 to 10 times its weight and dissolves on warming upto 30°C.

Edible uses of gelatine are in ice cream, jellies and soft chocolates as foaming agent, making of capsules, as binder in tablets and as plasma extender in blood transfusion. Commercially, it is used as sizing agent in textile and leather industry, photography etc. Glue is an important adhesive in plywood, furniture, sand paper, gummed tape etc.

Preparation of Crushed Bones for Gelatine

Crushing of bones is done to save the freight before packing to the manufacturing units or export:

i. *Selection of bones* This process involves selection of mainly six long bones-femur, tibia, metatarpus, humerus, radius and ulna, metacarpus etc.

ii. *Cutting* Knee cap and knuckles are removed with the help of a motor saw. It exposes the marrow to the direct action of hot water.

iii. *Heating or cooking* The bones are put in water which is heated gradually. Heating is continued at 85°C for about 6 hrs. Fat and other adhering material rises to the top and is skimmed off.

iv. *Washing* After removal of fat, the bone are washed with warm water.

v. *Drying* These bones are dried in sun on wire netting. In wet weather, drying is done in hot room.

vi. *Crushing* The bones can be crushed by a simple stone crusher to a size ranging from 1-2 cm cubes. Crushed bone pieces which pass through 0.2 cm mesh are considered bone meal.

Manufacture of Gelatine and Glue

Principle

Gelatine is produced by the action of hot or boiling water on collagen or ossein by the process of hydrolytes:

$$\text{Collegen} + H_2O \xrightarrow{\text{heat}} \text{Gelatine}$$

Procedure:

i. *Washing* Defatted and uniformly crushed bones to a size of 1-2 cm cubes are washed with water. Similarly, glue stock is washed and soaked with water. It is decreased by prolonged exposure to sun or limed (saturated $Ca(OH)_2$ solution, 10% by weight) for several weeks to remove non-collagenous material and fat.

ii. *Demineralization* Thoroughly washed defatted crushed bones or degreased glue stock, is demineralized by soaking in 4-10% HCl for 1-2 days. It is washed with water to yield a clean, soft stock or ossein.

iii. *Extraction or cooking* It is done by controlled hydrolysis to recover different grades of gelatine. Extraction takes places in several runs of 3-5 hours at successively higher temperatures.
60°C yields highest quality gelatine.
65-70°C yields medium quality gelatine.
80°C yields low quality gelatine.
100°C (boiling) yields glue.

The residue is pressed and dried for use as livestock feed or fertilizer.

iv. *Filteration* The liquid extracts of higher grades are pressure filtered to increase clarity.

v. *Concentration* The liquid extracts are vacuum evaporated to yield a gelatine of 30-40% concentration.

vi. *Drying* Concentrated gel of gelatine is spread on a thin sheet of a large drum which is heated by steam. Dried gelatine is removed in few minutes with the help of a knife.

The gelatine may be sold in sheets, broken to flakes or powdered as per the requirement of the user industry.

Collection of long bones
↓
Washing and cutting
↓
Degreasing (heat treatment or organic solvents)
↓
Crushing (1-2 cm cubes)
↓
Demineralization (4-10% HCl for 1-2 days)
↓
Thorough washing
↓
Extraction or cooking by controlled hydrolysis
(at 60°C, 65-70°C, 80°C, Boiling)
↓
Filtration (under pressure)
↓
Concentration (evaporation)
↓
Drying (at 8-12% moisture)
↓
Packaging

Hide and skin trimmings, ear pieces, poultry skin, sinews, Tendon, horn pith and casings can be used to prepare glue. *These are known as glue stock.* Zinc sulphate is used as a preservative to extend its keeping quality.

BONE MEAL

Bone pieces of less than 2 mm size constitute bone meal. Sterilized bone meal a good source of phosphate supplement in livestock feed. Animals deprived of adequate phosphorus in feed and fodder suffer from osteophagia, osteoporosis, rickets etc.

Bones that are obtained directly from slaughter house are called green bones. These have high contents of moisture, fat and protein. Desert bones become deprived of any left over meat or bone marrow fat. They are light in weight and contain only calcium, phosphorus and lesser minerals with dried ossein.

Collection of desert bones and their conversion to bone meal is an economically viable proposition. It is also important from sanitary point of view. It can provide employment to very poor

and illiterate people with meager means at their disposal. It will invariably bring about improvement in the livestock. It should be noted that sterilization of such bones is a must.

Flow Diagram for the Preparation of Bone Meal

Collection and cleaning of bones
↓
Sterilization
(Double jacketed bone digester, steam at 60 lb psi for 2 hrs)
↓
Draining
↓
Drying under sun
↓
Milling
↓
Packaging

The yield of bone meal is one third to that of raw bones (1:3). Quality of bone meal is determined by the presence of phosphorus and calcium that should ideally be present in the ratio of 1:2. The average composition of bone meal is:

Calcium	: 30.5%
Phosphorus	: 15.5%
Protein	: 7.0%
Fat	: 1.0%

Cattle lick—generally prepared from bone meal consists of:
66 parts Bone meal
33 parts Red oxide salt (contains iron)
1 part Copper sulphate, potassium or sodium iodide,
Cobalt nitrite, sulphate or chloride or other trace elements

BONE ASH OR CALCINED BONES

In places where no other option is available, collected bones can be burnt on grills or by trench firing to yield spongy and brittle bones and pieces which can be ground to ash. Being a good source of phosphorus, it serves as a fertilizer. In areas where soil is deficient in phosphorus, it enriches the soil resulting in better crop production.

UTILIZATION OF HOOFS (SHIN BONES)

Shin bones separated from hoofs can be utilized for the preparation of *'Neats foot oil'*. Pure neats foot oil is a pale yellow liquid which does not solidify or dry even at freezing temperatures. So, it is used as a lubricant for delicate machinery in aeroplanes, ships, costly watches etc. which are likely to be exposed to low temperatures. It fetches a very high price but yield per animal is only a few milliliters. The production of neats foot oil is economic only in places which are adjacent to the large abattoirs. It involves the following steps:

i. *Collection and cleaning of the hoofs* The hoofs are cut off from the feet, fresh from the killing floor and washed to free from blood, dirt etc.
ii. *Scalding* The hoofs are immersed in boiling water for sometime and shell is removed with the help of a hammer. Thus, shin bones are fully exposed.
iii. *Cooking or extraction* Shin bones are cooked in water at 85°C for about 8 hours in open tank or large kettle. The oil will come floating on the top.
iv. *Purification* Floating neats foot oil taken in another kettle and again heated at 85°C for 8 hours. The impurities are allowed to settle for 2 hours and the oil is filtered out.
v. *Dehydration* The moisture is removed from neats foot oil by heating at 100°C for nearly 2 hours.
vi. *Packaging* It is done in suitable containers.

UTILIZATION OF HORNS

Animal horns differ in shape, size, colour etc. according to species, breed, sex and several other factors. Horn consists of two parts—horn pith and horn proper. Horn pith forms the core or the inner part. As described earlier, it forms a very good raw material for the preparation of gelatine or alternatively, it can be utilized for the production of bone meal.

Scalding horns in water at 60°C for sometime and then hitting with a hammer separates the two parts of the horn. Outer part or horn proper is mostly keratin. Several articles like buttons, combs, handles, decorative pieces etc. can be prepared from horn proper, if it is subjected to regulated heating, pressing and cutting.

Horn and Hoof Meal—A Good Fertilizer

Rendering unit for the production of bone meal can also be utilized atleast for few days in a month to prepare horn and

hoof meal which is a very good nitrogenous fertilizer. Horns and hoofs of cattle, sheep, goats and pigs are digested in the renderer for 8 hours, dried and milled to a fine powder. The meal is not used as livestock feed supplement due to its unpalatability and very poor digestibility. However, horn and hoof meal enriches the fertility of the soil because of its high nitrogen content.

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Glandular Byproducts

Glands are an important byproduct of animals primarily slaughtered for meat. Our country is still importing life saving drugs worth several hundred crores, which could be derived from glands. It is very important to make proper and full utilization of this costly raw material and reduce our dependence on the imported antidiabetics, endocrines, hormones and intermediates, adrenaline and other sympathomimetics. Although active principles of several glands are now being synthesized, the potential of abundantly available animal gland can not be ignored.

Guidelines for Collecting Glands

1. Glands should be collected from healthy animals.
2. Glands should be excised from slaughtered animals within 15-20 minutes.
3. Glands should be immediately chilled in metal containers without allowing any direct contact with ice.
4. Fat and connective tissue surrounding the prechilled gland is trimmed.
5. At every stage, contact of glands with water must be avoided because active principle may leach out in water.
6. Clean glands are quick frozen at –18°C to –20°C in covered containers.
7. Packing of the glands should be done individually without leaving any air spaces.
8. Alternatively, acetone dried powder of the gland may be prepared. This is the best way of preserving the glands for a long time while retaining its biological activity

Table 18.1: Yield and commercial value of glands

Gland	*Number of animals required to obtain 1 kg of glands*		
	Cattle/buffalo	*Pig*	*Sheep*
Pancreas	5	15	40
Ovaries	100	125	1400
Thyroid	80	110	250
Parathyroid	2200	–	3500
Pituitary	600	4000	350
Adrenal	60	250	350
Bile (liquid)	6	–	55
Bile (conc. 75%)	55	–	500
Testes	4	–	–

Calculated from Mann (1962)

The glands like pancreas, pituitary, thyroid, adrenal etc. can be preserved for a long time by preparing its dried powder in acetone. In this process, the enzymes are inactivated and thus shelf life is increased. Fatty material which generally interferes with the extraction of most proteins is also eliminated. We can directly extract desired protein from the acetone dried powder with usual solvents. The requirement of solvent is also minimized.

Preparation of Acetone Dried Powder of Gland

i. Collect the gland within 15-20 min. of slaughter.
ii. Remove connective tissue, blood vessel, adhering fat etc. Immerse in 4 volumes of chilled acetone for 3 hrs.
iii. Cut into small pieces, immerse in 3 volumes of chilled acetone for another 2-3 hrs.
iv. Mince the tissue in a mincer and again treat with 3 volumes of chilled acetone.
v. Dry and grind into a fine powder.

IMPORTANT GLANDS AND THEIR BIOCHEMICALS

Pancreas

It is mixed type of gland. It remains attached to liver embedded in fat. *Insulin* is main hormone, an antidiabetic which is extracted from the b-cells of this gland with the help of acidified methanol. One kg of fresh bovine or pig pancreas yields about 150 mg of crystalline insulin with an activity of 25 IU/mg of insulin. Other biochemicals extracted from pancreas are:

Pancreatin (extract of pancreas)
Trypsin
Chymotrypsin
Amylase
Glucagon (a-cells)

Pancreas also contains several enzymes which are used in Industry by the tanners or cleaners. For this purpose, glands can be preserved for a week or so with some application of salt. These are packed in drums with a cover of salt.

Adrenal Gland

This gland has got two parts – outer cortex and inner medulla. Cortex of adrenal gland yields *corticosteroids* which are used in the treatment of Edison's disease, to overcome shock in surgery and as a non-specific treatment. Medulla of adrenal gland yields *adrenaline* (epinephrine) and *nor-adrenaline* hormones. Yield of adrenaline is 0.2% on fresh weight basis and 1% on dry weight basis. It is extracted in water or alcohol.

Thyroid Gland

These are two maroon coloured bodies situated on either side of trachea. Acetone dried powder of this gland is used to extract *thyroxine* hormone with the help of Barium hydroxide. Yield of the hormone is 0.08%.

Parathyroid Gland

These glands are located near thyroid gland. *Parathormone* is extracted from them. It is used in the prevention of tetany and increase to the rate of calcium excretion.

Pituitary Gland

It is very small gland and has to be carefully collected.
Anterior lobe:
i. Growth Hormone (GH)
ii. FSH
iii. LH
iv. ACTH
v. MSH
Posterior lobe:
i. *Oxytocin* can be extracted in 2% acetic acid. It initiates the uterine contraction. It also has milk ejecting factor.

ii. *Vasopressin* is an important vasoconstrictor. It is also an antidiuretic hormone.

Stomach Glands

i. *Rennet or Rennin*—enzyme is prepared from the 4th stomach (abomasum) of milkfed and unweaned buffalo or cow calf. The abomasum is cut out, packed and frozen without any washing.

ii. *Pepsin*—enzyme is extracted from the mucosal lining of the hog stomach. This part is distinguished by pink-red colour, wrinkles and fold. The glands are situated between the folds. The hog stomach is cut and the glandular linings are pulled away from the stomach wall and frozen as quickly as possible. These can be preserved in 1% H_2SO_4 or the acetone dried powder of the glands can be stored at room temperature but very low temperature is required during its preparation. Peptone can be prepared from the remainder part of the hog stomach.

Gall Bladder

It is attached to liver and contains bile, a dark golden-greenish viscous fluid of bitter taste. Bile is slightly alkaline (pH 7.15) and has a specific gravity of 1.025. It can be used as detergent in slaughter houses by dissolving in warm water (1:5 v/v).

Dried bile has medicinal use as a substitute in secretion deficiencies. One kg of bile can be obtained from the gall bladder of 6 buffaloes or 55 sheep/goats, stored frozen and should be treated as follows:

Bile + Acetic acid → Mucins are precipitated
↓ Filter
Filterate
↓
Concentrate to syrup
(20% moisture)
↓
Dry to powder form
(maximum 6% moisture)

Ovaries

These are collected only from mature animals i.e. those having corpus lutea. Ovaries are irregular, lobulated in appearance with large follicles projecting like small grapes. All ovaries with cysts are discarded.

Sex hormones—*estrogen* and *progesterone* are extracted from ovaries.

Testes

These are collected from mature animals that have been passed in antemortem examination and postmortem inspection.

i. Sex hormone—*testesterone* is extracted from testes.

ii. *Hyaluronidase* enzyme is made from bull or ram testes. It is a very good spreading factor. It is used in several drugs to intensify their effect.

PREPARATION OF INSULIN FROM PANCREAS

Grind hard frozen glands in acid alcohol (acidified methanol) to extract insulin
↓
Concentrate at temperature below 40°C
↓
Separate by salting out, crude insulin goes to the top as a cake
↓
Dissolve crude insulin in alcohol, then add diethyl ether, isoelectric pptn pH 4.5
↓
Crude insulin is purified by crystallization using zinc salt or picric acid
↓
[Insulin picrate can be stored for a long time.
Insulin can be recovered by gel filteration]

PREPARATION OF HEPARIN FROM CATTLE LUNGS

Minced lung tissue
↓
Autolyse at 30°C for 24 hrs. Toluene is added to avoid putrefaction.
↓
Add ammonium sulphate + NaOH mixture.
$(NH_4)_2SO_4$
Heat to 50°C, maintain for 1 hr for extraction
↓

Precipitate with alcohol for 20 hrs
↓
Subject to enzymatic hydrolysis with trypsin centrifuge to gel clear solution
↓
Precipitate crude heparin with 2 volumes of alcohol
↓
Purify by gel filteration
[For biology activity-conduct sheep plasma assay]

PREPARATION OF LIVER EXTRACT

Raw minced liver
↓
Suspend in water at 85°C
↓
Digest at neutral pH with pepsin for 3 hrs
↓
Inactivate enzyme at 90°C for 30 min.
Coagulate proteins
↓
Digest in filter pressed
Clear filterate is concentrated to syrupy consistency
↓
Drying is vacuum to powder
[The yield of the extract is 15% of the raw weight. It is indicated in anaemia]

Increased modernization of slaughter houses is improving the scope of utilization of glandular byproducts. Although many synthetic preparations have been evolved, products from natural resources are still cheaper source. Efforts are being made to create adequate facilities for the timely collection, immediate freezing and create trained manpower for preliminary processing of these glands in our country. Various research laboratories are equipping themselves to develop new processes and improve the existing technologies.

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Optimal Harvesting of Hides and Skins

Hide refers to the skin obtained from fully grown large animal which is comparatively large, thick and heavy. In cattle and buffaloes, it should be more than 30 lbs or 13.62 kg. The integral body covering of sheep, goat, pig and even that of a very young calf is called *skin*. It is comparatively small, thin and light. Hides and skins are one of the most important byproducts of the animals. The yield of this byproduct varies in different species. In cattle, the average hide yield is 7 percent of live weight, whereas in sheep and goat the average yield of skin is 11 percent of live weight. As per FAO estimates (2001), India produced the following quantities of fresh hides and skins:

Hides and Skins	*Weight (MT)*
Cattle hides	466000
Buffalo hides	510000
Goat skins	128880
Sheep skins	52380
Sheep skin with wool	524

Tannery is the ultimate destination of hides and skins where these are converted into leather. Inspite of the advent of various similar artificial or synthetic material, leather is superior because of its:

i. excellent flexibility due to high tensile strength over a wide range of temperature and moisture range,
ii. high tear and puncture resistance due to in-built fibrous network,
iii. ability to breath which provides cooling in hot weather and insulation in cold weather and
iv. moulding ability.

In the past, India had been one of the largest exporters of hides and skins. However, a few decades back, the policy of exporting only value added material was adopted and presently,

we are exporting finished leather to many countries. Interestingly, there is a marked difference in sources of hide and skins in our country.

Source of Hides

20-25% from slaughtered animals
75-80% from fallen animals

Source of Skins

80% from slaughtered animals
20% from fallen animals

A photomicrograph of hide (Fig.19.1) will show that it is composed of three different layers:

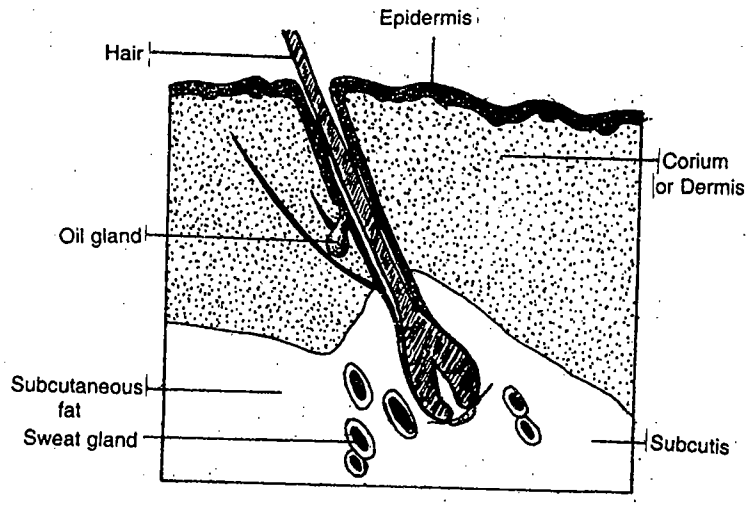


Fig 19.1: Cross section of a hide (diagrammatic)

i. *Epidermis* which has a outer pigmented surface and tubular invaginations for hair follicles.
ii. *Corium or dermis* which has elastin, reticulin and collagenous fibres. The inner portion of corium is interwoven with bundles of collagen. It is this part which makes up the leather.

iii. *Subcutis* which is a loose membrane network and contains fatty deposits.

CLASSIFICATION AND NOMENCLATURE OF HIDES AND SKINS

There are very many ways of classifying hides and skins:

According to weight

Origin	*Weight of green hide in lbs*	*Designation*
Unborn calf	–	Slunk skin
Immature calf	9-15	Calf skin
Calf	15-25	Kip skin
Heifer	25-30	Heifer skin
Cow	>30	Cow hide
	<53	Light cow hide
	>53	Heavy cow hide
Steer	32 to 48	Extremely light hide
	48-58	Light steer hide
	>58	Heavy steer hide
Bull	60-100	Bull hide

According to branding

Hides branded on butt or side	Texas or Colarado hides
Unbranded hides	Native hides

According to time of flaying

Flaying from slaughtered animals	Slaughtered hides
Flaying from fallen animals	Fallen hide or Renderer or Murrain hides

According to quality of flaying

Flaying by highly skilled workers	Big packer hides
Flaying by less skilled workers	Small packer hides

In India, buffalo hides are generally classified as
Buffalo (buff) calf skin
Buff heifer hide
Buff hide
Buff bull hide

Goat and sheep skin classification is done on the basis of weight and length in our country (Baba and Bhote, 1966):

Class	*Weight (kg)*	*Length (cm)*	*Region*
Big goat skin	1.3	96.5 and above	Punjab, Kashmir-Punjab border, Orissa, UP
Medium goat skin	0.91	96.4	Rajasthan, Maharasthra, Andhra Pradesh, Kerala, Karnataka
Small goat skin	0.86	76.2	Eastern part of India

It may be noted that sheep skins are mostly of medium size without much regional variation.

FAO Expert, Aten (1955) suggested a useful system of grading hides on the basis of degree of faults:

1. First grade — Shape or pattern regular and symmetrical, Minor scores and gouges may be over looked.
2. Second grade — Good shape or pattern. Reasonably free from knife damage i.e. upto 1/16 of area may show concentrated scores or gouges or 1/8 area may show dispersed scores or gouges. One or two cuts may be allowed on the edges of belly.
3. Third grade — Irregular shape or pattern. Upto ½ of the area damaged by knife showing cuts, scores or gouges.
4. Reject grade — Irregular shape or pattern. Extensive damage on the back or butt.

FLAYING

Flaying refers to the skinning operation performed by skilled workers following a uniform pattern. Being a valuable byproduct, improper flaying will lower its value. In modern large animal abattoirs, skinning of buffalo and cattle is done on pritch plates. These are steel plates 20 cm diameter having 120 cm length and 60 cm width anchored to the floor with bolts. The animal is lowered onto this plate for skinning and is held in position by a pritch bar—a metal rod. On a smooth floor, skilled workers can very well accomplish this task. If this facility is not available, it can be efficiently done on a skinning cradle made up of steel tubings or smooth round timber rods.

Flaying of Buffaloes and Cattle

1. With the help of a flaying knife, hide is first opened from the neck or slaughter incision and continued straight alongwith the middle of dewlap and belly to the middle of the tail.
2. Each leg is opened by making an encircling cut between knee and hoof joint. The cut on the foreleg is continued to the breast bone and that of the hindleg upto scrotum or udder to meet the longitudinal cut as mentioned at point 1.
3. The hide is than separated from the carcass by cutting from the lower part of the breast towards the neck on one side and towards the navel on the other side. This is done by careful use of knife.
4. Now with the help of tail grip and gambrel, which are connected to the hook of the hoisting gear, the hind legs are hung and hide is removed from the tail. Carcass is further raised and hide is pulled off the back to the hump and then to shoulder and neck separating the thick subcutaneous tissue with help of a knife. The hide is severed from the carcass by giving a cut behind the horns.
5. A properly flayed hide has
   i. rounded rumps
   ii. equal width from center line of the back to the belly edge on each side
   iii. medium length in the shanks
   iv. regular dewlap
   v. square outline

Flaying of Fallen Buffaloes and Cattle

Fallen hides are flayed from animals, which have died naturally due to old age, disease, drought etc. Fallen hides are often of inferior quality as coagulated blood remains in the capillaries of hide causing discolouration or staining and at times some putrefaction. Flaying also becomes difficult due to harder connective tissue which may result in knife cuts, scores and gouges. However, early and careful flaying can minimize the damage.

The dead animal is put on the wooden flaying cradle (Fig.19.2). Such a cradle can be easily made by keeping two smooth wooden logs of 150 cm length and 15 cm diameter at a distance of 30 cm and joining them with cross pieces of the same diameter with nails underneath.

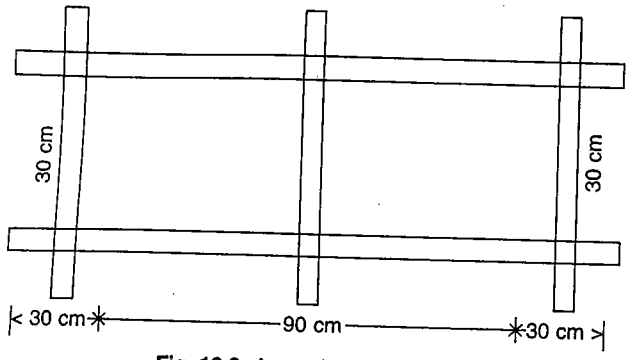


Fig. 19.2: A wooden flaying cradle

Case-on Flaying of Sheep and Goat Skin

This technique is quite popular in tropical and subtropical countries. It has the advantage of giving skins with minimum cuts and excellent keeping quality.

Procedure

1. After bleeding the carcass, a small incision is made on the inner side of the hind leg, about hock joint.
2. A narrow steel rod or a smooth wooden rod is now inserted in this incision and pushed under the skin towards the grain for about 45 cm. (This rod tears or loosens some of the connective tissue between the skin and body, thus minimizing the use of knife which could cause cuts, scores or gouge marks).
3. After removing the rod, a hand cut is made around the incision hole and air is blown inside with the help of bicycle pump or car foot pump.
4. Due to inflow of air, carcass is uniformly inflated like a balloon.
5. Now, the carcass is lifted from the ground and hung up by the hind leg on the hook or tripod.
6. The skin is then removed from the carcass in a cased form.
7. The following cuts are made in this technique :
   i. Circular cuts around knee and hock joint.
   ii. Cuts on scrotum, udder etc.
   iii. A circular cut on the neck to severe the head skin.

Now the fingers followed by fist are inserted into the incisions on the hind legs. The skin is pulled downward and knife is used only when it is absolutely necessary. When it reaches chest, the skilled worker holds the skin with both the hands and rigorously pushes with his foot to remove it from the carcass. The only disadvantage in this technique is that asymmetrical skins are obtained many a time. This defect can be prevented by careful flaying a small part of belly by knife (Fig. 19.3).

Defects that Affect the Quality of Hides and Skins

Antemortem Defects

1. *Poor substance*	due to genetic factor or species variation
2. *Defects due to diseases*	especially due to pox, mange, warble fly, dermatitis etc
3. *Grain damage*	due to thorn or wire fence scratches or branding, abscesses, abrasions, tick damage, bacterial action

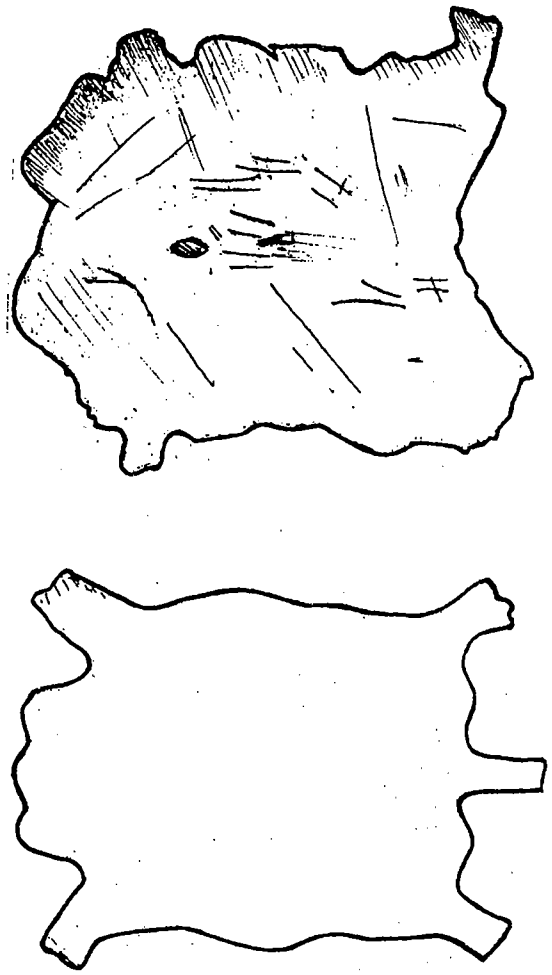
Postmortem Defects

1. *Bad shape or pattern*	due to ground drying, fallen hides etc
2. *Bad flaying*	due to knife cut
3. *Curing defects*	due to salt burn, improper application of salt
4. *Smoke damage*	

Preservation of Hides and Skins

Hides and skins are quite prone to spoilage due to bacterial or enzymatic decomposition due to the presence of neatly 60% water and contamination with blood, manure etc. Hence, steps should be initiated to preserve them immediately after flaying. The first step in this direction is thorough washing to remove blood, dirt etc. which hasten putrefaction. This is done with the help of scrubbing brushes on a special platform under running cold water and putting them on the draining racks. The hides are subjected to trimming to remove those parts which have no value as a leather and to conform to standard pattern.

An important antemortem care is the provision of enough water and withdrawal of feed for atleast 18 hours. It facilitates flaying and avoids useless cuts on the skin. Hoisting the animal



Improperly flayed hide

Properly flayed hide

Fig. 19.3: Properly flayed hide

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also improves bleeding and keeps the skin clean. Skins [illegible] and long-haired goats are washed on the flesh side only.

The term *green hides and skins* refer to those, which have only been flayed, fleshed, trimmed and washed. Such hides should be delivered to the tanners as early as possible, otherwise steps be taken for their short term preservation and proper storage in the premises. The basic principle in their preservation is to reduce moisture either by providing free air circulation among them or by application of salt or ideally a combination of both. There are various methods for the preservation of hides and skins. A suitable method will depend on the prevailing climatic conditions such as temperature, relative humidity, period of storage etc. There are two main methods of preservation :

1. Air drying
2. Hide curing

Air Drying

It is an age old method for the areas where relative humidity is low It is done in three different ways.

i. *Ground drying* It consists of stretching out the hides with their flesh side up. Though it is cheapest and easily adoptable for rural people with fallen hides, there is problem of blemishes, which become apparent on tanning. There may be problem of hair-slip and blisters due to incipient putrefaction of epidermis and hair follicles. However, in summer months, the hides become very wrinkled and fetch only half the value of salt cured hides and skins.

ii. *Suspension drying*: It is simple, cheap and effective way of drying in tropical countries. It allows free circulation of air, sun-rays strike the hides and skins obliquely and cooling of hides take place rapidly. The hides and skins become light after drying, so transportation is cheap. Suspension drying can be done in several ways:

iii. Frame drying is done on an angled frame which is appropriately oriented to the sun. The frame may be in the form of hoop, tripod or bamboo square. Line drying is more suitable for sheep and goat skins. Here skins are spread on horizontal cords with their flesh side up. In tent drying, the hides are supported over the ground in the shape of a tent by cords or wires. In any case, hides and skins take a minimum of seven days to dry.

In this case, drying is performed with the help of uniform and finally ground salt. It should be done immediately after flaying by either of the following two ways:

i. *Dry salting* is widely used in tropical countries. Salt packing helps in initial removal of moisture, the remaining being removed by exposure to air. Ideally, the store should have a temperature of about 15°C, a relative humidity of 85-90% with good ventilation and slatted platform. It involves stacking the hides flesh side up and applying fine salt (2-3 mm) evenly. The quantity of salt applied matches with the weight of the each hide. Stacking should not be more than one meter in height. The moisture drains onto the floor. Such hides retain only 12% water.

ii. *Wet salting* is done by preparing a curing solution consisting of 23 kg pure salt and 62 kg of water for 100 kg of hides. A saturated brine solution can also serve the same purpose. The hides are soaked in this solution in a pit of 1.25 meters for a few days. The time varies from 48 hrs for fleshed hides and 2 weeks for unfleshed hides. The hides are put on the slatted platform for draining. Such hides are left with only 35% water.

Conditioning of Hides and Skins

The preserved hides and skins are first conditioned at the tanneries before tanning under the following steps:

i. *Washing and soaking* It is done for several hours in water containing zinc chloride and a mix of soda ash and borax. During this process, salt is removed alongwith proteins of blood and lymph. The hides absorb water and restore their original shape and dimension.

ii. *Fleshing* It is done on a convex wooden beam by scrapping the flesh with a serrated knife.

iii. *Liming and dehairing* A saturated solution of lime and 0.1% sodium sulphide remove and loosen hair and epidermal cells.

iv. *Washing and deliming* The hides are now washed with weak acid to neutralize the lime. A mild heat can also be used if necessary.

v. *Bating* The hides are treated with proteolytic enzymes (pancreatic juice mixed with saw dust) at pH 8.5. It results in soft and pliable pelt which can be tanned by vegetable tanning.

vi. *Pickling* It is done for chrome tanning wherein pelts are pickled in a bath of 1% sulphuric acid and 10% salt in water at a pH of 2 to 2.5 for 2 to 3 hours.

TANNING

Tanning is the conversion of hides and skins into insoluble and nonputrescible leather without destruction of the original structure. Leather posses many desirable properties such as flexibility, heat resistance, chemical resistance, abrasion resistance, dimensional stability etc. It can also withstand repeated wetting and drying.

Vegetable tanning is the traditional process and takes a lot of time. It involves immersion of hides and skins in the infusion made from the extracts of tannin bearing plants and barks. Some tanning producing plants are:

Avarum (*Cassia auriculata*)
Babul (*Acacia Arabia*)
Myrabalan (*Terminalia chebula*)
Konnan (*Cassia fistula*)

Delimed pelts Immersion →Tan liquor in vats

↓ Progressively stronger in concentration

↓

Tannin difuses and imparts uniform colour

With the use of strong tan liquid and revolving drums, the speed of tanning can be improved. This type of tanning is good for the manufacture of sole, belts, harness, saddlery and other heavy leathers.

Chrome tanning is the modern and quite popular technique of these days because it yields a soft, supple and strong leather which is permeable to air. It also takes less time. It can be done by any of the two processes: In single bath process, basic chromium salt (chromic sulphate or chrome) is directly applied in solution to the skin in gradually increasing strength whereas in *double bath process*, chromium salt is formed on the fibres by interaction of chemicals. Usually sodium bichromate is reacted with a reducing sugar maltose and sulphuric acid to get chromic sulphate. It is applied in 1.5 to 3% range initially at a pH 2.8 which is then increased to pH 3.5 to increase the affinity of the

collagen for the chrome. Chrome tanning takes place due to the formation of cross linkages between chrome ions and free carboxyl groups in the collagen side chains. Tanning operation is accomplished in 5-6 hours.

Post tanning operations are also necessary to get finished leather. *Setting out or wringing* is done to remove excess tan liquor or moisture by passing the hide between two large rollers. *Splitting and shaving* is done to adjust the leather thickness for the desired ultimate use. *Dyeing* of the leather is done to produce the desired colour. *Fat liquoring* is done to adjust the firmness or softness of the leather by lubricating the fibres with oil. It also increases the tensile strength. *Staking* refers to the softening and making the leather more pliable. It decides the final temper of the product in combination with fat liquoring. *Buffing* is done to smooth the grain surface of leather for better appearance and to diminish the blemish. A buffed leather is called corrected grain which is otherwise called full grain. *Glazing* is done on chromic leather after seasoning and drying. A glass cylinder clamped to the end of a moving arm is made to roll on the leather. Heat generated by friction softens the wax in the finish and produces a continuous plastic coat of high lusture on the grain surface.

As the leather gets ready, it is graded for temper, uniformity of thickness, colour etc. The graded leather is properly packed for convenience and to fetch a better price for this noble byproduct.

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Wool

Wool is one of the important byproducts obtained from sheep. In comparison to hair, it is more elastic, flexible and curly. A wool fibre has two distinct layers of cells. The outer protective sheath has flat, irregularly shaped scales similar to that of fish which overlap each other. These scales are loosely attached to the inner layer in comparison to hair. The outer layer of scales is very important. It influences the quality of wool because of its shrinking, strengthening and felting conditions. In the living condition, the outer wool scales are with wool sweat (grease) or yolk called *suint* secreted from special glands to keep the fibre in good condition. In fact, *condition* denotes the degree of grease or oil present in the wool. A part of suint is water soluble and can be removed during washing. The inner layer of wool fibre called cortex consists of long fibrils cemented together. In hair, it encloses the air-filled medulla. Thus, hair has less strength than wool.

In the Indian subcontinent each geographical region has a distinct type of sheep. In the north-western India, the Bikaneri type breeds are small and coarse wooled. Sheep belonging to peninsular India are either coarse wooled (e.g. Deccani) or hair type (e.g. Mandya), whereas that of Himalayan region are wooled. Most of the Indian breeds have a low fleece yield, the annual total yield of greasy wool remains in the range of 1-1.5 kg. Wool produced from the tropical breeds of sheep is coarse and highly suitable for carpet manufacture. Thus, India is one of the chief carpet wool producing and exporting countries of the world. A large quantity of carpet wool is exported to the developed countries.

Shearing of wool is generally done twice a year in India with the help of special shearing scissors. It is a highly skilled task and should be given to experienced workers on contract basis. Sheep may be washed a few days before shearing, although it may not remove impurities like twigs and small stones. Shearing of soiled wool is sometimes referred as *dagging*. The entire coat of wool shorn from a sheep at one time is called *fleece*.

The term wool refers to the whole fleeces which contain mainly three types of fibres. *Fine wool fibres* generally have no medulla or hollow core and keep on growing e.g. Merino fibre has no medulla. *Hairs* are continuously growing long fibres with medulla in part of their length. *Kemps* are those coarse as well as short fibres which cease growing at intervals and are shed into the fleece (Fig. 20.1). Such fibres have medulla throughout their length. In the wool trade, fleeces itself are classified are *Merino (or fine), crossbred or medium, lusture long wool and carpet.*

Quality Characteristics

The important quality characteristics of wool are colour, fibre length, fibre diameter and clean dry yield. These characteristics account for nearly 80% value. The colour of wool is very important because presence of any pigment markedly decreases the possibilities of dyeing. The ideal colour is white or creamy white with conspicuous lusture. Canary staining or yellowing of wool is a general problem in Indian wool. It is caused by bacterial action at high pH and low grease content. Fine wool has a fibre diameter of 17-18 mμ (micron) whereas in coarse wool (Table 20.1) it may be nearly twice (35-38 mμ). Wools less than 30 mμ fibre diameter (non-medullated) are only suitable for clothing textiles. The coarser wools are generally used for carpet manufacture. To determine the proportion of fibres of each type, a representative sample containing atleast 200 fibres of a fleece are taken and sorting is done against a black background such a black velvet. All fibres showing chalkiness to the naked eyes are counted as hairs, unless those are kemps. Within the laboratory, medullation can be assessed by immersion of wool sample in xylol and examination under high magnification.

Table 20.1: Some characteristics of main types of wool

Wool type	*Use*	*Fibre diameter (mμ)*	*Clean dry yield (%)*
Fine wool	High quality, light weight woolen clothes	17-20	65-70
Medium wool	Good quality, heavier woolen cloths	22-24	70-75
Carpet wool*	Carpets	25-32	80-90

*Carpet wool should contain a mix of wool fibre and medullated fibre (hair) in the ratio of 65:25 (w/w). Kemp fibre should not exceed 4%.

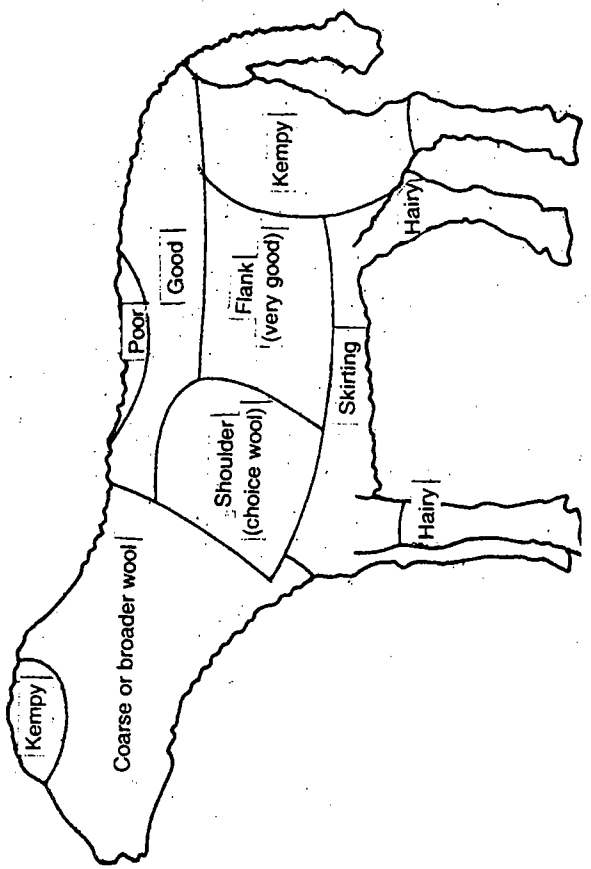


Fig. 20.1: Parts of the fleece

Fibre length is an important parameter from spinning point of view. In temperate climate, it is the growth of staple during 12 months of age. A fine wool is generally less than 6 cm in length whereas wool more than 10 cm comes in coarse grades. Clean dry yield refers to the yield of pure wool removing natural grease, vegetable material (grass seeds, burr etc) and soil. These impurities can range even upto 20 percent. Further, strength of the fibre is also an important property. It is affected by low plane of nutrition and chronic infection especially helminthiasis. In field conditions, it is measured by holding the ends of the staple between the fingers and giving a sharp tug. In fine wool, *crimp* which refers to the waviness of wool fibre is also considered as an important characteristic. It is measured in number of crimps per unit length. As far as elasticity is concerned, a fine wool fibre can stretch as much as 70% beyond its original length before breaking. Some defects generally encountered in wool are hairiness, impurities, skirting, lack of uniformity and cotts (pressing together of coarse fibres)

Shearing and Pulling

The shearing should not expose the sheep to temperature stress. It should not be done in winter or rainy season and early suckling or late pregnancy. The best time for shearing is shortly before the onset of summer. Rams are shorn before mating. The first shearing should be done at about 8 months of age. The most common shearing faults are inflicting the cuts or wounds to the sheep. There may be double cutting, thereby decreasing the length of the fibre. Shearing should neither be too near nor too far from the skin.

Pulled wool is obtained from the skins of slaughtered and dead animals by any of the three processes :

i. *Sweating* is the controlled putrefaction of skins in damp chambers at 20°C for 48 hrs. It loosens the hair follicles.
ii. *Painting* is a very good technique. A dehairing agent such as sodium sulphide is applied for loosening the hair follicles.
iii. *Liming* involves immersion of the pelt in lime water.

Post Harvest Handling

After shearing the fleece, poor quality material is separated and good quality material is grouped or classed. There are several main types of wool. *Lox* refers to wool contaminated with dung and urine alongwith wool from head, brisket and lower parts of the limb. *Skirting* refers to poor quality wool from bellies and edges. *Backs* refer to poor quality wool alongwith the back due to penetration of dust, whereas *brands* refer to wool from the areas stained with marking ink. The remaining material is classed on the basis of fibre length and diameter. The classified or grouped wool is pressed into bales. It should be stored at dry places.

Carpet wool is often a blend of coarse and medium wool and their proportion depends on the buyers preference or demand. In fact, carpet wool should have good resilience and ability to withstand hard use. Coarse outer fibres are necessary to give strength and a fine bulky undercoat is required to provide fullness and resilience. It should have 85% true wool by count or 65% by weight and 15% non-kempy medullated fibres (mostly hairs) by count or 35% by weight. Kemp is treated as a defect. More than 85% of the fibres should have a diameter of more than 25 mμ.

Quality Control of Carpet Wool

With a view to improving the quality of wool and establishing well recognized standard for its sale, the Government of India introduced a scheme for compulsory grading of wool before export. **Wool Grading and Marking Rules, 1961** define the quality of various types of wool, prescribe the grade specifications and lay down methods of marking, packing etc. Agricultural Marketing Advisor to Government of India issues Certificate of Authorization to the exporters.

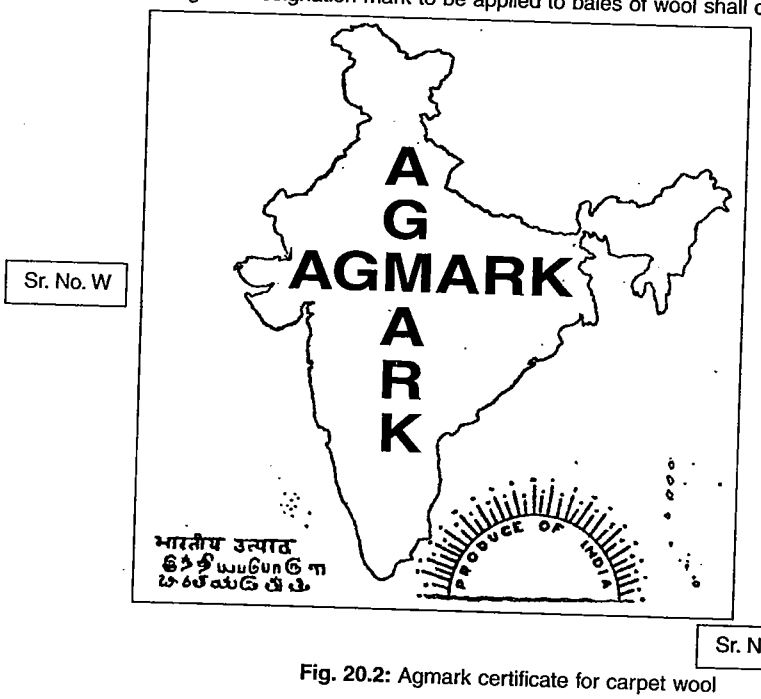
The quality of wool offered for grading under one lot should be uniformly prepared, properly cleaned and be reasonably dry in feel. It should not contain excessive extraneous matter like dirt, sand, moisture, raw fleece cuttings etc. except for a few unavoidables.

The wool is press packed in bales with complete covering of new gunny cloth and secured with sufficient number of iron hoops, tightly placed around the bale of customary commercial weights of 100 kg to 200 kg. A certificate of grading (Fig. 20.2) is issued by officer on behalf of Agricultural Marketing Advisor to Government of India.

Grade designations indicate the characteristics and quality of wool as described in Schedules I to VI.

Schedule VII

The grade designation mark to be applied to bales of wool shall contain the following designs:



Place of packing

Date of packing

Signature of Inspecting Officer

Grade

Colour

Yield percentage

Vegetable matter content

Nett weight

Place and date of marking

Signature of marking officer

Fig. 20.2: Agmark certificate for carpet wool

Schedule I : Indian Clipped Wool

Schedule II : Indian Pulled Wool

Schedule III : Indian Tannery Wool (limed)
a. Wools other than South Indian Tannery and Aden type.
b. South Indian Tannery and Aden type wools

Schedule IV : Indian Mixed Wool
a. Clipped—Carded
b. Clipped—Pulled

Schedule V : Indian Hill Wool, Greasy
a. Clipped

Schedule VI : Indian Ginned Wool

Specifications of Schedule I are reproduced here for better understanding:

SCHEDULE I

Grade designation and definition of quality of Indian Clipped Wool

Grade designation	*Colour of fibre*	*Special characteristics*		
		Yield percent of wool	*Maximum vegetable matter*	*General characteristics**
(1)	*(2)*	*(3)*	*(4)*	*(5)*
Clipped White	White	a) Over 80%	3%	All grades shall be free from gin-ned, carded, pulled and limed wool, wool waste, bleached wool, processed wool and any other animal fibre and also from burrs, thorns, sticks, sand, dust, greasy wool and other extraneous mater except for a few unavoidables and shall be reasonably dry in feel, homogenous in character and properly cleaned. No wool having a yield less than the minimum prescribed under columns 3 will be allowed to be exported.
Clipped Tinged White	Tinged White	b) Over 85%		
		c) Over 90%	–do–	
		a) Over 77%		
		b) Over 80%		
		c) Over 85%		
		d Over 90%		
Clipped Pale Yellow	Pale Yellow	a) Over 74%	–do–	
		b) Over 77%		
		c) Over 80%		
		d) Over 85%		
		e) Over 90%		
Clipped Yellow	Yellow	a) Over 74%	–do–	
		b) Over 77%		
		c) Over 80%		
		d) Over 85%		
		e) Over 90%		
Clipped Coloured	Coloured	a) Over 70%	–do–	
		b) Over 75%		
		c) Over 80%		

*Clipped wool shall be marked as 'Clipped Wool' on Agmark Labels and Bales.
Note: If the vegetable matter contents are over 3% and upto 5% in White, Tinged White, Pale Yellow, Yellow Wools and are over 4% and upto 6% in Coloured Wool, the same shall be marked on the bales and indicated in the Certificate of Agmark Grading. If the vegetable matter exceeds the upper limits mention in the preceding sentence the lot shall be rejected for export.

twenty-one

Bristles

Bristles are stiff, wiry hairs of pigs, hogs or boars. These are used for making various types of brushes. Bristles are generally obtained from back, neck and tail because those growing on flank and belly are too short to serve the intended purpose. Pig bristles are coarse and stiff in nature. They taper from base to the tip which shows splits or flagged appearance. In fact, flagged tips make them highly suitable for paint and varnish work due to their paint holding property. Shaving brushes made from soft bristles are superior to synthetic shaving brushes due to their water absorption and holding properties.

China holds the leading position in the production and export of bristles followed by India and Russia. Bristles are mostly obtained from indigenous and crossbreed pigs in our country,. Indian bristles are known for their coarseness and stiffness, so these are stout and strong but have longer flag (almost 30% of their total length) that is required to be trimmed before dressing. This factor decreases their value.

Bristles are plucked from living pigs or boars either once or twice a year. These are also pulled out from slaughtered or fallen animals. These are also obtained by shaving slaughtered animals after scalding in some bacon factories. Bristles obtained from live animals are superior in lusture and resilience than those obtained from fallen animals.

Characteristics

Colour of bristles is one of its special characteristics. On the basis of colour, bristles are classed as white, black or grey. It may be noted that all those bristles which are neither wholly black nor white are also classified as grey. White colour bristles are available only in western Uttar Pradesh and Punjab. Grey is the predominant colour of Indian bristles.

The length of bristles is another special characteristics. Longer bristles fetch more price. The thickness also increases with length in a particular quality. Thus longer bristles are stiffer as well.

Bristles less than 44 mm length are called *shorts and riflings*, whereas those from 44 mm to 159 mm and above are put in as many as 19 grade designations as per Bristle Grading and Marking (Amendment) Rules, 1973. Due to tropical conditions, Indian bristles generally vary in length from 57 mm to 159 mm. Bristles obtained from wild boars in Punjab and Madhya Pradesh are of good length. Superior quality black bristles called *'Darjeeling bristles'* are obtained from wild boars frequenting foothills of Himalayas. These bristles are similar to Chinese variety except for more flag. Darjeeling is the main market for such bristles.

On the basis of thickness, bristles are commercially classed as:

i. *Extra stiff*
ii. *Stiff/Semi-stiff*
iii. *Soft*

Extra stiff bristles are obtained from wild boars and are thicker and stiffer than stiff/semi-stiff bristles. Soft bristles are thinner than stiff bristles. White, soft and lengthy bristles fetch maximum price.

Post Harvest Handling

Raw bristles have to be dressed before packing. In this process, these are first soaked in warm water containing washing soda for a few hours, scoured with detergent, then thoroughly rinsed with cold water and dried. The dried bristles are collected again with the precaution that root and flag ends do not get mixed at any stage. These are then tied into bundles.

The dried bristles are sorted into grades of specific lengths by 'dragging'. In this process, nearly 500g bristles are tied with a strap on a small wooden platform with their root or butt ends down. The longest bristles are first pulled out by hand, followed by next smaller ones and so on. The separate sizes are tied into bundles of about 100g each. These bristles are again 'solid dressed' and finally graded into bristles of uniform length and size. Kanpur is the biggest dressing centre in India, the other important ones being Jabalpur, Allahabad, Gorakhpur etc.

Bristles meant for use in shaving brushes or export have to be sterilized to eliminate the chances of possible presence of anthrax spores and bacteria. It is accomplished by autoclaving them at 25-40 lbs pressure for about 1.30 hours. The wet disinfected

bristles are dried in hot air oven at 60°C or so to regain their stiffness.

Packing

Bristles of same thickness and grade lengths are tied in bundles. Each bundle of bristles (except for short and riflings) is made to a diameter ranging from 38 mm (1.5 inches) to 51 mm (2 inches). These bundles are packed in clean and dry wooden cases to have a net content of 10 kg or more in multiples of 2 kg subject to a maximum of 46 kg for shipment. However, tin or aluminium cases may be used for packing bristles for transit by air freight. Bristles of different grade designation lengths of 121 mm or below are packed in separate cases from those of over 121 mm. The wooden cases and other containers are lined with water proof paper and contain sufficient quantity of insecticide such as DDT or naphthalene balls.

QUALITY CONTROL OF BRISTLES

Indian bristles are exported to several European countries. These are shipped to London in cases of 100 lbs net weight from Bombay and Calcutta seaports. Bristles of 121 mm or above fetch very good prices due to heavy demand, white bristles are sold at about 25% higher rate than dark bristles. It may be noted that fine, soft bristles with small flag are highly suited for making shaving brush and painting flat varnish. As a trade malpractice, grey and black bristles are sometimes with ox or horse tail hair or vegetable fibres or artificial fibres. Some traders go to the extent of reclaiming bristles obtained from old and used brushes.

Government of India formulated Bristles Grading and Marking Rules, 1969, to have a control on the quality of bristles exported from India. These rules were further amended to be named as **Bristles Grading and Marking (Amendment) Rules, 1973.** As per these rules, there are XI schedules, each dealing with different commercial type of bristles:

Schedule I : Extra Stiff Bristles (White)
Schedule II : Extra Stiff Bristles (Black)
Schedule III : Extra Stiff Bristles (Grey)
Schedule IV : Stiff/Semi-Stiff Bristles (White)
Schedule V : Stiff/Semi-Stiff Bristles (Black)
Schedule VI : Stiff/Semi-Stiff Bristles (Grey)

Schedule VII : Soft Bristles (White)
Schedule VIII : Soft Bristles (Black)
Schedule IX : Soft Bristles (Grey)
Schedule X : Special Grade Of Bristles
Schedule XI : Select Grade Of Bristles

Schedules I, X and XI are reproduced below for better understanding and convenience :

SCHEDULE I

[See Rules 2(ii) and 3]

Grade designations and definition of quality of bristles known commercially as* Extra Stiff Bristles produced in India

Grade designation	*Special characteristics*		*General characteristics*
	*Colour***	*Length@*	
1	*2*	*3*	*4*
159 mm and above	White	159 mm and above	The bristles shall be the natural product obtained from pigs, hogs or boars and shall be thoroughly washed, clean and dry in their natural colour with root-ends intact and free from damaged or reclaimed bristles obtained from brushes as well as from all foreign mater such as dust, dirt or ox or horse tail hair or vegetable fibres or artificial fibres.
152 mm	–do–	152 mm	
146 mm	–do–	146 mm	
140 mm	–do–	140 mm	
133 mm	–do–	133 mm	
127 mm	–do–	127 mm	
121 mm	–do–	121 mm	
114 mm	–do–	114 mm	
108 mm	–do–	108 mm	
102 mm	–do–	102 mm	
95 mm	–do–	95 mm	
89 mm	–do–	89 mm	
83 mm	–do–	83 mm	
76 mm	–do–	76 mm	
70 mm	–do–	70 mm	
64 mm	–do–	74 mm	
57 mm	–do–	64 mm	
51 mm	–do–	51 mm	
44 mm	–do–	44 mm	
Shorts and Riflings (less than 44 mm)	–do–	Less than 44 mm	

*Extra stiff bristles are such bristles as are obtained from wild boars and are thicker and stiffer than Stiff/Semi-Stiff Bristles.
**The following colour tolerances shall be allowed:
a. In grades 114 mm and over, non-whites shall not exceed 3%.
b. In grades 108 mm to 44 mm, non-whites shall not exceed 6.25%.
c. In shorts and rifling, non-whites shall not exceed 12.50 %.
@All bristles shall have root-ends and flag ends pointed in opposite direction and shall be 'solid dressed' which term shall mean such dressing as to contain:

i. A minimum of 85 percent of the specified grade designation length (called 'tops'), the balance being made up of the next two lower grade lengths and unavoidables, the maximum tolerance for the latter being 5 percent in the case of grade designations 102 mm and over.

ii. A minimum of 75 percent of the specified grade designation length (called 'tops'), the balance being made up of the next two lower grade lengths and unavoidables, the maximum tolerance for the latter being 8 percent in the case of grade designations 95 mm to 64 mm.

iii. For grades 57 to 51 mm, a minimum of 75 percent of bristles of the specified grade designation length (called 'tops'), the balance being made up of the next lower grade lengths and unavoidables; and

iv. For grade 44 mm, a minimum of 65 percent of the bristles of the specified grade designation length (called 'tops'), the balance being made up of the next lower grade lengths and unavoidables; and

v. The following tolerance limits in respect of types shall be permitted:

Grade designation	*Maximum percentage of bristles of the next lower type according to stiffness to be tolerated*
114 m and above	35
83 mm and upto and including 108 mm	40
44 mm and upto and including 76 mm	45

SCHEDULE X

[See Rules 2(ii) and 3]

Special Grade of Bristles

In addition to the requirements specified for General Grades in Schedule I to IX, Special Grades of bristles shall satisfy the following requirements, namely:

1. i. Subject to the tolerance limits specified below, all bristles shall be free from nits (egg capsules left by hog lice); and

ii. Bundles of bristles shall not contain more than 8 infested hairs per bundle upto and inclusive of 95 mm in length; and

iii. Bundles of bristles shall not contain more than 5 infested hairs per bundle exceeding 95 mm in length

2. The following colour tolerance limits shall be allowed:

i. In grades 114 mm and over non-whites (in the case of whites) and non-blacks (in the case of blacks) shall not exceed 1 percent;

ii In grades 108 mm to 44 mm, non-whites (in the case of whites) and non-blacks (in the case of blacks) shall not exceed 2 percent; and

iii. In shorts and riflings, non-whites (in case of whites) and non-blacks (in the case of blacks) shall not exceed 3 percent.

3. All bristles shall have root ends and flag ends pointing in opposite directions and shall be 'solid dressed' which term shall mean such dressing as to contain:

i. A minimum of 85 percent of the specified grade designation length (called 'tops'), the balance being made up of the next two lower grade lengths and unavoidables, the maximum tolerance for the latter being 3 percent in the case of grade designations 102 mm and over.

ii. A minimum of 75 percent of the specified grade designation length (called 'tops'), the balance being made up of the next two lower grade lengths and unavoidable, the maximum tolerance for the latter being 5 percent in the case of grade designations 95 mm to 64 mm.

iii. For grades 57 mm and 51 mm, a minimum of 75 percent of bristles of the specified grade designation length (called 'tops'), the balance being made up of the next lower grade lengths and unavoidables; and

iv. For grade 44 mm a minimum of 65 percent of bristles of the specified grade designation length (called 'tops'), the balance being made up of the next lower grade length and unavoidables.

SCHEDULE XI

[See Rules 2 (ii) and 3]

Select Grade of Bristles

In addition to the requirements specified for General Grades in Schedules I to IX and for Special Grades and Schedule X, Select Grade of bristles shall be free from nits (egg capsules left by hog lice) and

i. Bundles of bristles shall not contain more than 16 infested hairs per bundle upto and inclusive of 95 mm in length; and
ii. Bundles of bristles shall not contain more than 10 infested hairs per bundle exceeding 95 mm in length.

twenty-two

Utilization of Organic Waste Available through Animal Industry

Animal farm residues such as dung, droppings, urine etc. have to be gainfully utilized. Slaughter house waste such as ruminal contents, blood, urine, meat and fat trimmings etc. are also required to be used to derive maximum advantage. We have to devise ways and means for a thorough and innovative utilization of all the inedible parts of the animal. In fact, utilization of slaughter house waste will serve dual purpose—it will alleviate the pollution problem besides providing light and heating source to the abattoir itself.

In Indian villages and suburban areas, buffalo and cow dung is converted into dried cakes and used as a cooking fuel. Dung compost, when applied to soil, retains its humus content and maintains its fertility. Of the total quantity of dung produced from buffaloes and cattle, almost 2/3 is used as fuel and 1/3 as manure. Thus, cow dung and urine are rich in organic matter and nitrogen. Cow dung is also being used to give a thin covering to the *kaccha* floor for ages. This cover serves as a temporary plaster and is again applied after a week or so. In several parts of India, sheep flock owners are paid for grazing their stock on fields after harvest for the value of manure. Grazing sheep deposit faeces and urine contributing to the soil fertility and crop yield. The organic waste available through animal industry can be utilized in several different ways.

Production of Biogas for Light and Heat

It involves the anaerobic fermentation of faeces, urine and slaughter house waste to produce biogas (largely methane). This process also yields a high quality manure as a byproduct.

A biogas producing plant comprises of three components:

i. *Digester*—a tank wherein fermentation causes production of gas
ii. *Gasometer*—a gas storage tank
iii. *Pipes*—for gas distribution at desired points

A digester is made up of bricks as a circular cylinder. In small plants, gasometer is an inverted iron movable cover of digester itself which rises or descends according to the gas it contains. The entire assembly has to be water and gas proof. The receptacle or inlet feeds the digester with animal blood, urine, dung, ruminal contents, effluents etc. at regular intervals. These organic and nitrogenous animal wastes undergo anaerobic fermentation. As a result, biogas consisting of nearly 60% methane, 30% carbon dioxide and traces of hydrogen, carbon monoxide etc. are produced. Methane burns with a blue alongwith traces of orange flame and does not form any smoke. Impure methane as produced in this plant burns in a mixture of 93% air and 7% gas.

During filling of digester with waste material, water is so added that it remains almost 0.5 meter above the waste. Some old sludge is also added for rapid fermentation. Gas production will start after a week. A discharge hole is provided near the bottom of the digester for periodic removal of the digested or spent slurry which is an excellent compost fertilizer. Biogas plants have great potential in augmenting the development of rural industries.

Production of Compost Manure

In places where no better utilization of organic matter of animal origin is feasible, it can be composted to yield manure of reasonably good quality. This manure contains nitrogen, phosphorus, potassium and trace minerals essential for plant growth and improved soil structure. Compost making units at some distance from the slaughter house will utilize the blood, ruminal and intestinal contents, bedding waste from lairages, meat and fat trimmings, floor washings, hair, feather etc. Even condemned meat and offal, if reduced to chunks and spread out in the middle of the heap, can be used in compost making.

Compost pit or bunkers can be made up of bricks leaving enough open spaces in the walls. Coarse material such as maize or millet stalks or small twigs are laid at the bottom for proper ventilation. Then alternate layers of ruminal and intestinal contents or vegetable matter and slaughter house waste are laid upto 2 meters. The outer side and top of the heap is protected by clean soil or grass. In tropical climate, the temperature inside the heap will go upto 75°C killing the larvae and checking the entry of undesirable pests. In dry weather, it has to be wet periodically to allow aerobic bacterial decomposition. That is why proper ventilation of the heap is also essential. Some people use rejected pieces of corrugated sheets to ensure it. To achieve uniform decomposition, the material has to be turned atleast three times after 20, 40 and 60 days. In about 90 days, the material is fully decomposed into compost manure.

Production of Animal Glue

Hide trimmings and fleshing, pig skin, sinews, tendons scattered and weathered bones and their pieces can be used for the production of glue. Horns and hoofs, ears, lips, snouts, tail etc. can also be utilized for the same purpose. It requires simple equipment and people with very little education can be trained to produce this good value material. Glue is extensively used in match, paper and rug industries, plywood making, carpet sizing and production of imitation leather.

Production of Pig Feed from Paunch and Intestinal Contents

The average weight of the paunch contents of cattle, sheep and lambs are 27, 2.7 and 1.7 kg, respectively. The liquid fraction of this material can be separated from fibrous residue by using a press. The fibrous fraction can be used as a fuel whereas the protein in the paunch contents can be concentrated for utilization as pig feed. It has been suggested that this use will reduce the cost of disposal of paunch contents by about 50%.

Slaughter house waste posed a big disposal problem for a long time but it has now been realized that with the advancement of technology, this waste can be recycled to several different uses to derive financial and sanitary benefits. In this context, simple equipment should be designed and available manpower within the country should be engaged. It will create new employment and skills as well.

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